

Degree Project in Integrated Product Design:
Innovation Management and Product Development

Second Cycle, 30 Credits

Balancing Reactive Development and Proactive Innovation

The challenges of managing innovation in a large defense manufacturing firm and how a systems approach based on ISO 56001 can mitigate them

ADAM CEHLIN

Balancing Reactive Development and Proactive Innovation

The challenges of managing innovation in a large defense manufacturing firm and how a systems approach based on ISO 56001 can mitigate them

Adam Cehlin

6/18/2026

Master's Thesis

Examiner
Sofia Ritzén

Academic adviser
Mats Magnusson

Industrial adviser
Edris Safavi

KTH Royal Institute of Technology
School of Industrial Engineering and Management (ITM)
Department of Engineering Design
SE-100 44 Stockholm, Sweden

Abstract

Large firms typically face a paradoxical challenge: combining the efficient execution of current operations (exploitation) with the development of disruptive future technologies (exploration). For Saab Surveillance, a major defence contractor experiencing record-high order backlogs, this tension is especially pronounced. While top management has set ambitious strategies for proactive innovation, the organization has yet to achieve meaningful transformation. Investigating how to achieve this dual focus is therefore highly relevant: navigating this conservative industry requires new approaches, which the recently introduced ISO 56001 innovation management standard can provide.

To address this challenge, an exploratory case study was conducted. Qualitative data from interviews and quantitative data from a survey assessing ISO 56001 adherence were gathered and synthesized. These findings were then presented in a workshop to validate the results and jointly develop recommendations.

The study reveals that Saab Surveillance moderately fulfils the requirements of ISO 56001. While leadership intent and portfolio planning demonstrate strengths, gaps exist in operational support for innovation, innovation processes, and performance evaluation. A systemic misalignment was found between strategic goals and the practical reality of middle managers, who frequently must divert resources toward short-term efficiency to deliver on customer commitments. Furthermore, statistical analysis established a significant positive correlation between support for innovation and radical innovation performance.

Based on the synthesized data, the study recommends that Saab Surveillance focus on three main areas for change: establishing measurable, holistic innovation objectives; creating dedicated pathways connecting ideation portals directly with rapid-prototyping labs; and designing high-trust partnerships to overcome confidentiality silos. Combined with a systems approach to management, adopting these changes will allow the organization to better balance exploitation and exploration, improve the conversion of ideas into commercialized solutions, and increase ISO 56001 compliance.

***Keywords**: Innovation management systems, ISO 56001, Exploration and exploitation, Systems theory, Defence industry, Organizational ambidexterity*

Sammanfattning

Stora företag står ofta inför en paradoxal utmaning: att balansera ett effektivt genomförande av sin nuvarande verksamhet (exploatering) med utvecklingen av banbrytande framtida teknologier (utforskning). Saab Surveillance är en betydande försvarsaktör som de senaste åren upplevt rekordstora orderstockar, vilket gjort denna utmaning särskilt påtaglig. Trots att den högsta ledningen har satt upp ambitiösa strategier för nyskapande innovation, har organisationen ännu inte uppnått någon markant transformation. Att undersöka hur denna balans kan uppnås är därför högst relevant och kan kräva nya arbetssätt, vilket den nyligen introducerade standarden för innovationsledning ISO 56001 kan erbjuda.

För att undersöka detta genomfördes en explorativ fallstudie. Kvalitativa data från intervjuer och kvantitativa data från en enkät som utvärderade företagets kravuppfyllnad av ISO 56001 samlades in och analyserades tillsammans. Fynden presenterades sedan i en workshop för att validera resultaten och gemensamt utveckla rekommendationer.

Studien visar att Saab Surveillance till viss del uppfyller kraven i ISO 56001. Även om chefers driv kring innovation och projektportföljsplanering utgör styrkor, finns det brister i operativt stöd för innovation, sammanfogandet av innovationsprocesser och prestationsutvärdering. En systemisk obalans identifierades mellan strategiska mål och den praktiska verkligheten för mellanchefer, som ofta inte har ett annat val än att fördela resurser mot kortsiktig effektivitet för att kunna leverera på kundåtaganden. Vidare fastställde den statistiska analysen en signifikant positiv association mellan stöd för innovation och radikal innovationsprestanda.

Baserat på datan rekommenderar studien att Saab Surveillance fokuserar på tre huvudsakliga förändringsområden: att etablera mätbara och holistiska innovationsmål, att skapa dedikerade processer som kopplar idéportaler direkt till labb för snabb prototyptillverkning, samt att utforma ett ramverk för förtroendefulla samarbeten för att minska silos både internt och externt. I kombination med ett systemperspektiv kan införandet av dessa förändringar att göra det möjligt för organisationen att bättre balansera exploatering och utforskning, förbättra omvandlingen av idéer till kommersialiserade lösningar samt öka efterlevnaden av ISO 56001.

***Nyckelord**: Innovationsledningssystem, ISO 56001, Utforskning och exploatering, Systemteori, Försvarsindustri, Forskning och utveckling, Innovation*

Acknowledgments

This Thesis was conducted at KTH in collaboration with Saab in Stockholm. I would therefore like to thank Saab for providing me with this opportunity, and the devoted employees who contributed with fantastic insights. I would also like to extend my most sincere gratitude to Edris Safavi and Jonas Lagerqvist for your continuous support and engagement during this project.

Finally, I would like to acknowledge Mats Magnusson for his vital role as my academic supervisor. His guidance and continuous contribution of extensive knowledge and insightful advice gave me the tools I needed to complete this work. Furthermore, his cheerful attitude kept me motivated throughout the process.

As I write this, I reflect upon all the years I have spent not only at university, but also in school throughout my life. I am incredibly grateful for the opportunities that I have been given by my parents, the support I have received from my close family and friends, and the experiences I have made along the way. Thank you.

Stockholm, June 16th

Adam Cehlin

Table of Contents

1. Introduction **2**

2. Context **4**

2.1. History 4

2.2. Internal Context 4

2.3. External Context 4

3. Theoretical framework **6**

3.1. Combining Exploitation and Exploration 6

3.2. Systems Design: Applying A Systems Perspective to Innovation Management 8

3.3. Systemic Management: Key Factors Of An Innovation Management System 10

3.4. The Innovation Management System Standard: ISO 56000 Series 15

3.5. Research Questions 20

4. Methodology **22**

4.1. Qualitative Data: Exploratory Interviews 23

4.2. Quantitative Data: ISO 56001 Adherence Survey 26

4.3. Qualitative Data: Workshop For Validation And Recommendations 34

4.4. Secondary Data Collection: Literature And Internal Documentation 34

4.5. General Method Analysis 35

5. Results and analysis **38**

5.1. Qualitative Data: Exploratory Interviews 38

5.2. Quantitative Data: ISO 56001 Adherence Survey 45

5.3. Qualitative Data: Workshop Results And Analysis 55

6. Synthesis **58**

6.1. Convergent Findings 60

6.2. Areas Of Divergence 62

7. Discussion **64**

7.1. Main Findings and Answers to Research Questions 64

7.2. Theoretical Contributions 74

7.3. Practical Contributions 74

8. Limitations **76**

9. Future Work **78**

10. Works Cited **80**

11. Appendices **86**

11.1. Appendix A: Observation Validation 86

11.2. Appendix B: Quantitative Data 87

11.3. Appendix C: Interview Guide 96

11.4. Appendix D: Questionnaire 97

11.5. Appendix E: Generative AI Statement 101

11.6. Appendix F: R Code Used for the Quantitative Data Analysis 102

1. Introduction

Innovation is a word often associated with boldness and excitement. It has its roots in renewal, and has come to be defined in many ways, but the most widely agreed upon definition describes it as a new or changed entity, realizing or redistributing value (ISO, 2025). According to this definition, innovation is governed by two dimensions – novelty and value, and must therefore be perceived as both new and value adding by the founding organization and by relevant parties of interest. The term innovation is the final outcome, and it is preceded by a number of activities that support the development of new or valuable products, services, processes, models, methods and more. These activities include, but are not limited to, market analyses, internal research, ideation and conceptualization, prototyping and experimentation, external knowledge acquisition and partnerships, and idea collection forums (OECD/Eurostat, 2018). These activities can be supported by a number of internal and external sources, such as employees, academic institutions, and customers (Demircioglu, et al., 2019). Tying these aspects together is innovation management, a field that in recent years has grown to become more reputable and legitimized. Organizations are actively appointing innovation management professionals, and certification programs offered by private consulting firms, professional associations, and global certification bodies are rising in popularity (Kihlander, et al., 2022). These developments are similar to what the quality and project management disciplines have seen years prior, both of which are highly legitimate fields and professions.

However, innovation management is a vastly different field, and therefore requires unconventional methods (Jalonen, 2011). For example, unlike project and quality management, innovation management is not about minimizing risk. Instead, it is about actively managing it through proactive measures and experimentation to reduce uncertainty and validate assumptions, and being ready to pivot when new knowledge or circumstances change the playing field (Van de Ven, 2017). This requires specific management, tools and organizational structures, and the mentality around it can require long and continuous efforts of organizational transformation to achieve, due to its stark contrast to typical management (Jalonen, 2011). Additionally, the exploration that innovation management contributes to must be combined with the efficient exploitation of the existing business even though the two approaches have contradictory goals, creating a paradoxical managers dilemma (Benner & Tushman, 2003; O'Reilly & Tushman, 2004). This tension, and the effort required to manage it is among many reasons why organizations fail to achieve their innovation goals and growth targets. It may also be a contributing reason why 94% of executives are dissatisfied with their innovation performance, even though 84% of them state that innovation is critical (McKinsey, 2015). For the six percent that are satisfied and for the companies that do succeed, the benefits of managing the tension between exploitation and exploration are striking – for example, they are reportedly "nine times more likely to create breakthrough products and processes [...] while sustaining or even improving their existing businesses." (O'Reilly & Tushman, 2004). This shows the importance of not neglecting innovation efforts in the quest for operational efficiency, yet delivering on current customer commitments is equally critical.

In exploring this tension and the innovation management field, Saab Surveillance represents a compelling case study. Between 2022 and 2025, Saab AB's sales have grown by 46.9%, propelling their order backlog and value to all-time highs (Saab, 2025). Order bookings increased by 74 per cent in 2025, mainly attributable to large orders which grew by 117 per cent, and the company reported an organic growth of 25.6 per cent (Saab, 2025). Although this has had an immensely positive effect on market expansion efforts and revenue, the increased order backlog is straining the company's production capacity and delivery rate. It is noted that failure to deliver on customer commitments in terms of technology, cost and time is one of the

top risks for Saab in the coming years (Saab, 2024). This will be described as **Frontier 1: Exploitation**. Simultaneously, the top management of Saab are looking toward the future. As part of the *Target State 2024*, Saab aims to manage innovation as business critical, support its innovation performance, and have an innovation-friendly culture (Saab, 2025). The company is devoted to "investing in future capabilities" (Saab, 2024) by re-investing earnings from sales into R&D of the core portfolio and future potential innovation. More specifically, in its innovation strategy, the company highlights customer involvement in development, collaboration in development, speed to market, and achieving disruptive solutions as major targets. This will be referred to as **Frontier 2: Exploration.** The rationale behind pursuing this frontier is well summarized in a quote by Saab's Chief Technology Officer, Petter Bedoire:

> *"We have realized that we need to start thinking in a new way. For many years we have had customers who were very frugal and cautious. It has been more important to provide something at a very low cost than to deliver the best that is available on the market.*
>
> *Now we are in a completely different situation, where we must begin to take risks. We cannot wait for a customer to order something; instead we have to stay ahead and develop based on what we believe the customers want. It is a massive shift." – Petter Bedoire, SAAB CTO (Lindsten, 2025)*

Both frontier one and two are ambitious targets on their own, but Saab plans to pursue both simultaneously. Achieving this will demand a structured innovation management effort that explicitly manages the tension between stable, risk-averse exploitation and the uncertain, experiment-driven nature of exploration. A systemic, systematic, and effective framework that aligns innovation activities with organizational strategy, balances exploitation and exploration, and sets a new trajectory for the entire organization – something which the newly introduced ISO 56000 family of standards can potentially deliver. This master's thesis will therefore explore how Saab can reshape its innovation management with inspiration from the ISO 56000 series and systems theory in order to meet its goals, while staying true to its core purpose of "Keeping people and society safe".

2. Context

This chapter aims to introduce the reader to the contextual landscape of the company in focus, and the broader industry which it operates in. Contextual factors that have a significant impact on the company's strategies, governance and management will be described.

2.1. History

Saabs inception in the 1930's arose from a need of military resilience and independence for Sweden. The early developments of fighter jets were a remarkable achievement for a company of its size, and the company quickly became a crucial actor securing Sweden's defense capabilities. Since, Saab has continued the development of its aircraft and diversified into adjacent defense sectors such as signal processing, naval systems, and weaponry, becoming Sweden's primary defense company and one of the country's largest employers. These efforts have ultimately consolidated Sweden's defense industry into a single, dominant actor, and the company is now a global exporter of products, ranging from specialized components to whole aircraft systems.

2.2. Internal Context

In large parts of its history, Saab has operated in an insulated environment shielded from uncertainties, with a product development process characterized by clear requirements and incremental development, and with few but loyal customers. Although these customers have been loyal, their orders have been limited, often cost-conscious, and with rigid order requirements, leaving little room for creativity. Furthermore, due to the sensitive nature of its products and services, and the rampant prevalence of military espionage, the company is managed with strict security principles and guidelines. These internal contextual factors are typical in the defense industry, but have adverse effects on the company culture – it has bred a culture of reactive and cost-cautious development, and the sensitivity surrounding security has caused systematic silo-fication of business areas and units within them, thus hindering knowledge sharing and openness. As one employee put it – "*Ideas are born where thoughts meet*" – underscoring the importance of openness and knowledge exchange between units, which has proven to be difficult in the defense industry.

2.3. External Context

Externally, the defense industry has for the last 60 years had stalling development, especially in smaller economies (SIPRI, 2025). The increasingly diplomatic geopolitical climate and decreased risks of global conflict have caused nations to become more cautious about military spending and defense budgets, especially Sweden, one of Saabs primary customers. This is of great importance, since the industry is heavily dependent on government expenditure on defense. This military drawdown has led to decreased investment and development across the industry (Dehoff, et al., 2013). This is evident in Figure 1, which shows how the military expenditure as share of GDP in the United States, the United Kingdom, France, and Sweden has decreased, on average.

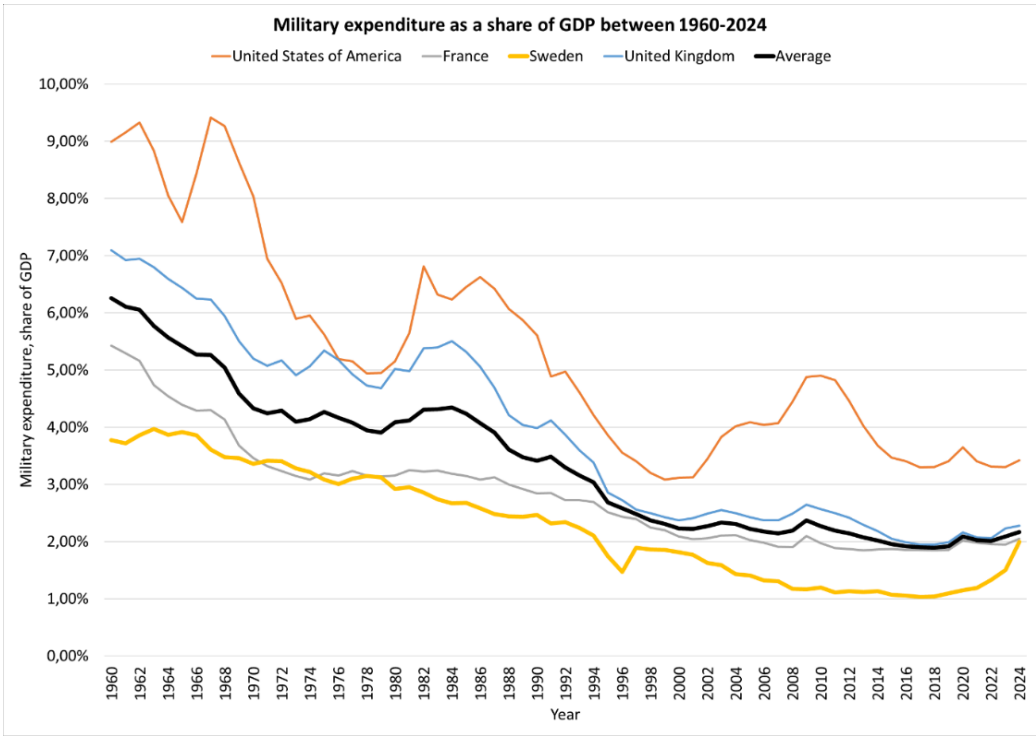


Figure 1: Visualization of military expenditure share of GDP vs. year, adapted from SIPRI, 2025

This trend, which has decreased the lucrativeness of innovation, has been persistent until recent years. With threats of global conflict looming, governments are increasing their military budgets and share of GDP devoted to military expenditure (Liang, et al., 2025). Sweden, among other nations, have vouched to devote 2.8% of GDP toward defense in 2026, a new record high representing an 18% increase (Government offices of Sweden, 2025). From the year 2022 and onward, the beginning of this rise is visible in Figure 1. In Europe, a quest for military independence and *strategic autonomy* is developing (Bran & Bălășoiu, 2025), potentially due to the historical reliance on the NATO alliance is being questioned. Authors are describing the situation as Europe's "sudden correction to historical defense underinvestment" (Caverley & Kapstein, 2025), which has caused a surge of orders to global defense actors like Saab, both in Europe and internationally.

The seasonality and instability of demand prevalent in the defense industry has a great impact on business strategy, increasing the importance of maintaining a portfolio where parts of the organization are divested or invested further into following market turbulence (Dehoff, et al., 2013). Such environmental turbulence can have a negative impact on core business performance, but paradoxically stimulate innovation as it motivates firms to take action (Turulja & Bajgoric, 2019). Firms must therefore establish systems to capitalize on the opportunities turbulence creates while divesting from no longer relevant parts — a process potentially facilitated by systemic innovation management.

3. Theoretical framework

The aim of this chapter is to present a state-of-the-art framework of systems theory, innovation management principles, and other theories relevant to the topic of this thesis.

3.1. Combining Exploitation and Exploration

Combining exploitation and exploration represents the concept that firms should both exploit their current knowledge, capabilities and offerings, i.e. *core* business in efficient ways in order to maintain operations and grow, while simultaneously exploring and capitalizing on future business opportunities (O'Reilly & Tushman, 2004). Exploring may entail improving current offers, and researching entirely new value offerings, often in new markets, with the end goal of improving business opportunities and staying ahead of competitors (O'Reilly & Tushman, 2004). It has been found that large firms giving attention to both exploration and exploitation achieve greater sales growth, while firms having an imbalance between exploration and exploitation instead show a negative correlation to sales growth (He & Wong, 2004). This emphasizes the importance for Saab Surveillance to seek this combination.

However, despite being researched and promoted for decades (March, 1991), companies face the dilemma of managing this tension yet today. This is often referred to as organizational ambidexterity, originating from the trained ability to have equal control over ones left and right hand. Thus, the term serves as an analogy for simultaneous management of exploiting the core business while exploring new opportunities. However, this theoretical exposition will refrain from using the term due to its connotations, and instead highlight exploitation and exploration separately to assess the dual focus that Saab seeks to realize through its Frontier 1 and Frontier 2 initiatives.

3.1.1. How Firms Can Support Both Exploitation and Exploration

Organizations must deal with the paradoxical, strategic contradiction that exists between exploitation and exploration. Although both agendas share the overarching goal of sustaining the organization, they differ dramatically in time horizon, strategic priority and the criteria for success. In practice, combining the two development areas is a highly difficult task, and if an organization lacks the enabling components, such efforts are highly likely to fail (O'Reilly & Tushman, 2011). These components include a compelling strategic intent to shift resources from exploitation to exploration, a shared vision and values that endorse the shift, senior-team consensus, separate structures for each agenda with appropriate management, and leadership that can make the required trade-offs between exploitation and exploration (O'Reilly & Tushman, 2011). More specifically, findings suggest that leaders play a significant role in achieving both exploitation and exploration (O'Reilly & Tushman, 2011). Consequently, separate organizational structures, performance-measurement systems, and resource-allocation rules for exploration and for exploitation must be set up, and simultaneously integrated so that they reinforce each other and generate synergistic value for the firm (Smith & Tushman, 2005).

3.1.2. Managing The Tension Between Exploitation and Exploration Through Dedicated Portfolio Management

An influential model proposed by Nagji and Tuff in the 2012 Harvard Business Review article *Managing Your Innovation Portfolio* discusses portfolio management as a key factor of combining exploitation and exploration. Their model presents a two-dimensional diagram with the axes of *where to play* and *how to win*, ranging from *existing, adjacent and new* markets, and *existing, incremental and new* products and assets, respectively (Figure 2). The diagram includes three zones: The first zone represents the *core* business, i.e. exploitation. The second zone represents developments *adjacent* to the core, and the third zone signifies *transformational* activities. Adjacent developments attempt to capture future solutions related to, but completely differentiated from the core business. Transformational development, on the other hand, aims for true transformation unrelated to the core business, although both adjacent and transformational activities can be considered exploration. The model can be described as a scale of innovation intensity: core activities correspond to largely incremental improvements, adjacent activities represent a moderate degree of radical innovation, and transformational activities the most radical innovations.

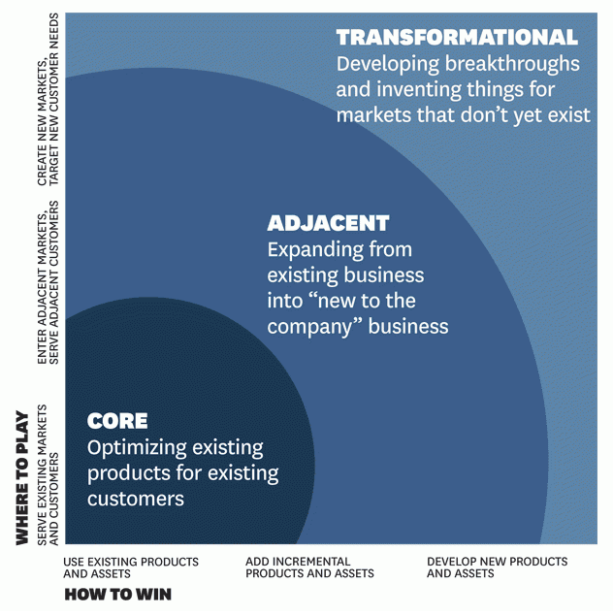


Figure 2: The Innovation Ambition Matrix from Nagji and Tuff (2012)

Aside from the model, the authors present a pattern in distribution of innovation budget which correlated to meaningfully higher share price among studied companies. These firms, dubbed *outperformers*, typically allocated "about 70% of their resources to core offerings, 20% to adjacent efforts, and 10% to transformational initiatives." (Nagji & Tuff, 2012), further finding that "returns from innovation investments tend to follow an inverse ratio, with 70% coming from the transformational realm." (Nagji & Tuff, 2012). This distribution provides a concrete exemplification of how to combine exploitation with more explorative efforts.

The model and findings have had an influential role in innovation management research, and are a point of high discussion within organizations striving for innovation. It underscores the need to refine the existing asset portfolio to sustain a competitive advantage, while simultaneously investing in forward-looking development that has the potential to deliver future financial benefits, as seen from the results of the study. Consequently, firms must pursue

both incremental and radical innovation to remain competitive, raising the question of how an IMS can be designed to continuously support and combine these two complementary approaches. Which leadership practices to employ, what structures and processes to follow, and how resources shall be allocated between innovation projects are critical considerations, to which the ISO 56000 series could offer useful guidance (Karstegl, et al., 2025). These questions introduce problematic trade-offs and decisions that organizations must face.

3.2. Systems Design: Applying A Systems Perspective to Innovation Management

Systems theory has its roots in a wide variety of knowledge domains, including biology, physics and social sciences (Mele, et al., 2010). The theory stipulates how complex systems, such as organisms and organizations, can be studied, interpreted, and managed by adopting a more holistic perspective where attention is shifted from parts to the whole (Jackson, 2003). The theory emphasizes interactions (Mele, et al., 2010). They can exist externally between the system and the environment it resides in, as described in *Open Systems Theory* (OST) (Katz & Kahn, 1978), or internally between system elements, explored in the *Viable System Approach* (VSA) (Golinelli, 2005).

In organizational management, VSA is especially relevant, as it includes both the internal and external perspective – describing internal component interactions as *sub-systems*, and external connections between organizations and other actors in their context as *supra-systems* (Golinelli, 2005). Internally in the sub-system, elements are structurally related, but ideally systemically interact in sharing resources and knowledge in order to reach a common goal (Golinelli, et al., 2002). It distinguishes between simply having different system elements in a structure, versus actual interactions between those elements that drive alignment and change. This highlights the importance of both internal and external interactions: The capability of exchanging resources within the system and with the environment around it, such as other firms, is crucial for survivability in continuously changing contexts. This contextual response aspect of system is also corroborated by Ashby's 1956 theory about *requisite variety*. It states that for an organization to control or adapt effectively to its environment, the organization must possess at least as much variety as the variety of conditions, disturbances, or demands it encounters. If the organizational system's internal variety, like its number of distinct structures, processes, or responses is lower than that of its surroundings, there will be situations it cannot cope with or learn from (Ashby, 1956). As a result, "viable organizations are able to survive as a result of changes (adaptations, transformations and restructuring) in their logical and physical aspects … and in their specific configuration over time" (Golinelli, et al., 2002), highlighting the intrinsic need for rejuvenation and, in the context of this report, the importance of innovation across the system.

To anchor systems theory closer to organizations and their operations, Slack and Lewis' (2011) operations strategy model can be considered. It describes how decision areas within operations (*Capacity, Supply Network, Process Technology, and Development and Organization*) align with key performance objectives (*Quality, Speed, Dependability, Flexibility, and Cost*). Although the model is a tool for evaluating critical areas for achieving operational performance and efficiency, it implicitly describes key aspects of a system through its 'fit' evaluation criteria. A 'fit' is achieved when operations resource capabilities align with the requirements of its markets (Slack & Lewis, 2011), and hinges on the core dimensions Comprehensiveness, Coherence, and Correspondence. With the perspective that an organizations' operations are part of, and acts as a system, Slack and Lewis' model can be extrapolated and applied when

evaluating innovation management systems. The dimensions will be described further in this section.

3.2.1. Comprehensiveness – How Is The System Managed In A Holistic Way?

The first dimension concerns to what extent the strategy considers all decision areas and key performance objectives (Slack & Lewis, 2011). If the strategy lacks comprehensive breadth across all operational decision areas, and all performance objectives, crucial factors may go unnoticed and eventually threaten operations. In systems-theoretic terms, comprehensiveness can be evaluated through the sub- and supra-system of the organization. To be comprehensive, a viable sub-system must have the necessary components, and clear interaction pathways between all of them - for example that that resources, information and knowledge shall be shared across teams and units. At the same time, the supra-system of suppliers, customers, and other outside actors that interact with the organization must also account for every relevant external exchange, ensuring that the system interacts with all relevant stakeholders. This relates to how the system must possess enough complexity to cope with changes in its environment, as per requisite variety (Ashby, 1956).

3.2.2. Coherence – Are Strategic Updates Coordinated Across Different Subsystems?

The second dimension, coherence, relates to how to maintain an evolving operations strategy. Like all organizational areas, the strategy must continuously be iterated to ensure its alignment with internal and external opportunities and threats (Slack & Lewis, 2011). However, when such changes are made, strategic coherence must be considered: When the strategy is changed in one function, other functions that also are influenced must adapt to follow the change. This alignment can prove challenging due to the numerous decision areas affected (Slack & Lewis, 2011), but ensures continuous comprehensiveness and coherence in the strategy. From a systems perspective, coherence relates to VSA, where the system benefits from the exchange of information and resources within itself and with its environment. Therefore, coherence requires the whole system to remain synchronized to prevent any part from diverging from the whole, to ensure that systemic interactions within the sub-system are preserved. Adaptability is emphasized, with the focus that the system shall have the possibility to re-align in an integrated manner when new opportunities or threats appear, again tangential to the Viable System Approach.

3.2.3. Correspondence – Are Actions Aligned With The Core Strategic Values?

The third dimension stresses that strategic decisions must be reflected in the prioritized performance objectives. When responding to changing contexts through the coherent re-alignment described previously, the system must continuously consider its strategic weighting of objectives for correspondence. In other words, if an organization places the greatest weight on flexibility, the ways that it works with capacity planning, supply-network design, process-technology selection and development must all support that priority. Choosing low-cost, off-the-shelf suppliers when flexibility is the dominant goal would therefore break the correspondence, because such suppliers typically cannot provide the rapid re-configuration or customization that a flexible operation needs (Slack & Lewis, 2011). Connecting to systems theory, correspondence affects both the sub-system and the supra-system, whose resources

and behaviors must be configured for every interaction to support the organization's chosen performance goals.

The same principle is held for an innovating organization. If speed is a top-ranked innovation objective, the IMS must provide the tools like rapid-prototyping equipment and organizational mechanisms like agile, short-cycle working methods that enable quick idea testing. If the IMS instead includes strict requirements specifications, the system will be more geared toward quality maximization, and be misaligned with its performance priority. Ultimately, the correspondence is then lost.

3.3. Systemic Management: Key Factors Of An Innovation Management System

Aside from the aforementioned components of the systems approach, the ability of an organization to innovate is dependent on several interconnected factors, such as, leadership, resources, culture, structures, processes, and more. These components and the links between them are essential in managing innovation (Karlsson & Magnusson, 2019).

Early innovation management literature regarded components of innovation management, such as processes, strategy and employees as mutually exclusive and individually contributing to performance (Smith, et al., 2008). The relationships between these components and the performance benefits achieved through their combination (Bititci, et al., 2011) were given less attention (Smith, et al., 2008) – An inter-factor, systemic management view has been missing from the innovation-management of organizations (Smith, et al., 2008). But with the evolution of the field, this more holistic view has become increasingly common. Much of this development has been promoted through an influence of systems theory (Midgley & Lindhult, 2021), and applying such an approach to innovation management has been praised for introducing a new perspective on managing innovation. Despite this, the perspective has also been disputed due to its sometimes abstract nature that may offer low applicability for practitioners (Teece, 2018).

Nevertheless, the systems approach is one of the drivers behind Innovation Management Systems, around which the new ISO 56000 standard series is designed. According to ISO (2025), an IMS is "a set of interrelated or interacting elements of an organization to establish policies and objectives, as well as processes to achieve those objectives [...] with regard to innovation". ISO's wording not only mentions the elements and processes that constitute an IMS, but also highlights the critical importance of their interconnectedness and interactions. The definition highlights the comprehensive and holistic nature of IMSs, and mirrors the Viable Systems Approach's core premise: that both the interactions of internal elements and the system's external exchanges with its environment are essential for the organization to achieve its objectives (Golinelli, 2005). Using systems thinking can give increased attention to the linkages between system elements (Karlsson & Magnusson, 2019), and the consideration of new perspectives through systems-focused questions, such as: "can we change the relationships between the parts of the system [...]? If we do, how would this impact its emergent properties?" (Midgley & Lindhult, 2021).

Therefore, in assessing an innovation management system, it is not only relevant to consider that foundational processes, tools and methods are employed – one must also consider the interactions between these elements (Kihlander, et al., 2024), and the interactions between the IMS and its environment (Chen, et al., 2025). The graphic below provides an exemplification of this concept

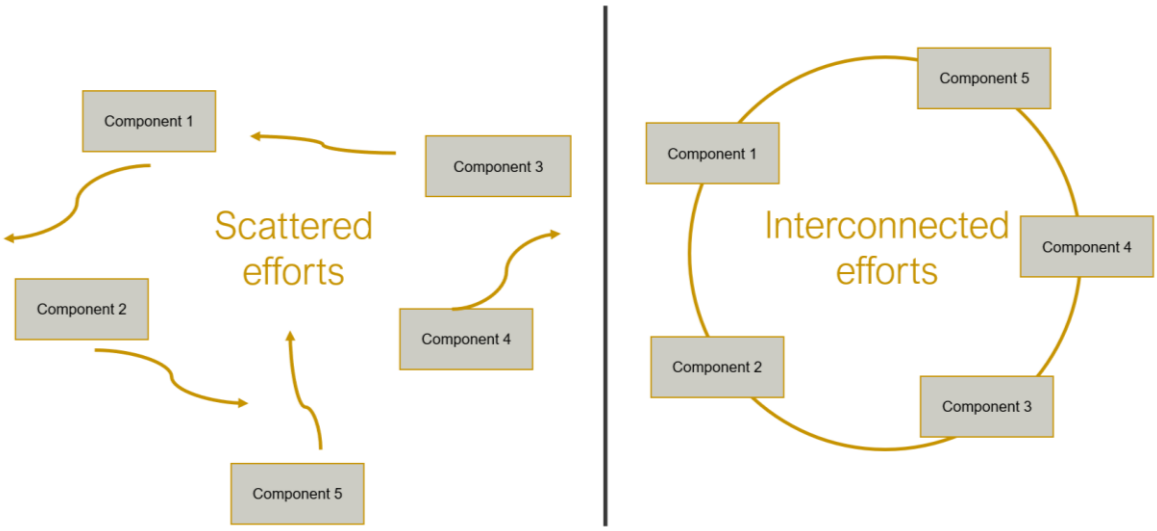


Figure 3: Visualization of the interactions between innovation management system components

To differentiate between the internal elements (like structures and methodologies) versus the interactions between them, the present report argues for a distinction of factors present within IMSs: systematic and systemic factors. The literature features conflicting definitions of these terms, but the following descriptions will guide the continued analysis:

- Systematic factors refer to the concrete methods, tools and established processes (Cambridge Dictionary, 2026) that enable the generation and development of innovation
- Systemic factors, conversely, are defined by Cambridge Dictionary (2026) as "relating to or affecting the whole of a system, organization, etc. rather than just some parts of it". Consequently, these factors concern the degree to which systematic elements are interconnected, the quality of their interactions, and the extent to which the IMS permeates the organization in a holistic manner.

As an analogy, the systematic aspects can be described as having the right tools, whereas the systemic factors act as the driving force behind, and glue between the tools. With poor systemic management, a firm may have implemented systematic processes and working methods for innovation, but still experience dissatisfactory innovation performance and outcomes. Determining what elements contribute to systemic management is therefore crucial for assessing the systemic nature of an IMS, but because it is an emerging field, a consensus regarding the most influential drivers of it does not yet exist. The present report argues that Leadership, Strategy and Openness are core elements of systemic management, based on the following analyses.

3.3.1. Leadership – Aligning The Organization Toward A Common Goal

Leadership is a crucial aspect of all management systems, and is one of the key aspects that advances organizations. This also holds true when pursuing innovation (Van de Ven, 1986; ISO, 2024), hence why leaders must actively practice management styles that promote innovative behavior.

Among established leadership practices are transactional and transformational leadership. Transactional leadership entails clear performance expectations, the monitoring of task execution, and the provision of rewards or corrective action based on the degree to which followers meet said goals (Bass, 1985). Typical transactional mechanisms include "contingent rewards", i.e. financial, material, or immaterial compensation upon the completion of certain goals, and "active management by exception", meaning that deviations from standards or rules are systematically scrutinized (Bass, 1985). In contrast, transformational leadership relies on intellectual stimulation, inspirational motivation, and the leader acting as a moral- and charismatic role model, going beyond self-interest for the sake of the group (Bass, 1985). These leadership styles have since their formalization been studied with regard to innovation and shown significant results; Transactional leadership is more strongly associated with exploitative innovation, since it is associated with clear performance expectations, monitoring of task execution, and rewarding reliable, incremental improvements to existing products or processes (Jansen, et al., 2009). Transformational leadership, on the other hand, promotes exploratory innovation by encouraging employees to question existing assumptions, imagine novel solutions and pursue new strategic directions (Jansen, et al., 2009). It also directly empowers followers and cultivates an innovation-supportive climate (Le & Lei, 2019), thus contributing to fostering an innovative organizational culture (Jung, et al., 2003; Sattayaraksa & Boon-itt, 2018). This type of culture is the antecedent of developing innovation capabilities which lead to high innovation performance (Prajogo & Ahmed, 2006), highlighting the importance of transformational leadership. Therefore, the understanding of transactional and transformational leadership can enable organizations to design leadership development programs and behavioral interventions that deliberately shape the type of development and innovation they wish to achieve. In other words, by matching the appropriate leadership style to the desired innovation outcome, companies can more effectively direct their innovation efforts.

This underscores leaders' significant role in pursuing both exploration and exploitation. As discussed previously, it requires vastly different management styles and elements for success. To achieve this and prevent failure, organizations may reshape senior-team composition by removing managers who resist the pursuit of innovation while running the core operations, and create separate organizational structures for the exploratory and exploitative units, appointing dedicated managers to each unit who report directly to the CEO (O'Reilly & Tushman, 2011). They may also link executive incentives to the performance of both the core and the exploratory units, and repeatedly communicate a clear rationale for pursuing both exploration and exploitation (O'Reilly & Tushman, 2011). These actions show the importance of having leadership aligned with innovation – to have authority make these necessary decisions and inspire many more to follow. Driving these changes requires perseverance and unity in the leadership that can steer the firm through the inherent tension between exploitation and exploration, eventually contributing to both present performance and future renewal. This is highlighted in ISO 56001, which states that "ultimately, the effective implementation of the innovation management system relies on the leadership and commitment by top management and other leaders at all levels in the organization." (ISO, 2024). It may therefore be argued that the foundation of innovation performance rests on the leaderships' actions, underscoring their importance.

Taken together, these factors are among the reasons why leadership is considered an essential aspect of an IMS, and is included in the ISO 56001 and related standards (ISO, 2024).

3.3.2. Strategy – Establishing A Common Purpose For The Organization

An innovation strategy is a central component of an Innovation Management System (IMS) and plays a significant role in the ISO 56000 family of standards. Its purpose is to guide innovation efforts by establishing a clear rationale for undertaking innovation activities and by outlining the desired outcomes and value creation for the organization and its stakeholders (ISO, 2025). It is a core component (Kihlander, et al., 2024) of an IMS that ensures that every innovation initiative is linked to the broader strategic direction, correlating with Slack & Lewis' *Coherence* dimension (2011). It also promotes the alignment of resource allocation, governance structures and performance management processes defined in the IMS (ISO, 2025). With this alignment, the organization can achieve consistent decision-making, prioritization of high-impact opportunities, and effective tracking of progress toward the organization's innovation goals. Conversely, without a clear strategy, organizations risk "overly scattered and unstructured work", not realizing "the full potential from their innovation activities", and limiting "the effects of the resources spent on innovation" (Kihlander, et al., 2024).

However, the outcomes of a well-defined innovation strategy extend beyond more disciplined resource allocation and end-to-end development processes. When applied effectively by the leadership, it can also generate benefits such as a shared vision, heightened employee motivation, and greater organizational cohesion (Smith, et al., 2008) – the results of which can contribute to innovation performance. In applying *requisite variety* (Ashby, 1956) to innovation management, Van de Ven (1986) was one of the first scholars to adopt a systems-based perspective in innovation management. His analysis indicated that to achieve requisite variety, every member of the organization must continually monitor and respond to changes in the external environment to increase the systems' adaptability and responsiveness to change, also enabled by *redundant functions* where employees understand their role in the greater innovation system, and act holistically in their respective local units (Van de Ven, 1986). This "think globally while acting locally" (Van de Ven, 1986) approach could act as a stimulator of interactions and enabler of systemic innovation management. This analysis is relevant yet today, and has implications on the management of innovation: it cannot hinge on the tireless efforts of a few individuals (Kihlander, et al., 2024). Instead, it must permeate the entire organization and color every department, and multiple processes must effectively interface each other.

Thus, with a clear strategy effectively implemented by the leadership, the organization can be aligned toward its purpose of innovating (Smith, et al., 2008). In turn, this may contribute to the systemic management of the IMS, since the employees will actively drive the necessary interactions between system elements, acting as the glue from the previously mentioned analogy. It is for this reason that strategy is deemed an essential aspect of systemic management.

3.3.3. Openness – Promoting Collaboration And Interactions In The System

The third driver of systemic management is openness. Systems theory repeatedly vouches for both internal and external openess' crucial role in the prosperity of the system (Katz & Kahn, 1978; Golinelli, 2005). It has been found that "limited openness may result in lost opportunities, which negatively affect performance" (Lichtenthaler, 2009), emphasizing its role for system performance and survivability. Openness acts to integrate elements and stakeholders in the sub- and supra-system, and therefore hinges on certain capabilities needed to manage complex business landscapes (Ashby, 1956). It promotes the interactions between these elements, thereby contributing to systemic management as a result. Collectively, these

interactions enable change and adaptability in the system, which increases survivability (Golinelli, et al., 2002). Openness can, however, be managed in two areas: Internally, and externally.

Internal openness relates to preventing silo-fication, having transparent process pipelines, and overall interconnectedness in the sub-system. For example, if different managerial processes, such as culture, decision-making, and knowledge management (Le & Lei, 2019) are managed with an integrated managerial approach, where the practice of activities like communicating change, checking financial performance, and providing feedback overlap across the processes, higher interconnectedness can be achieved (Bititci, et al., 2011). Organizations with higher levels of interconnectedness show higher performance across several areas, such as "revenue growth; market share growth; profitability growth; cash flow growth; value-added productivity growth; customer satisfaction; new value streams; investments; employee satisfaction and morale" (Bititci, et al., 2011). A concrete example from Bititci et al.'s case study was a performance scoreboard that was used simultaneously to inform staff, guide decision-making and monitor financial results, thereby linking managing performance, communication and decision-making (2011). These benefits are strongly dependent on leadership (Bititci, et al., 2011), since managers who actively facilitate collaboration and connections across internal units through identifying critical linkages within the system contribute to the systems effectiveness (Teece, 2018). Openness is also strongly tied to knowledge sharing, and reportedly "[…] helps NPD team members from different functions to share ideas and find new solutions" (Sattayaraksa & Boon-itt, 2018). These observations underline that internal openness, by enabling transparent pipelines and cross-process integration, is a key component for building a systemic, holistic innovation management system.

External openness relates to engaging in collaboration with stakeholders in the supra-system, driving knowledge and information exchange. This may be exemplified by establishing partnerships and starting projects with research institutions, universities and other companies, all of which are common sources of innovations (Demircioglu, et al., 2019). It may also involve open innovation (OI), where firms do not have to depend solely on their own R&D and commercialization capabilities. Instead, they may build on discoveries made by other organizations in inbound OI, and share their own internally created intellectual property with external partners for commercialization or further development in outbound OI (Chesbrough & Crowther, 2006). Despite confidentiality concerns, the defense sector has begun to move toward open innovation to seek external ideas from, for example, civilian-sector start-ups (Reuven & Shamir, 2025), one reason being the significant performance benefits of such collaboration. Inbound OI, like horizontal or vertical technology collaborations, have shown a strong relationship with performance dimensions like "perceived sales growth, market share growth, return on investment, and profit level relative to […] major competitors" (Wang, et al., 2015). Concretely, this may entail collaborations with suppliers and competitors, and gathering market information from customers (Wang, et al., 2015), the latter providing performance gains especially for high-technology firms (Wang, et al., 2015), such as Saab. On the other hand, outbound OI has shown a correlation with increased return on sales (Lichtenthaler, 2009), often attributed to technology out-licensing (Dodgson, et al., 2006). Controversially, it has also been indicated that markets with greater competitive intensity for technology show greater performance benefits from adopting outbound open innovation policies (Lichtenthaler, 2009), suggesting that Saab could stand to benefit from such methods.

Both the internal and external openness perspective highlights the potential performance benefits of systemically integrating the system within itself, and with its environment.

3.3.4. Towards An Application of Systems Design and Systemic Management

Ultimately, an innovation management system (IMS) can be defined as "systemic" when it integrates a holistic strategy, supportive leadership, and openness. Furthermore, the three principles derived from Slack and Lewis' operations strategy model are equally valuable to consider when designing a system, echoing Kihlander et al.'s (2024) call for "more comprehensive, coherent, and consistent approaches for managing innovation activities".

The present report argues that by considering the Slack and Lewis' principles when designing the system, and by actively managing the system with attention to the systemic management elements, greater performance can be achieved. It is therefore essential to assess how these factors are applied. One method of doing so is through a standardized benchmark. The ISO 56000 standards fulfill this role by operationalizing the systems approach into concrete requirements, providing a robust framework to evaluate the IMS's strengths and shortcomings. The next section will therefore describe the standards and their role in inspiring an updated innovation management system for Saab Surveillance.

3.4. The Innovation Management System Standard: ISO 56000 Series

The ISO 56000 series are new standards in the ISO collection. They are written by the Technical Committee ISO/TC 279 that was founded in 2013, with 58 participating and 20 observing members (ISO, n.d.), and features multiple standards that describe cornerstones of managing innovation, each with a different role and purpose. All current standards are briefly described in the overview presented in Table 1.

Table 1: A summary of the ISO 56000 collection of standards

Standard	Description
ISO 56000:2025	Fundamentals and vocabulary: The core concepts, definitions and a common language for innovation management. Primarily descriptive and informative.
ISO 56001:2023	Requirements for an Innovation Management System (IMS): Specifies the mandatory elements an organization must establish, implement, maintain and continuously improve to run an effective IMS.
ISO 56002:2022	Guidelines for the implementation of an IMS: Offers practical guidance, recommended processes and tools for applying ISO 56001 in practice.
ISO 56003:2021	Tools and methods for innovation partnership: Outlines techniques for selecting, structuring and managing collaborative innovation relationships.
ISO 56004:2022	Assessment of an IMS: Defines a framework for internal/external audits, performance measurement and maturity assessment of an organization's innovation system.
ISO 56005:2021	Intellectual-property (IP) management: Gives guidance on protecting, managing and exploiting IP generated through innovation activities.
ISO 56006:2021	Collaborative innovation: Focuses on establishing and sustaining ecosystems, networks and partnerships that enable joint innovation.
ISO 56007:2023	Metrics and performance indicators: Proposes a set of quantitative and qualitative metrics to monitor, analyze and improve the effectiveness of an IMS.
ISO 56008:2024	Leadership and governance: Establishes the leadership responsibilities, governance structures and strategic alignment required to embed innovation at board level. Newly released.

ISO 56009:2024	Risk-based approach to innovation: Integrates risk-identification, assessment and mitigation into the innovation lifecycle, aligning with broader enterprise risk-management frameworks. Newly released.

The goals of the ISO 56000 standards are to give organizations a shared language and a coherent framework for managing innovation. The documents aim to enable companies to build and sustain innovation capability, create financial and non-financial value, and improve competitiveness, resilience and societal contribution (ISO, 2025; ISO, 2024). They also seek to help organizations understand core concepts, apply innovation-management principles, assess performance, and continually improve their innovation processes, thereby aligning innovation activities with strategic objectives and stakeholder needs (ISO, 2025; ISO, 2024).

3.4.1. Key Dimensions

The standards hinge on seven foundational dimensions: Context of the Organization; Leadership; Planning; Support; Operations; Performance Evaluation; and Improvement. These serve as a broad, comprehensive base upon which certain innovation management aspects relate to. Additionally, several factors impact the system across these dimensions, such as organizational culture, through e.g. risk propensity, willingness to experiment and openness to collaboration, but also other elements like people, technology, finance and governance (Smith, et al., 2008). Nevertheless, the seven dimensions are described as follows (ISO, 2025).

---

i. **The context of the organization** represents the internal and external factors that affect an organization's ability to innovate, the key stakeholders behind innovation, and the scope of the management system.

ii. **Leadership** is the responsibility of top management to demonstrate commitment, establish an innovation policy and strategy, allocate authority and resources, and promote a culture that enables creativity and risk-taking.

iii. **Planning** translates the strategic direction set by leadership into concrete actions. It stipulates that the organization identifies opportunities and risks, sets innovation objectives, determines what structures and portfolios are required, and outlines how these will be achieved.

iv. **Support** describes how the necessary resources, like competent personnel, financial and material assets, awareness-raising, communication, tools, methods, and the management of knowledge and intellectual-property, shall be provided to enable the effective operation of the system.

v. **Operations** cover the key elements of innovation initiative execution, including appropriate processes such as opportunity identification, ideation, concept validation and solution deployment.

vi. **Performance Evaluation** ensures the system's effectiveness is continually measured against planned targets through innovation performance indicators, considering the organization's vision, strategy, policy and objectives.

vii. Finally, **Improvement** builds on the results of the performance evaluation to drive continual enhancement of the innovation-management system, addressing gaps and deviations in context, leadership, planning, support and operations.

These ISO 56000 dimensions reflect established best practices that are widely recognized in the literature as foundational to innovation management. For example, broadly allocating resources toward innovation projects has been shown to be crucial for innovation performance (Agrawal, et al., 2018), such as new product sales (Klingebiel & Rammer, 2014), and allocating time and resources for idea generation is part of the overall innovation stimulus that drives innovation capacity (Prajogo & Ahmed, 2006). Furthermore, performance evaluation of innovation is essential for organizations to develop effective strategies and make informed decisions about where to direct innovation efforts. Without appropriate monitoring systems, organizations risk misallocating resources and losing sight of whether their activities translate into meaningful outcomes (Taques et al., 2021). Measurement may also serve as a necessary trigger for managerial action, a guide for strategic direction, and a mechanism to intensify innovative behaviors over time (Nilsson & Ritzén, 2014). While the discussion above highlights specific areas such as resource allocation and performance evaluation, it is important to acknowledge that all seven dimensions are interdependent and equally vital for an innovation management system.

3.4.2. Characteristics Of the Standards

A common theme of the standards is that they are deliberately open ended. As an example, in one section of ISO 56000 it is stated that a possible action for value realization is to "… establish objectives, processes, systems and performance indicators for value capture and realization" (ISO, 2025). This statement identifies the elements that can support value realization in innovation, but leaves the actual implementation and choice of methods up to interpretation. This openness of the standards can be seen as both a strength and a weakness: the lack of prescriptive detail allows organizations to tailor innovation activities to their own needs, yet it may be perceived as vague by practitioners according to critique. In a 2021 review, Tidd noted that the ISO 56002 standard rests on solid empirical support from the innovation literature for most of its foundational dimensions (*Context of the Organization, Leadership, Planning, Support, Operations, Performance evaluation and Improvement*). However, the author found that the operations (formerly known as *process*) element in particular is weakly supported, and that the standard's linear operations model, lack of concrete tools, and limited attention to sector-specific or contextual variations all reduce its practical usefulness.

Because implementation studies of the ISO 56000 standards remain scarce, it is difficult to empirically evaluate how effective their adoption is, or how well-founded Tidd's (2021) arguments are. Hyland and Karlsson (2021), who participated in drafting ISO 56002, argue that the aspects Tidd (2021) identified as shortcomings largely are the result of intentional design choices, because the standard is meant to be a guidance document rather than an implementation manual. They further argue that complementary standard documents (e.g., ISO 56003-56007) will address these gaps, also underscoring the standard's role as a necessary development in a previously stagnant field that previously lacked a common, profession-wide framework (Hyland & Karlsson, 2021).

3.4.3. Standardization And Systematization – The Dark Horse of Innovation?

Despite the limited base of literature empirically examining the potential drawbacks or benefits of standardizing and systematizing innovation management, ISO themselves claim potential benefits, such as:

"The processes and support needed to achieve innovations can be managed more effectively and efficiently as a system of interrelated and interacting elements. Reasons for managing innovation activities as a system include:

a) guiding the organization to focus on the most important and relevant innovation activities, initiatives and support given its strategic direction;

b) enabling top management to establish a relevant innovation ambition and vision and to optimize the use of resources;

c) creating awareness of innovation activities internally and externally based on a common framework and vocabulary;

d) facilitating the evaluation, including identification of weak elements, bottlenecks and unintended consequences, and benchmarking of the innovation management system in order to drive improvements;

e) being compatible, with the possibility of integration, with the other management systems of the organization."

(ISO, 2025)

To evaluate these claims, further empirical investigation is required to determine the long-term effects of the frameworks (Karstegl, et al., 2025), like whether they genuinely improve innovation outcomes, or merely impose bureaucratic constraints. Forthcoming research suggests that using ISO 56000 series principles and frameworks not only can provide a common language and understanding for innovation management, but also legitimacy to decisions taken by managers in the field (Lerche & Magnusson, 2025). In turn, the improved collective understanding and efforts toward innovation management could improve organizations' innovation outcomes. In another recent publication, a case study showed that the guidance of ISO 56002 enabled companies to develop an innovation management system (Kihlander, et al., 2024), which could act as an intermediary for achieving innovation performance. This provides an early indication of potential benefits of following the ISO 56000 series, but can not be considered conclusive evidence, nor validating of ISO's claimed performance gains.

However, insights can be drawn from the national innovation standards established in Spain (2006) and Portugal (2007), which served as precursors to the current ISO frameworks. The Spanish UNE 166 000 series was the first standardized innovation management system (SIMS) framework, and originated from a need to formalize research, development and innovation (R&D&I) activities. Specifically, its goal was to help companies structure projects, protect generated knowledge, and secure government tax incentives for innovation (Mir & Casadesus, 2011). By providing a framework for auditing innovation work, these standards allowed for a more rigorous verification of R&D outputs. Following their implementation across numerous Spanish firms, subsequent research has indicated that this standardized

approach has positively influenced various facets of innovation performance. Quantitative analyses have shown that UNE 166 002 has a strong, positive impact on innovative capabilities and business performance (Mir, et al., 2016) through promoting the development of both incremental and radical innovation (Gimenez Espin, et al., 2025). It has also been shown to be a driver of product, process, organizational and marketing innovation (Martinez-Costa, et al., 2019).

While the comprehensive benefits of the standards are well-documented, identifying the specific mechanisms that drive these performance gains remains challenging. For example, how certain performance gains map to individual elements of UNE 166 000 has not been established, potentially due to the complex and emergent nature of innovation management systems (Chen, et al., 2025). The positive effects of individual items of the standards showed low statistical significance in Giménez-Espin et al. (2025) and Martinez-Costa et al.'s (2019) studies, and Mir et al.'s study treated having the standard or not as a single binary variable (2016). Despite this, a generally agreed upon picture is that the benefits arise from standards' ways of (i) reducing uncertainty and risk, (ii) guaranteeing the necessary resources, tools and documentation, (iii) creating clear innovation policies and measurable objectives, (iv) promoting an organizational culture that rewards idea generation and protects the results, and (v) linking the innovation pipeline from idea capture, to project selection, and finally exploitation, and to the firm's strategic goals (Giménez-Espín et al., 2025; Martínez-Costa et al., 2019; Mir et al., 2016).

The ISO 56000 series includes similar elements to those found in the UNE 166 000, but lacks the same empirical evidence of its efficacy. It can therefore be cautiously inferred that ISO 56000 series implementation could provide similar performance outcomes, although their real-world effectiveness can only be confirmed through further empirical study, similar to the long-standing ISO 9001 or ISO 14001 families whose effectiveness eventually became supported by extensive implementation research.

3.5. Research Questions

The present report will analyze Saab Surveillance's innovation practices through four research questions. The questions were developed drawing on the staples of systems and innovation management theory, and three guiding questions presented by Saab Surveillance: (i) What is the new ISO standard about? *(Scope, purpose, key concepts, and what problem it aims to address.)*, (ii) how does SAAB currently follow its innovation process, and to what extent does this align with the ISO standard? *(Are there similarities, gaps, or informal practices that resemble the standard?)*, and (iii), how could adopting this ISO standard affect Saabs innovation ecosystem? *(Potential benefits, challenges, impact on collaboration, governance, speed, or flexibility.)*. The four questions are presented below:

1. *To what extent are ISO 56001 requirements fulfilled by Saab Surveillance's innovation management practices?*
2. *To what extent is innovation at Saab Surveillance managed with a systems approach according to the dimensions Comprehensiveness, Coherence and Correspondence?*
3. *To what extent are the three systemic management dimensions Leadership, Openness and Strategy fulfilled at Saab Surveillance?*
4. *How can insights from ISO 56000 and systems theory guide improvements to Saab Surveillance's innovation management system in order to ensure the fulfillment of innovation strategies?*

4. Methodology

The purpose of this chapter is to outline all methods used to gather and analyze information and data, to provide transparency for the reader. Figure 4 visualizes the methodology and data collection process.

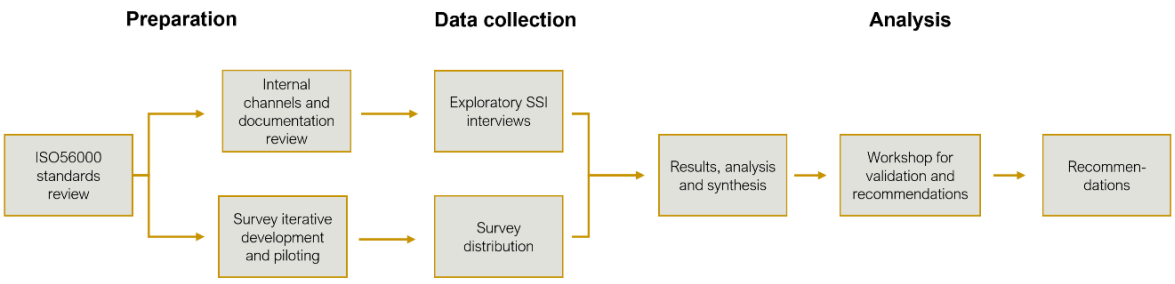


Figure 4: Visual diagram of the Methodological process

To reinforce the validity of the data, evidence was triangulated from multiple sources following a mixed-methods approach; data was collected through interviews, a targeted survey, a workshop, and review of internal documentation. Furthermore, Case-Study Evaluation Template (CASET) principles for case study quality (Goffin, et al., 2019) were considered in the design of the method. CASET was developed as a systematic checklist for assessing the methodological rigor of case-study research in the innovation-management field. It consists of ten binary quality criteria organized under four overarching stages of the research process (Goffin, et al., 2019):

I. Research design (theoretical justification, pilot testing, and purposeful sampling),
II. Data-collection (triangulation, external validation of evidence, and transparency of the collection procedures),
III. Data-analysis (inter-coder agreement, systematic case presentation, and theoretical interpretation), and
IV. Post-hoc reflection on validity and reliability.

Each criterion receives a score of 1 if it is satisfied and 0 if it is not, yielding an overall quality rating ranging from 0 to 10. Goffin et al. (2019) applied CASET to 818 published case studies and found a mean overall score of 3.05, indicating that many studies were missing several of the checklist items. By using CASET as a design guide, many frequently neglected elements in the literature were deliberately addressed. Table 2 provides an overview of the criteria that were addressed, and the methodology behind it.

Table 2: A summary of the CASET criteria, a binary value indicating their inclusion, and an explanation of how criteria were addressed

CASET criterion (Goffin et al., 2019)	Addressed (Y/N)	How it was addressed
Theoretical foundation – explicit justification for a case-study approach	1	Articulated why a mixed-methods design is the most suitable for exploring the present research.
Pilot study – pilot testing of instruments	1	A pilot study was conducted for the quantitative data collection
Theoretical sampling – purpose-driven case selection	0	Selection of particular case was explained, but the case was not deliberately selected with the topic in mind
Triangulation – ≥ 2 data sources	1	Integrated three sources: (1) semi-structured interviews, (2) survey data, and (3) a workshop
Review/validation of evidence – external verification	1	Presented preliminary findings to select interviewees for fact-checking, and feedback was incorporated and documented
Transparency of data collection – instruments disclosed	1	Full interview guide, survey questionnaire and quantitative data analysis code are available in the appendices
Inter-coder agreement – multiple coders with reliability stats	0	Only one researcher coded all data.
Case presentation – systematic tables/evidence trail	0	A systematic code trail was not developed, since the Gioia methodology was bypassed.
Case interpretation – linkage to theory & propositions	1	Taking the analysis beyond describing data or identifying themes, also focusing on theorizing
Reflection on validity & reliability – discussion of threats	1	Dedicated subsection discusses methodological limitations, internal and external validity, and reliability

The data collection started with open-ended, semi-structured interviews, conducted to inductively develop theory surrounding the company's innovation management system. Then, a survey based on ISO 56001 requirements and the findings of the initial exploratory interviews was developed for three specific target groups: operational level employees (OE), middle managers (MM), and senior leadership (SL). The survey sought to map Saab Surveillance's adherence to ISO 56001. Finally, a workshop was held to allow the organization to validate the results from the two data streams, and to discuss potential causes behind the shortcomings of the innovation management system identified in the data.

4.1. Qualitative Data: Exploratory Interviews

Twelve initial exploratory interviews with eleven unique interviewees were conducted to identify areas of interest and to gain an understanding of the company's innovation management structures. These interviews followed a semi-structured format, with respondents being allowed to speak freely about their domains of expertise and opinions on innovation management, but being directed by questions when needed. The interview guide is available in Appendix C, section 11.3. The interviewees were selected based on their experience in different innovation functions, and iteratively according to the inductive grounded theory

method. The diverse roles of the interviewees ensured that multiple, complementary perspectives from the Surveillance organization were brought to light, and supported the exploration. Interviews were predominantly conducted digitally, by a single interviewer, and typically ranged from 30 to 60 minutes, depending on the subject relevance and the interviewees' availability. The interviews were transcribed manually during the session to avoid potential confidentiality breaches associated with recording and automatic transcription. The list of interviews can be found in Table 3.

Table 3: A list of interviews conducted and descriptions of the participating interviewees

Inter-view	Interviewee / role	Area of expertise and focus of the interview
1	Head of department Member of Surveillance Innovation Group	Leading a department Reviewing innovation ideas from Saab Surveillance
2.	Innovation leader Former innovation leader at other large, high-technology firm	Shaping innovation activities
3.	Innovation Manager	One of first RISE Innovation Management Certified. Shaping innovation activities
4.	Lead, Surveillance Innovation Group	Reviewing innovation ideas from Saab Surveillance
5.	Lead, Innovation and Technology Council	Managing early research studies, reviewing selected ideas from innovation group
6.	Interim manager, software development. Former innovation coach	Used to provide innovation coaching to employees. Now leading software development team.
7.	C-suite manager	Driving innovation culture Developing processes to adopt for innovation Internal marketing of innovation management
8.	Investment Director, Saab Ventures	Incubation and venturing of start-ups, board advisory on start-ups, idea management
9.	Innovation Management Team	Idea management portal member, higher management advisor
10.	Lead, X Innovation Lab (same as interviewee 4)	Software prototyping, driving innovation culture, Driving internal projects in innovation lab
11.	Lead Mechanic innovation lab. Full time hardware project member	Hardware prototyping, driving innovative working methods as a mechanical design engineer
12.	Project manager of business development, Lead of novel 3D printing for Saab Surveillance Stockholm	Driving research and innovation projects aside of main project tasks Finding inspiration and motivation for innovation

Post interviews, the findings of the interviews were thematically analyzed and linked to various concepts. Although the qualitative data analysis was inspired by it, the Gioia methodology (Gioia, et al., 2013) for thematic analysis was bypassed in favor of adopting a more exploratory approach. Several observations were formalized based on the findings, resulting in a set of 19 condensed observations that were shared with the interviewees (N=11) for validation. Their

responses were scored on a scale of disagree (=0) to maybe (=0.5) to agree (=1), allowing an evaluation of the observations to be made, in line with CASET validation criteria. These observations are presented in the results and analysis, section 5.1.7, and in the appendix section 11.1.

4.1.1. Qualitative Data Methodology Analysis

Efforts were made to provide theoretical rigor to the qualitative findings. The study presents qualitative findings through summaries and direct quotes, but as mentioned, the Gioia methodology was not followed. This may have reduced the analytic quality of the interview findings, in turn granting less validity to the results. Nevertheless, the survey provided complementary data which was used to complement the observations and assumptions of the interview findings.

Interviews were summarized in writing once completed to highlight the principal discussion points, and to capture the immediate impression that the interviewer formed about the interviewee (Wengraf, 2001). This approach may be considered a shortcoming since field notes may not be replayed like recordings (Tessier, 2012), and due to potential subjective inferences made in the summaries. However, the transcription and summaries allowed nuanced observations to be made, since factors like interviewee hesitation or confidence when speaking about a certain topic could be captured (Wengraf, 2001). Such context has high relevance for the analysis of qualitative data, since a topic can be described, or the interviewee may show agreement with a topic, but their inflection may reveal disingenuity or acquiescence bias. Furthermore, writing the summaries immediately after each interview also helped to minimize retrospective bias that could have arisen if the summaries had been prepared only after all interviews were completed (Wengraf, 2001).

4.2. Quantitative Data: ISO 56001 Adherence Survey

A survey was designed to gather quantitative data from the company regarding its innovation management and adherence to ISO 56001 requirements. The survey served the purpose of: (1), supporting the findings of the initial interviews through convergent triangulation, (2) broadly auditing Saab Surveillance's adherence to the ISO 56001 innovation management system requirements, and (3) examining how different aspects of the ISO standards present in the organization affect innovation performance. The survey comprised a set of Likert-scale items measuring perceptions of innovation and innovation-management capabilities, with an introductory section assessing the background and role of the respondent. The questionnaire was deliberately organized into three distinct sections, each tailored to the role of the respondents, and skip logic allowed three groups of respondents to receive three different sets of questions. Before respondents selected their role, brief definitions of the roles were presented to ensure that each participant answered the set of questions that best matched their position. Table 4 shows the definition of the roles, and the way questions were tailored to the groups.

Table 4: The three hierarchical roles surveyed in the questionnaire, a definition of their roles, and a summary of the questions they were asked

Group	Definition	Questions
Operational level employees (OE)	*You work in one or more projects without decision making authority over your own time allocation and project plan.*	Were asked about their day-to-day activities and their attitudes toward innovation.
Middle / Project managers (MM)	*You manage one or more teams, have responsibility over maintaining budgets, and you have decision making authority over project plans and employees' time.*	Received items that linked these operational issues with the implementation of policies and coordination across teams.
Senior leadership (SL)	*You have decision making authority surrounding strategy, resource allocation and multiple middle managers.*	Answered questions focused on strategic and managerial decisions, such as long-term vision, cultural efforts, and resource allocation.

This design served two complementary purposes. First, it allowed the wording of each set of items to be phrased in language that is familiar to the target group (Hinkin, 1998). Second, it helped to mitigate response bias associated with the role of the respondent: senior leaders and operational employees tend to "have widely differing opinions about the health of their organizations" (De Smet, et al., 2025), and managers tend to overestimate their management effectiveness (Song & Meier, 2022). This gap has shown to cause dissatisfaction in the workplace among employees (Song & Meier, 2022), and indicates that different levels in organizations report or experience different realities. Therefore, by capturing perspectives from all three organizational levels, the design provides a more balanced view of operations.

While the survey questions were tailored to three recipient groups, they shared a consistent structure based on the main dimensions of an innovation management system as outlined by ISO 56001, like *context of the organization*, *leadership, etc.* (ISO, 2024). One or more items were used to evaluate the adherence to an ISO 56001 dimension. This design was employed in order to examine adherence to requirements within each ISO dimension, and the questions were formulated to provide broad and unbiased coverage of the standard's IMS requirements. This structure was adapted through consultation with *Amplify* AB, an innovation management

consultancy specializing in innovation management assessment and ISO 56001 auditing, and validated with a co-author of the ISO 56000 series before distributing the survey. Furthermore, to ensure comparability, a 5-point Likert scale was maintained throughout the survey. A sixth, "*I don't know"* option was also included to prevent inaccurate responses. The following table (Table 5) exemplifies this survey design structure. The full questionnaire is available in the appendix, section 11.4.

Table 5: Example of survey questions addressed to the three respondent groups

Factor	Operational Employees (OE)	Middle / Project Managers (MM)	Senior Leadership (SL)
Operations– Q3 **Deploy solutions** **(Clause 8.3.6)**	We take multiple steps to achieve the effective commercialization of solutions, such as implementing production and distribution plans, promoting them to customers, and considering the IP of solutions. *I don't know* *Strongly disagree* *Disagree* *Neutral* *Agree* *Strongly agree*	We take multiple steps to achieve the effective commercialization of solutions, such as implementing production and distribution plans, promoting them to customers, and considering the IP of solutions. *I don't know* *Strongly disagree* *Disagree* *Neutral* *Agree* *Strongly agree*	To what extent does senior leadership take steps to ensure the effective commercialization of solutions, such as implementing production and distribution plans, promoting them to customers, and considering the IP of solutions? *I don't know* *Very limited extent* *Limited extent* *Neutral* *Large extent* *Very large extent*
Performance Evaluation – Q1 **Measuring and improving outcomes** **(Clause 9.1)**	We use defined key performance indicators (KPI) to follow up the outcomes of innovation and R&D projects. *(e.g. financial, customer value, societal outcomes)* *I don't know* *Strongly disagree* *Disagree* *Neutral* *Agree* *Strongly agree*	We use defined key performance indicators (KPI) to follow up the outcomes of innovation and R&D projects. *(e.g. financial, customer value, societal outcomes)* *I don't know* *Strongly disagree* *Disagree* *Neutral* *Agree* *Strongly agree*	To what extent does senior leadership use R&D and innovation KPIs to follow up innovation outcomes and performance? *(e.g. financial, customer value, societal outcomes)* *I don't know* *Very limited extent* *Limited extent* *Neutral* *Large extent* *Very large extent*

Although all three groups answered a question relating to the same sub-requirement of ISO 56001, the wording of the question and response anchors sometimes differed across groups. For Operations – Q3, the operational-level employees and middle manager scale was anchored "Strongly Disagree, Disagree, Neutral, Agree, and Strongly Agree", whereas the senior-leadership item was anchored "Very limited extent – Limited extent – Neutral – Large extent – Very large extent." This design was repeated throughout the survey, where a lower proportion of "disagree to agree" items were presented to senior leaders in favor of alternative anchors. Conversely, operational staff were given the more familiar "disagree to agree" anchors, and middle managers a balanced variety between the two anchors. The intention of this difference between the groups was to mitigate acquiescence bias among senior managers, i.e. the propensity of respondents to agree with statements (Baxter, et al., 2015), and to keep the questionnaire engaging. This may also have mitigated the satisficing bias among senior leadership, since new response anchors required engagement and discouraged them from straight-lining, i.e. selecting the same option for multiple questions (Baxter, et al., 2015).

4.2.1. Actions Taken to Strengthen the Quantitative Data Collection

To further improve the strengthen the data collected through the survey, various recommendations from Mismetti et al. (2025) and Cycyota & Harrison (2006) were followed. To ensure that the content of the survey was valid and understandable for the participants, items were designed with the results of the qualitative interviews in mind. For example, some interviewees indicated a limited familiarity with innovation-management terminology. Interviews also indicated firm-specific language to describe innovation concepts, such as "low or high TRL" to describe radical or incremental innovation, respectively. Therefore, complex innovation management jargon was purposefully limited, and firm-specific language was included in items in order to increase content validity (Mismetti, et al., 2025). Then, once a first draft of the survey had been completed, the items were piloted with six participants representing the three organizational levels. Their feedback was used to refine wording and anchor options, improve clarity, and reduce the length of the instrument (Mismetti, et al., 2025) in order to improve engagement and answerability (Hinkin, 1998).

Furthermore, efforts were made to reduce respondent biases. To avoid common source bias, data was gathered from three layers of participants (Mismetti, et al., 2025), as highlighted previously, allowing the comparison between managers' perceptions and employees' operational realities. Then, for the distribution of the survey, participation was presented as voluntary and respondents were assured that their answers would remain anonymous and confidential, in line with recommendations (Mismetti, et al., 2025). Efforts were made to increase the response rate, especially for respondents in senior leadership positions. Conventional response-enhancement techniques such as giving advance notice and personalizing the cover letter or questionnaire were not opted for, since they have shown to have little impact on executive response rates (Cycyota & Harrison, 2006). Instead, measures were taken to increase topic salience, a factor that has shown to increase their response rates. This was done through linking the cover letter of the study to current strategic outcomes relevant to senior managers, and emphasizing the survey's timeliness with regard to recent industry developments. Specifically, the importance of the innovation management discipline, and highlighting how it can enhance the company's future capabilities were addressed. An endorsement from KTH, the authors university, and the authors' manager at Saab was also included. These steps were designed to make the topic appear more immediately relevant and worthwhile, and to provide legitimacy, thereby encouraging participation (Mismetti et al., 2025; Cycyota & Harrison, 2006). The instrument was distributed electronically. Finally, to promote methodological openness and transparency, design decisions were described in detail, and surplus information like the analysis code was made available in the appendices, also cited as a best practice by Mismetti et al. (2025).

In sum, although not all improvement actions were incorporated due to practical constraints, deliberate actions were taken to improve the survey. The survey was created and distributed through careful item development and pilot testing, including bias-reduction techniques and transparent reporting, and adhering to the best-practice recommendations of Mismetti et al. (2025).

4.2.2. Limitations Of the Quantitative Data Collection

Despite these improvements, the quantitative data collection was subject to several limitations that should be taken into account when interpreting the results. These constraints arose partly from design decisions, and partly from organizational obstacles beyond the author's control.

Best practice suggests that at least four items be used to assess the homogeneity of a single construct (Hinkin, 1998). However, due to the extensiveness of ISO 56001, assessing all sub-

requirements with four or more items would have produced an excessively long questionnaire, likely heavily depressing the response rate by increasing dropout rates (Hinkin, 1998) or causing satisficing bias. Additionally, budgetary constraints from the organization resulted in a further need to decrease the number of questions. Therefore, the questionnaire was iteratively condensed through the piloting process: First from approximately 70 items to about 40, and ultimately to 16 items for operational employees, and 20 items for middle managers and senior leadership, excluding the introductory control items. In the reduction, a balance was sought between completion time and empirical depth. To achieve this, question areas most critical to the innovation management system at Saab, such as *Leadership, Operations* and *Support* were prioritized. Consequently, the survey was delivered with an average of two to three items per corresponding ISO dimension. The fulfillment of any single sub-requirement should therefore not be inferred from this data, since measuring a construct with fewer items reduces the ability to capture it reliably (Hinkin, 1998). Because of this structure, the survey more sufficiently provides statistical evidence to evaluate adherence at the dimension level (i.e., the aggregate of the three items representing each ISO dimension), rather than at the level of individual sub-requirements. However, the reduction in granularity could mask subtle differences in how individual sub-requirements are met, potentially limiting the depth and comprehensiveness of the data on overall ISO 56001 adherence. This is especially true for the ISO dimension *Improvement*, which was represented by only a single item – the need to condense the survey led to a prioritization of other question areas, particularly within *Operations* and *Support,* thus limiting the evaluation of the *Improvement* dimension.

The survey was also not delivered with a randomized item order. This was a deliberate choice employed to preserve the logical flow of the ISO 56001 dimensions, but may have contributed to priming effects (Mismetti, et al., 2025), i.e. where respondents are influenced by similar, preceding questions, which can artificially inflate inter-item correlations and skew the distribution of scores.

Sampling was also an area with lacking methodological rigidity. Due to constraints from the organization, a sufficiently large, diverse, and randomized respondent group could not be attained. Hinkins (1998) finds that approximately 150 responses per response group should be sufficient for exploratory analyses like this report. However, other findings indicate that the number of respondents necessary for statistical significance should be put in relation to the number of items included in a 1:4 or 1:10 ratio. Employing a 1:4 ratio would require the survey to attain circa 64 responses from employees, and 80 responses from middle managers and senior leadership. Due to the average response rate of 28%–34% in senior leadership and 45%–56% found in employee and manager populations (Cycyota & Harrison, 2006), an ideal population size would be around 130 individuals for employees, 160 for middle managers, and 270 for senior leadership. However, due to internal budgetary limitations, these could not be achieved. Instead, the survey was limited to a maximum of 120 total respondents. As a result, the analyses were severely under-powered, meaning that the detection of subtle relationships among the ISO-56001 dimensions were not possible. Consequently, any conclusions drawn from the data should be treated as exploratory rather than definitive. Among these 120 recipients of the survey, the desired randomizing and spread across business units was not achieved either due to internal security regulations regarding the sharing of data across business units. Additionally, the uneven distribution between respondent groups is recognized as a limitation. In turn, this limited the representativeness of the sample and introduced bias toward the business units that were included, thereby reducing the external validity of the findings and limiting their broader applicability to Saab Surveillance.

The budgetary constraint that imposed limitations on the sampling and survey length also affected other crucial areas of the instrument. Although bias mitigation techniques were

included when designing the survey, an initial effort to curb retrospective bias through inviting participants to reflect on specific past events anchored to clear temporal landmarks (Mismetti, et al., 2025) was abandoned when the questionnaire was shortened, leaving more generic wording in the items. The constraint also affected the possibility of asking the same question in different ways using reverse-coded items, since it would have increased the length of the survey. In turn, this may have led to satisficing bias and straight-lining (Baxter, et al., 2015), and potentially common method biases (Mismetti, et al., 2025). Furthermore, the decision not to include attention checks is acknowledged as a limitation of the current study (Mismetti, et al., 2025), but was also result of the survey length reduction.

Another potential limitation was the variation in anchors among the items provided to the different respondent groups. Because the survey is highly self-reported, a level of self-perception bias is to be expected (Mismetti et al., 2025), and anchors were therefore varied in order to mitigate this bias in managers. However, the varied anchors may have introduced a methodological limitation. Scales that share the same number of points but employ different verbal anchors are not necessarily equivalent in terms of the psychological distance between adjacent categories (Spector, 1976; Casper, et al., 2020). For example, the perceived interval between “Disagree” and “Neither agree nor disagree” may differ from that between “Limited extent” and “Neutral.” Consequently, comparisons of mean scores across groups may be biased because the underlying response scales may lack direct comparability.

4.2.3. How the Quantitative Data Was Analyzed

Quantitative data was analyzed using R code in RStudio to develop descriptive and statistical data. After closing the survey, raw data was exported to an excel document, and then imported into R studio to start the analysis. The following table (Table 6) summarizes the statistical methods used, and the R packages used in the analysis. The full code can be found in appendix F, section 11.6.

Table 6: The statistical analysis methods utilized in the quantitative data analysis

Statistical method	***R packages used***
Cronbach Alpha	readxl, psych, dplyr
Descriptive data	readxl, dplyr
Harman	readxl, psych
T-test	readxl, dplyr
Spearman correlation	readxl, dplyr
Pearson correlation	readxl, dplyr

Across all analyses, responses coded as "I don't know" were treated as missing and excluded listwise. The remaining Likert values were recoded by subtracting one, shifting the scale from its original 2–6 range to 1–5 to exclude the “I don’t know” anchor. This resulted in item-level sample sizes that vary across items and respondent groups, but ensured that items with higher rates of "I don't know" responses do not carry disproportionate influence on average item scores. Furthermore, for the purpose of this exploratory analysis, respondents’ agreement with the items that map onto the ISO 56001 requirements were treated as a provisional indicator of compliance with the standard. The findings should be interpreted cautiously for three main reasons:

1. ***Coverage***: the questionnaire does not encompass the full set of ISO-56001 requirements; consequently, the measured items represent only a subset of the standard.

2. ***Construct validity***: the wording of the items may not capture the precise meaning or practical implementation of the underlying requirements, introducing potential measurement error.

3. ***Sample size and composition***: the data were collected from a relatively small sample of employees within Saab Surveillance, limiting statistical power and the generalizability of the results to the broader organization.

In light of these constraints, the results are best viewed as an initial indication rather than definitive evidence of ISO-56001 adherence. Below follows a detailed exposition of the statistical analyses conducted.

Harman's Single-Factor Test to Evaluate Common Method Bias

Since MM and SL respondents answered both the ISO dimension items and the innovation performance items within the same survey, their responses to the independent and dependent variables were collected from the same source. This raises the risk of common method bias (CMB), whereby observed correlations between predictors and outcomes may be inflated by shared response tendencies rather than reflecting genuine relationships. To assess the extent of CMB, Harman's single-factor test was conducted separately for MM and SL respondents, as each group answered a distinct set of items. For each group, both ISO dimensions and innovation performance outcomes were submitted to a principal component analysis constrained to one factor. The proportion of total variance explained by this single factor was then examined: if one factor accounts for the majority of variance (conventionally > 50%), CMB is considered a potential concern. The test was run between items where a single respondent had answered both, and any items lacking sufficient valid responses were excluded prior to analysis. All calculations were performed using the psych package in R.

Cronbach Alpha to Assess Internal Consistency of ISO Dimension Items

To assess the internal consistency of items within each ISO dimension, Cronbach's alpha was computed by combining responses from all three respondent groups (OE, MM, SL) into a single item matrix spanning all 35 respondents. The psych package in R was used to perform the calculations. For each area, parallel items administered to each group were arranged so that each column represents one item position, populated from the corresponding role-specific survey column. This rests on the assumption that parallel items across hierarchical roles measure the same underlying construct, differing only in the perspective from which it is assessed. Alpha was calculated excluding respondents with missing responses on any item within a given area. Areas consisting of a single item, such as Planning Portfolio, Planning Collaboration, and Improvement, were excluded by definition, as alpha requires a minimum of two items. Alpha values were interpreted according to conventional thresholds: ≥ 0.70 acceptable, 0.60–0.69 questionable, 0.50–0.59 poor, and < 0.50 unacceptable.

Descriptive Statistics to Summarize ISO 56001 Adherence Across Organizational Roles

The descriptive data was developed to provide an overview of ISO 56001 adherence. Remaining Likert scale values were recoded so that valid responses began at 1. Respondents were categorized into three groups by organizational role: Operational Employees (OE), Middle Managers (MM), and Senior Leadership (SL). Per-item descriptive statistics (n, M, SD) were computed separately for each group and pooled across all groups, with pooled means weighted proportionally by group sample size. Standard deviations are reported using Bessel's correction (n – 1), which produces an unbiased estimate of population variance – particularly important given the modest sample size relative to the broader organizational population. Area-level means were computed as n-weighted averages of item-level means. All analyses were conducted in R using the readxl and dplyr packages.

Spearman and Pearson Correlation Analyses to Examine the Relationship Between Perceived ISO-56001 Fulfilment and Innovation Performance

To analyze the potential relationship between fulfillment of the different ISO areas and perceived innovation performance, Spearman and Pearson correlation analyses were conducted. The Spearman analysis served as the primary method, with Pearson being utilized as a robustness check, exploiting the complementary properties of the two methods. Cases where only one method reached significance are reported but interpreted with additional caution. This approach was opted for in favor of linear regression, since the number of respondents was not sufficient relative to the number of predictor variables required for such an analysis. Prior to analysis, items belonging to respective ISO dimensions were averaged per respondent to form a single composite score for each ISO area. The two Planning items were separated, however, as they targeted two distinct requirements more accurately described individually. Consequently, the analysis featured eight predictors: Context, Leadership, Planning Portfolio, Planning Collaboration, Support, Operations, Performance Evaluation, and Improvement. These independent variables were each tested against four dependent variables: radical innovation, incremental innovation, customer-centricity, and innovation speed. In total, 32 dimension-specific statistical tests were conducted. The potential relationship between overall ISO fulfillment and innovation performance was also explored by constructing a composite IMS score, calculated as the unweighted mean of the seven ISO area scores for each individual respondent. For this composite, the two Planning items were recombined into a single Planning area prior to averaging, preserving adherence to the groupings stipulated by the ISO 56000 series. This composite was subsequently tested against the same four performance dimensions, bringing the total number of statistical tests to 36. Significance was assessed at $p < .05$ and $p < .01$.

Furthermore, this analysis was conducted with a limited and extended sample. Given their greater insight into strategic and performance-related aspects, only MM and SL respondents received innovation performance questions. Therefore, the limited analysis included only MM and SL respondents ($n = 17$), who provided both ISO area scores and innovation performance scores from their own responses. Then, the extended analysis also incorporated OE respondents by attributing innovation performance scores from the MM and SL respondents within the same department, to the OEs. The performance scores were computed as the mean of all valid managerial responses per department per performance outcome. OE respondents were included only if their department had both an MM and an SL respondent who completed the survey, and only if they themselves had provided valid responses across all seven ISO areas. One OE respondent who had only a senior leader and not a middle manager in their department was excluded to maintain consistency in how the attributed performance score

was constructed. Both Spearman and Pearson were run for each predictor-outcome pair across both the limited and extended analyses.

It is recognized that this method may constitute a limitation, and the findings from both analyses must be interpreted with caution for several reasons. First, the overall sample size is modest, which limits statistical power and increases the risk of both Type I and Type II errors across the 36 tests conducted. Second, in the extended analysis, all OE respondents within the same department receive an identical attributed performance score derived from their managers. This may artificially suppress within-group variance on the dependent variable, inflates the effective sample size, and introduces non-independence among observations. The extended analysis should therefore be treated as exploratory and interpreted alongside the limited analysis rather than in isolation. Lastly, no correction for multiple comparisons, such as the Bonferroni correction, was applied. This decision was made on the basis that the eight ISO area predictors are likely intercorrelated in the system, as are the four performance outcomes, meaning the 36 tests cannot be treated as independent, which is a core assumption of the Bonferroni procedure. Applying it under these conditions would risk rendering the correction overly conservative, increasing the risk of Type II error and masking potentially meaningful relationships. Furthermore, given the exploratory nature of the analysis and the modest sample size, the objective was to identify candidate relationships for future investigation rather than to confirm hypotheses. Consequently, results flagged as significant should be interpreted as indicative rather than confirmatory, until a larger and more representative sample can confirm or disprove the findings.

Welch's Independent-Samples t-tests to Compare Innovation Management System Perceptions Between Hierarchical Levels

To examine whether perceptions of the organisation's innovation management system differed between hierarchical levels, pairwise group comparisons were conducted between OE, MM, and SL respondents. Welch's independent-samples t-tests were employed for all comparisons, as this variant is more robust for samples of unequal size and potentially unequal dispersion. Tests were conducted at two levels of aggregation. At the item level, each individual ISO 56001 survey item was compared across the relevant group pairs (OE–MM, OE–SL, MM–SL); items exclusive to MM and SL were compared only for that pair. At the dimension level, a composite score was computed for each respondent as the row-wise mean of their valid responses within each ISO dimension, and these scores were subjected to the same pairwise procedure. A Bonferroni Alpha threshold was applied to control for the potential false positives that may arise when conducting multiple statistical tests simultaneously. It was applied separately to the item-level and composite test families. The item-level family comprised 52 tests, yielding a corrected threshold of $\alpha=.05/52=.00096$; the composite family comprised 25 tests, yielding $\alpha=.05/25=.002$. Results are reported against both thresholds: findings passing the Bonferroni-corrected threshold are treated as robust, while those significant only at the unadjusted $\alpha=.05\alpha=.05$ are flagged as indicative and interpreted with caution.

Welch's t-tests to Compare the Fighter Core Capabilities Unit (Dept 2) with All Other Business Units (Dept Rest)

To analyze whether different departments within Saab Surveillance manage and perceive innovation differently, Welch's t-tests were used to compare the business unit Fighter Core Capabilities (*FCC, coded as Dept 2*) with all remaining units. The sample was heavily skewed toward Dept 2, while the other units each had only a few respondents, so a fully disaggregated department-level analysis was not feasible. Instead, responses from Dept 2 ($n \approx 17$) were

contrasted with a pooled "Rest" group combining units 1, 3, 4, 5, 6, and free-text department entries ($n \approx 16$), among them the business unit *Design*, providing a more stable and interpretable comparison between the dominant unit and the rest of the organization. The same recoded Likert data as in the previous analyses were used: "I don't know" responses were treated as missing, and the remaining response categories were shifted downward by one scale point. Department membership was derived from one of the control items of the survey. Using this classification, the existing ISO item-level and ISO 56000 dimension composite scores were compared between Dept 2 and Rest with Welch's independent-samples t-tests. In addition, innovation performance outcomes (radical innovation, incremental innovation, customer-centricity, and speed to market) were compared in two ways: (1) using managers' own scores only (MM and SL), and (2) in an extended analysis where OEs from the departments were assigned attributed performance scores based on the mean of their department's MM and SL ratings. To account for multiple testing, Bonferroni corrections were applied separately to four families of tests: ISO items (n=16), ISO dimension composites (n=8), performance outcomes including attributed OE scores (n=4), and performance outcomes based on MM+SL only (n=4). Within each family, the nominal $\alpha = .05$ level was divided by the number of tests in that family to define the corrected significance threshold. Results are reported with both the Bonferroni-adjusted thresholds and the unadjusted $\alpha = .05$; only findings that meet the corrected threshold are treated as more robust, while raw-significant results are noted but interpreted cautiously.

4.3. Qualitative Data: Workshop For Validation And Recommendations

The post-survey workshop for validation and recommendations featured seven participants, three of which participated in the interviews, two who responded to the survey, and two who did not participate in previous stages. Three participants joined on site, and four partook remotely, and the workshop lasted for approximately 60 minutes. The workshop was organized around several key themes that emerged from the discussion of the innovation-management findings: the perceived tension between leadership and support, internal and external collaboration and how it is affected by confidentiality measures, how to measure innovation, the combination of exploitation and exploration, and strategic scope. Overall, participants endorsed the findings, acknowledged the strengths and weaknesses that had been highlighted, and added further nuance to each area. As with the interviews, no audio recording was made to protect confidentiality; instead, the discussion was transcribed in real time. After the workshop concluded, the notes were synthesized into a written summary that captures the key discussion points and records the interviewer's immediate impressions of the participants. The findings from the workshop contributed greatly to the final recommendation in the discussion chapter, thanks to the specific insights supplied by participants.

4.4. Secondary Data Collection: Literature And Internal Documentation

The primary objective of this study was to analyze the innovation management system of the target company through empirical data. Consequently, a systematic literature review (SLR) was deemed outside the scope of this research. Instead, a targeted literature search was conducted to compare, analyze and develop findings as they emerged. Searches were performed using the KTH library database, utilizing keywords related to innovation

management and associated domains, such as 'innovation management resource allocation,' 'innovation performance drivers,' and 'innovation performance evaluation,' among others. Priority was given to highly rated peer-reviewed journals; lower quality peer-reviewed sources were included only when necessary to address gaps in the existing literature.

Internal documentation was retrieved via searches of the company intranet using terms such as 'innovation management,' 'innovation strategy,' and 'innovation portal.' Additionally, specific documents were provided by interviewees when relevant topics arose during the interview process

4.5. General Method Analysis

This section describes various elements of the general method and analyzes them to assess strengths and limitations.

Quality

By addressing the CASET principles presented by Goffin et al., 2019, the quality of the methodology was improved systematically. This methodological decision was taken to strengthen the credibility and validity of the findings; studies that omit any of the criteria tend to receive lower overall quality ratings, attract fewer citations, and make weaker theoretical contributions (Goffin, et al., 2019), highlighting the importance of adopting the criteria. The inclusion of the items described (See Table 2) positions the current work well above the benchmark.

Even though the design does not incorporate purposive (theoretical) sampling, inter coder reliability checks, or a formal case presentation matrix, the elements that are present directly address many of the frequently neglected quality dimensions. These three factors were not addressed due to inherent limitations of the study. The company chosen for the case study was not purposefully selected, and the choice was instead more opportunistic in nature. The author received the assignment upon request by the company, which in turn may negatively impact the transferability of the findings (Gioia, et al., 2013). The prevailing study therefore fails the *theoretical sampling* criterion of CASET (Goffin, et al., 2019). *Inter coder reliability checks*, i.e. having “multiple researchers code the data independently” (Goffin, et al., 2019), was not possible to perform because a single researcher was responsible for data collection, interpretation and analysis. Consequently, the reliability of, and confidence in the qualitative interpretations may be reduced (Goffin, et al., 2019). The mixed-methods design, however, helps to offset this limitation, as the quantitative component is not subject to the same single-coder constraint. Finally, in analyzing the qualitative data, a formal, systematic coding method was not employed, leading to the unfulfillment of the *case presentation* criterion.

Grounded theory approach

The overall methodological process followed the classic grounded-theory cycle of collection, coding, and comparison, continuing data collection until theoretical saturation was reached (Lim, 2025). Initially, open-ended interviews were conducted to gather qualitative data. Interview participants were chosen through theoretical sampling; after each interview, emerging concepts guided the selection of the next participants who could elaborate on the most promising areas, allowing the theory to develop directly from the data (Lim, 2025). During this qualitative phase, the literature review was deliberately postponed so that existing

theories would not influence the emerging categories. This delay helped maintain neutrality while the core concepts were still forming (Lim, 2025).

Mixed method approach

Gathering qualitative data before developing a survey for quantitative data collection was integral to obtaining relevant information, in-depth insights and a holistic perspective (Lim, 2025). The exploratory interviews captured specific details about innovation work and organizational context which were used to shape the design of the survey. For example, the decision to survey three respondent groups at different organizational levels stemmed from an observed discrepancy in their perceptions of innovation and of their innovation-management competence. Without this context, the findings of the quantitative data would have remained generic and far less likely to address the core issues (Molina-Azorin, 2012). These benefits of the present study's mixed-methods approach is supported by the literature: it has been indicated that mixed-methods research can contribute to greater breadth and richer interpretation of data, especially useful for assessing complex innovation management (Bracio & Szarucki, 2020). Thus, the combination of qualitative exploration and quantitative testing, together with documentation review provided a robust, triangulated data set, as opposed to a purely qualitative or quantitative study.

5. Results and analysis

This section outlines the results of the data collection and incorporates an analysis of the results in the same section. The purpose of the section is to reveal the main findings of the qualitative and quantitative data collection, respectively.

5.1. Qualitative Data: Exploratory Interviews

The exploratory interviews resulted in multiple findings relating to innovation management and performance across different functions of Saab Surveillance, and Saab Group. Several areas that currently are holding back the company's innovation efforts and diverging the company from adhering to ISO 56001 requirements were highlighted. The following areas were identified as key areas of Saab Surveillance's innovation work, both as strengths and weaknesses.

5.1.1. Unlocking Employee Innovation Potential

Innovation at Saab Surveillance most frequently occurs in project teams of customer delivery projects (interview 7). Development is guided by skilled engineers while customers are involved to create solutions tailored to their needs. This innovation usually results in more incremental outcomes, but still represents fruitful efforts that shall be recognized as a key strength of the organization. Multiple interviewees therefore see the potential in utilizing the capabilities of the employees to achieve both incremental and radical innovation (Interview 2, 7). Saab Surveillance seek to capitalize on this using various innovation functions that serve to generate ideas and foster creativity. Among these are the innovation portal (Interview 1, 4, 5, 9), and multiple innovation labs (Interview 10, 11).

Innovation portal

The innovation portal allows employees to submit ideas that are then inspected by innovation councils. Ideators receive a hybrid budget that includes both time and money, and in practice, both elements become financial costs for the organization. The hours they log for innovation are not billed to their regular projects, and are instead charged to dedicated innovation resources (Interview 5). Promising ideas are allocated an initial 24 hours of billable time for development, and upon passing a second review, 300 000 SEK is awarded (interview 9). These funds aim to result in a further improvement of the concept. In a final step, ideators may receive extended budgets aiming to start an actual project surrounding the idea (interview 1, 9). Typically, ideas submitted to the portal involve more radical solutions (Interview 1), since e.g. incremental improvements typically occur organically in projects (Interview 7). The review steps are hosted by separate bodies – the first step by an innovation group featuring both employees and managers, whereas the second and third steps are controlled by a team with more senior managers. The participants of these councils do not manage the idea evaluation full time (interview 1, 9).

The innovation portal has received praise for giving employees an outlet for their ideas, but also critique: So far, submitted ideas reportedly seldom reach a commercialization stage, and follow up of ideas and performance measurement has been limited (Interview 1, 5). The effectiveness of its process has also been questioned, with some arguing that the short time

allocation rushes ideators (Interview 2), that improved coaching for developing ideas and driving innovation efforts is necessary (Interview 2, 6), and that projects sprung from ideas lack ownership (Interview 1). For example, a portal manager suggested that submitted ideas are awarded a sufficient budget for initial development (interview 5), whereas multiple employees indicated that the allocated resources are limited, leading ideators to focus less on testing the solution and instead trying to appease the rating council to receive more time (Interview 2, 11). The tension between core operations and enabling innovation was also highlighted. One manager of the technology council indicated that the ideation portal primarily serves the role of promoting innovative ideas, and that the current order backlog is what needs to be prioritized (Interview 5). The interviewee further cited that the organization struggles finding time to invest in the ideas (Interview 5). This is supported by feedback from employees indicating a perceived lack of transparency regarding the tracking and feedback loop of ideas submitted to the portal (Interview 11).

Innovation labs

The innovation labs employ a less hierarchical structure, and instead act as forums for testing concepts, creating early-stage prototypes, and promoting an innovative culture (Interview 10, 11). The software innovation lab is the older of the two, having been founded in 2019. The lab has worked for many years to raise their credibility and internal awareness, through e.g. hackathons, lectures and projects. The labs' goal is to promote future technologies like AI, and rapid development methods, and internal marketing has therefore been pivotal for success (Interview 10). The lab has grown to 4 to 5 full time positions, and an expansion to 7-8 people is underway as of 2026. They are partially separated from the core organization, allowing them to more rapidly test new technologies without confidentiality constraints. So far, most projects have been experimental and lacked implementation in the core business, but recent efforts are seeking practical applicability (Interview 10). For example, a concept from the software innovation lab was connected with a relevant project where it could have delivered meaningful value. However, in that case, the project owners could reportedly not prioritize new ideas and opted out of the collaboration (Interview 10). The hardware innovation lab is a younger organization, having started only three years ago. They have no full-time employees, and thus have a smaller outreach. However, the founder aims to promote innovative ways of working and to ignite the creative "flame" present in employees (Interview 10). He seeks to grow the team to meet the demand of physical, rapid-prototyping for new products (Interview 10). For example, the lab recently exemplified how what used to be a bureaucratic multi-year process of CAD design, supplier procurement and stage-gate process project could be fast-tracked using fast failure working methods and minimum viable products (MVP)'s (Interview 10). In sum, the lab aims to promote simple, hands-on innovation and to explore the hardware-software intersection, where innovative hardware solutions can enable the implementation of future software.

Thanks to these factors, the labs represent a key asset for the organization. Through their missions, the labs directly promote Saab Surveillance's innovation strategy of faster development and proactively seeking to meet the future demands of customers. Thus, they also contribute to increasing the correspondence of the system. Nonetheless, in common between the two labs is that they seldom collaborate with ideators from the innovation portal (Interview 10, 11). Employees developing their ideas in the portal have no direct connection to tools like the innovation labs. Instead, the continued refinement of ideas selected in the portal is more ad-hoc (interview 1). These challenges in interconnectedness between system components may cause synergies between creative ideators and supportive innovation lab professionals to be lost (Mele, et al., 2010), in turn lowering the system performance (Bititci, et al., 2011).

5.1.2. Tensions In Resource Allocation For Innovation

As described previously, the company has stipulated processes for allocating resources toward ideas that enter through the innovation portal (Interview 1, 4, 5). Employees can also reportedly seek resources outside the innovation portal, e.g. through the innovation labs (interview 11), or even through informal means (Interview 12). One interviewee explained that they on multiple occasions bypassed the formal funding portal entirely, instead turning to their personal network. By leveraging relationships and networking skills, they obtained budget from project managers who had spare funds available (Interview 12). One might argue that this approach allows innovation to occur more organically, accelerating decision-making because it represents a more flexible method of obtaining funds for explorative projects. However, this may also disfavor less connected employees and cause the resource allocation to lack strategic direction (Taques, et al., 2021). In parallel, for organization-driven innovation projects, funds are often deliberately earmarked for either exploitation, or more explorative development (interview 1). This is a strength of the innovation system, since it seeks to ensure that incremental and radical innovation can be pursued simultaneously. Nevertheless, an interviewee indicated that funds intended for more radical innovation often are diverted to units seeking operational efficiency rather than being channeled to the dedicated innovation unit (Interview 2). They also suggested that this may be due to the limited performance evaluation for innovation (interview 2). These factors can hamper innovation efforts (Kihlander, et al., 2024), and indicates a culture geared toward incremental improvement, but also the operational reality where shorter-term project delivery targets can naturally create a resource-allocation conflict with longer-term technology exploration.

These challenges appear to be related to the company's broader perspective and culture regarding resource allocation, especially time. This applies to daily operations, but also the innovation portal. Reportedly, the time for development allocated to promising ideas is not guaranteed, since it depends on the priorities and approval of the ideators' managers (Interview 5). As for the daily operations, a common sentiment was again that the company does not have time for long-term focused, radical innovation efforts with only its internal resources (interview 3, 5, 7, 11). This is concurrent with the supported observation that *Time for innovation work is not prioritized or protected in a mandated and structured way*. Instead, interviewees expressed that internal innovation resources are best spent on operations and incremental improvements to the current catalogue (Interview 5, 7), whereas external collaboration should be responsible for radical innovation (Interview 3, 7). The two latter observations are a common structure to pursue both incremental and radical innovation, but requires a clear strategic distinction of where each type of innovation will be pursued (March, 1991; O'Reilly & Tushman, 2004). The current structure where the organization pursues radical innovation both internally and externally may potentially limit the overall effectiveness of both innovation pathways (Kihlander, et al., 2024). Furthermore, since incremental and radical innovation requires vastly different leadership styles (Jansen, et al., 2009), this puts a large strain on leaders who must navigate the tension between exploitation and exploration. They face the pressure of both fulfilling operational efficiency, while allocating time for long term focused innovation ideas and projects that have limited impact on their immediate priorities. One employee described this tension, citing that

> *"there's nothing that can take priority over delivering to the customer [...] [but] we must make innovation a core part of our project work if we want to remain a high-tech company in the future."*
>
> (Interview 11)

5.1.3. Communication Pathways Are Not Always Clear

When discussing the tensions reported in resource allocation, communication challenges were also mentioned. An interviewee indicated that even though a budget had been allocated toward explorative innovation, colleagues displayed no awareness of it (Interview 2). This suggests that certain communication pathways could be improved, both within and across projects, although it is advised that this perspective is verified through further inquiry. Nevertheless, it has been found that "a lack of relentless communication of the [exploit and explore] strategy [by senior leaders] can further undermine cooperation" (O'Reilly & Tushman, 2011), highlighting the importance of communication and the role of leadership in ensuring it.

5.1.4. Collaboration Efforts Are Present, But Could Be Improved

Saab proudly operates with a triple helix approach, where government, industry and academia jointly collaborate. External collaboration enabled by supra-system interactions with universities, start-ups, research institutes, and others is seen as essential to generate new ideas and speed up development (Interview 1, 2, 3, 5, 8, 9, 10). To meet this end, Surveillance conducts technology scouting within an innovation and technology council (Interview 9). The company also actively pursues low TRL research projects in collaboration with government agencies, the Swedish armed forces, and research institutes who occasionally help fund projects (Interview 1, 7). The institutes may also require that a university be involved, further contributing to external collaboration (interview 1). In fact, most explorative innovation efforts start with university research collaborations (Interview 2, 5). This is a merit of the innovation system (Demircioglu, et al., 2019) – the organization actively stays informed of technological trends, seeks collaborative relationships, and learns from its environment.

Saab Surveillance also currently collaborates with Saab Ventures, a separate organization that delves into incubation, business coaching, and venture capital (Interview 8). Ventures utilize inbound open innovation (OI), where start-ups are invested in and actively managed through Ventures' employees' expertise. Outbound OI projects also occur, with defense technology being spun-off in start-ups for civilian applications, or patents from the core organization being exchanged with start-ups for an ownership share (Interview 8). Eventually, these start-ups may be acquired by Saab Surveillance or a competitor, or developed to become a supplier or customer of Saab. The mission of Ventures is not to serve Saab's core organization, but to grow the companies they invest in, which is why they operate independently (interview 8). The organization has had numerous successes, both independently and through collaborating with Surveillance (interview 8), and therefore represents a fruitful channel for innovation. Furthermore, in the last years, Saab Group have also sought to create internal start-ups through accelerators and innovation incubators – separate from Saab Ventures (Interview 7). These occur through Innovation Hubs. For example, in one project, a cross-functional team of young, skilled, and motivated employees were provided an office separate from the core organization and its security rules. The team was given a clear mission and purpose. Through iterative working methods like build, measure, learn, the project has shown promising results, with a prototype being tested early 2026 (Interview 7).

These examples exemplify the structures and methods that the greater Saab organization has in place to drive collaborative innovation. They represent effective methods of achieving more explorative and radical innovation, either through the development methods of the Innovation Hubs, or the commercialization expertise of Saab Ventures. The physical separation of innovative units from the main business seen in these examples, has shown to lead to more

explorative outcomes (Smith & Tushman, 2005). Surveillance may therefore be inspired by, or leverage these actors to achieve more radical innovation.

However, it was also indicated that external collaboration has potential for improvement. Some interviewees saw that a "not invented here" culture and lack of belief in external partnerships exists among employees (interview 3). A senior manager said that Saab's partnerships are underutilized, suggesting that Saab should "work with Sweden's innovation eco-system" to a greater extent (Interview 7). This is corroborated by an innovation manager, who saw the potential of further external collaboration to achieve innovation speed and performance, but also suggested that Saab's confidentiality measures hinder this (Interview 2), affecting both internal and external partnerships. For example, external collaborators like universities only receive partial information about the research (interview 1), potentially limiting the applicability of their findings.

Internally, confidentiality measures, established to protect the organization, have reportedly led to silo-fication of ideas and resources in the sub-system: both in projects, but also across business units and business areas. For instance, at the project level, an innovation budget within a project was increased to promote collaboration with other Saab Surveillance functions, but this effort was suspended to prevent possible leakage of classified information (interview 2). Thus, development may be stalled since the utilization of budgets intended for innovation is hindered (Klingebiel & Rammer, 2014), and the possibilities of cross-functional knowledge sharing are limited. It also indicates a strained strategic correspondence in the system, since the goal of improving collaboration (as per the innovation strategy) was prevented. Furthermore, this can be seen the innovation idea submission portal. Only council managers can view, comment, and grade ideas (interview 1, 2), potentially limiting the synergies that can be achieved if ideators can share and build upon each others' ideas (Sattayaraksa & Boon-itt, 2018). At the organization level, business areas (BA) within the Saab Group rarely share knowledge or capabilities either (Interview 12). Although the BAs develop vastly different technologies, innovation frequently arises from the recombination of current solutions, arising from unforeseen synergies. The wider organization could therefore stand to benefit from more open internal collaboration.

5.1.5. A Culture Geared Toward Operational Excellence

It was frequently cited that Saab *should* work more innovatively, with shorter development cycles, and take more risks as vouched for by top management (Interview 7; Saab, 2025), but that the broader company struggles with it. Multiple interviewees highlighted historical reasons behind this, citing that the procurement process of Saabs main customer, the Swedish Defense Materiel Administration (FMV) has created a market-pull and fulfill-orders mentality throughout the company (Interview 2, 3, 7, 11). In turn, this has caused a lack of proactive work to anticipate what customers might demand in the future and developing it today (Interview 7). It has also caused a culture that favors perfection and complete certainty (Interview 2), suggesting a low tolerance for risk and experimentation.

The gap between top management strategy endorsement and the operational cultural reality is validated by the observation that senior leaderships' explicit endorsement of innovation does not translate into higher perceived support for innovation among middle managers. For example, allocating time away from project work and customer deliveries is sometimes frowned upon by colleagues (interview 11), despite the company's new strategic goals. This suggests that the coherence of the innovation strategies could be improved. Without mitigation, this may limit innovation outcomes and prevent the organization from meeting its innovation goals.

5.1.6. Lack Of Shared Vocabulary

In conducting the interviews, it became evident that managers and employees, even those working with innovation efforts, lack a shared vocabulary. Among higher level professionals working in same team, there were inconsistencies in their interpretation of innovation management concepts highly relevant to their tasks (Interview 1, 4, 5). This could potentially lead to misunderstandings when conducting innovation and when taking strategic decisions. Adopting ISO 56000 could be used to mitigate this shortcoming, since one of its key reported benefits is the creation of shared vocabulary.

5.1.7. Validation Of Interview Observations

These findings presented in this chapter were condensed into a list of 19 observations, and then validated with the eleven interviewees. Four responded, the summary of which is shown below in Table 7 and Table 8. The full results are presented in the appendix, section 11.1.

Table 7: Average rating of validated observations from the four respondents

Respondent role	Innovation manager	Project / middle manager	Operational employee	Senior leadership
Average score - higher indicates agreement	0.92	0.72	0.88	0.64

Although the observations describe complex areas that cannot be accurately represented by the short 0 to 1 scale, the results provided an indication toward the validity of the observations. The four respondents together represented the three levels of employees identified for the quantitative data collection, but was skewed toward management roles. Consequently, the findings may reflect the perspectives of managers more strongly than those of operational staff, which in turn may impact the result through the observed divergence between the senior-leadership and operational-employee responses. However, because the validation data set comprises only four participants, it lacks the statistical power required to draw reliable conclusions. The aim was not to achieve statistical significance, but simply to obtain an indication of the observations' validity, and should therefore be treated as such.

Table 8: Compilation of the most and least supported observations from the qualitative data validation

Rank	Most supported	Least supported
1	O7: Time for innovation work is not prioritized or protected in a mandated and structured way. (It depends a lot on managers' willingness to innovate)	O15: There is a lack of support for innovation, like coaching, tools, and processes.
2	O9: The organization lacks formal, measurable innovation objectives at the operational level	O2: After ideas have been evaluated, there is no formal continuation to the process to develop or commercialize the idea(s)
3	O12: Little personal coaching or mentorship is given to employees developing their ideas in the innovation portal	O6: Line managers are reluctant to allocate time for their employees to carry out innovation.
4	O1: Cross BU visibility of ongoing projects and research is low; people often do not know what other Business Units are working on, limiting synergies.	O17: There is lacking systematic corrective-action after an innovation project fails or stalls.
5	O4: Senior leaderships' explicit endorsement of innovation does not translate into higher perceived support for innovation among middle managers	O18: Deep insights do not commonly precede projects / catalyze the starting of projects. (deep insights can be knowledge from technical and market research that lead to the creation of many ideas)
6	O10: The core organization lacks a culture geared toward exploration and innovation. (fail fast approaches, testing, prototyping, etc)	O3: The idea generation and development process lacks integration with other Saab functions. (innovation labs, Saab ventures, operational projects, etc.)

5.2. Quantitative Data: ISO 56001 Adherence Survey

This section presents the results of the quantitative data collection. It revealed exploratory patterns regarding managing innovation systems, and descriptive data about the company's adherence to ISO 56001. Results are reported first and immediately followed by interpretation. Data is organized by the type of test performed. For a detailed description of the analysis method, please refer to section 4.2.

5.2.1. Response Rate

The survey was sent out to 120 potential respondents, receiving 35 responses for a response rate of 29% (Table 9). OE received 80 invitations and generated 17 responses (21 %); MM received 18 invitations and generated 11 responses (61 %); SL received 22 invitations and generated 7 responses (32 %). The MM group attained the highest response rate of 61 %, which could suggest that the survey's content, length, and communication resonated well with this audience. SL resulted in a 32 % response rate, which is considered moderate. While it provides a useful snapshot of executive perspectives, two-thirds of senior leaders surveyed did not contribute, limiting the depth of strategic insight that can be extracted. The OE group, however, delivered a low response rate of only 21 %, meaning that the voices of the workforce are under-represented. Nevertheless, the generally low number of recipients and respondents across the three groups remains a limitation. Consequently, any organization-wide inferences must be treated with caution. The response rates of SL and MM were consistent with the findings of Cycyota & Harrison (2006), but the OE group delivered a lower response rate than expected.

Table 9: Overview of the number of recipients and respondents of the questionnaire

Target group	*Recipients*	*Respondents*	*Response rate*
Operational Employees (OE)	80	17	21%
Middle Managers (MM)	18	11	61%
Senior Leadership (SL)	22	7	32%
Overall	120	35	29%

5.2.2. Harman's Single-Factor Test to Evaluate Common Method Bias

The Harman's single-factor test explained 36.0% of total variance among MM respondents and 43.5% among SL respondents, both below the conventional 50% threshold. A high percentage would suggest that one underlying influence, such as a shared tendency to respond in a certain way regardless of question content, is driving most of the answers. The values obtained instead indicate that variance is distributed across multiple factors, suggesting that respondents were genuinely reacting to the content of individual items rather than a single common source. Common method bias (CMB) is therefore not considered a primary concern in either group, and correlations between ISO dimension scores and innovation performance outcomes were deemed interpretable within both groups.

Table 10: Result of Harman's single factor test, presenting the variance of each target group compared to the threshold

Group	*Variance*	*Threshold*
MM: ISO dimension vs Performance	36%	50%
SL: ISO dimension vs Performance	43,5%	50%

It should be noted that Harman's single factor test is a limited diagnostic that can indicate the presence of CMB but cannot fully rule it out, and the results should therefore be treated as a preliminary check rather than a definitive assessment. Alongside this assessment, the internal consistency of each ISO area was examined using Cronbach's alpha to evaluate the reliability of the composite scores used in subsequent analyses.

5.2.3. Cronbach Alpha to Assess Internal Consistency of ISO Dimension Items

The results of the Cronbach Alpha analysis are presented in Table 11.

Table 11: Result of the Cronbach Alpha analysis applied to the items within each ISO 56000 dimension

ISO Area	*No. of items*	*N respondents*	*Alpha*
Context	2	33	**0.648**
Leadership	3	32	**0.771**
Planning_Portfolio	1	NA	**NA**
Planning_Collab	1	NA	**NA**
Support	3	32	**0.726**
Operations	3	25	**0.554**
Performance	2	23	**0.832**
Improvement	1	NA	**NA**

Items grouped under the Leadership, Support, and Performance dimensions each yielded $\alpha > 0.70$, which meets the commonly accepted threshold for internal consistency and indicates acceptable construct validity for those scales. The *Context* dimension produced an α slightly below 0.70, while the Operations dimension showed a markedly lower value. These findings suggest that the items composing the Context and especially the Operations dimension do not reliably measure a single underlying construct, representing a limitation that should be acknowledged in all subsequent analyses. The particularly low reliability of the Operations scale may however be attributed to its item composition: the three items address distinct phases of the innovation process (ideation, concept validation/testing, and commercialization), which are conceptually heterogeneous and therefore not expected to co-vary strongly. The analysis then progressed toward assessing the ISO 56001 compliance.

5.2.4. Descriptive Statistics to Summarize ISO 56001 Adherence Across Organizational Roles

To assess ISO 56001 adherence for Saab Surveillance, the response values for items within ISO dimension question areas were pooled and averaged across the three respondent groups. This

produced average Likert scores for each ISO dimension, which were subsequently converted into percentages and displayed in the following radar chart (Figure 5).

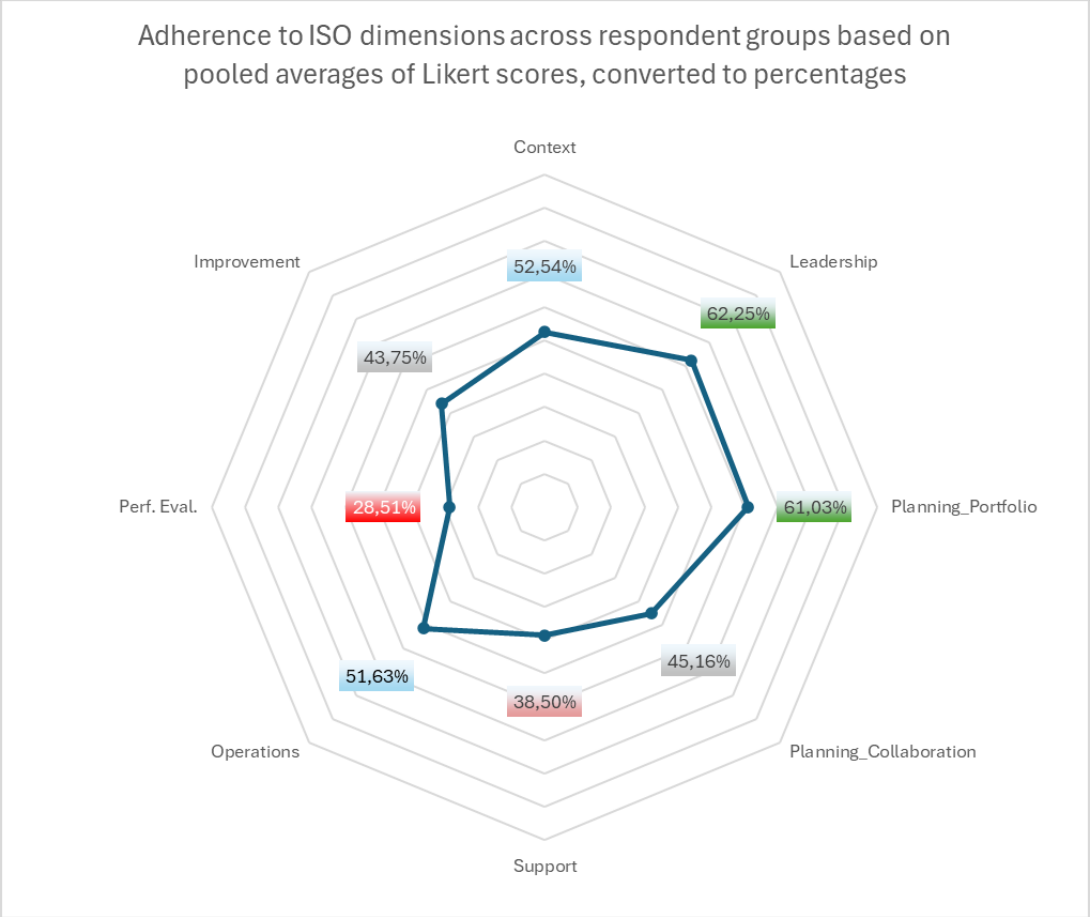


Figure 5: Radar diagram showing the adherence to ISO 56000 dimensions, rated on a 0-100% scale based on survey results.

Respondents show varying levels of agreement relating to the seven ISO dimensions. The highest score of 62.25% was achieved within the Leadership dimension, derived from items surrounding cultural acceptance for innovation and leadership encouragement. More specifically, the individual items, relating to tolerance for failure, risk taking, and promoting an innovative culture earned scores of, 61%, 55%, and 71% respectively. The planning dimension, which was separated into the areas innovation portfolio management and innovation collaboration, received mixed results – portfolio received the higher agreement of 61.03%, whereas collaboration was scored 45.16%. Then, around 50%+ agreement was shown within the areas *context* and *operations*, concerning topics like contextual awareness and innovation process, respectively. *Improvement* (43.75%) and *support* (38.50%) both resulted in scores under 50%, and finally, *performance evaluation* received the lowest agreement of 28.51%. A data table including a full table of results can be found in appendix B, section 11.2.1. These descriptive results provide an initial guide for further investigation, highlighting that Saab Surveillance shows moderate maturity in innovation-encouraging leadership, cultural support, and portfolio management, but requires attention in collaborative practices, resource support, and systematic performance measurement. The uneven alignment further suggests a

limited comprehensiveness of the IMS, since not all areas prescribed by the ISO 56001 requirements are adequately addressed. Nevertheless, these results require further inquiry with more questions and a wider population to determine whether the same status can be observed across the Saab Surveillance organization.

Four innovation performance dimensions were also investigated, and are presented in Figure 6 below. The results showed that between the four dimensions, incremental innovation had the most positive score of 67% (3.67, SD = 0.84). Perceived customer involvement in developing solutions also had a positive inclination, although to a lesser degree (3.18, SD = 1.01). On the other hand, radical innovation scored 46% (SD = 0.73), toward disagree relative to the "neutral" score of 3 (50%), and speed to market was the metric with the lowest perceived performance, with only a 32% agreement (2.28, SD = 0.83). This indicates that Surveillance is successful in generating incremental innovation, like the continuous development of existing products, and moderately successful in engaging customers. On the other hand, it also shows that generating radical solutions and quickly commercializing these are areas of potential improvement.

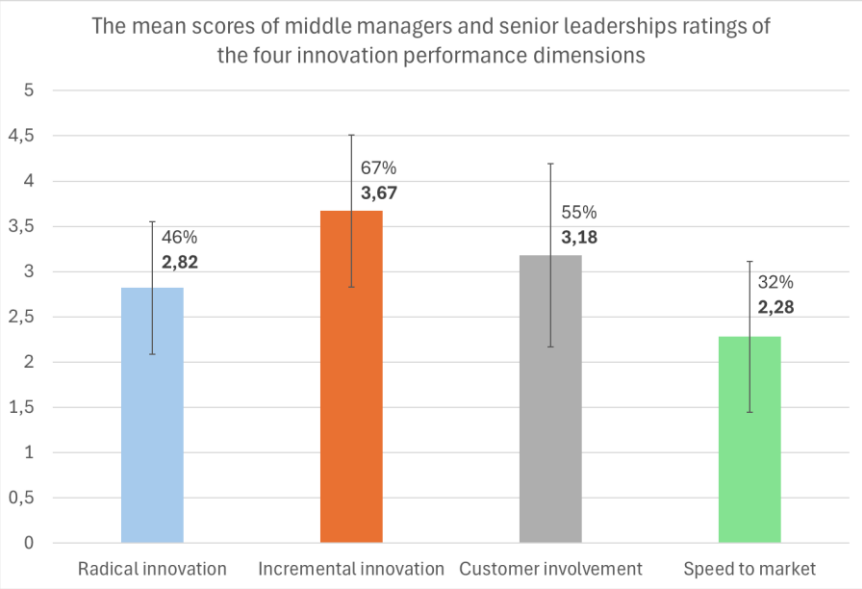


Figure 6: Visualization of descriptive results showing the mean scores of middle managers and senior leaderships ratings of the four innovation performance dimensions

5.2.5. Spearman and Pearson Correlation Analyses to Examine the Relationship Between Perceived ISO-56001 Fulfilment and Innovation Performance – Extended Sample

The Spearman and Pearson correlation analysis revealed areas of statistical significance. Out of the 36 tests comparing separate ISO dimensions and the overall IMS to perceived innovation performance, five resulted in significance (Table 12). An excerpt of the data is shown below. For the comprehensive data table, please refer to appendix B, section 11.2.2.

Table 12: Result of the Spearman and Pearson correlation tests, featuring the extended sample of OE, MM and SL respondents

DV	*Predictor*	*n*	*Spearman r*	*Spearman p*	*Pearson r*	*Pearson p*	*Significance*
Radical	Support	29	0.629	3.00E-04	0.593	7.00E-04	** Both sig (p<0.01)
Radical	Improvement	27	0.485	0.0104	0.396	0.0411	** Both sig (p<0.05)
Customer	Support	29	0.379	0.0425	0.318	0.0927	* Spearman only (p<0.05)
Radical	Overall_IMS	29	0.382	0.0407	0.382	0.0409	** Both sig (p<0.05)
Customer	Overall_IMS	29	0.387	0.0383	0.28	0.1409	* Spearman only (p<0.05)

Three tests showed significance in both the Spearman and Pearson analysis, all with the radical innovation performance as the dependent variable. The first was with the predictor support, the second with the predictor improvement, and the third with the predictor consisting of the overall IMS. A number of findings stand out, including:

1. First, that having organizational support for innovation – such as time, budget allocation, and tools – is positively associated with radical innovation performance (Spearman p = 0.0003, Pearson p = 0.0007), a relationship supported by literature (Klingebiel & Rammer, 2014). It shall be noted that determining whether this result generalizes beyond the studied firm would require additional evidence and analysis. Nevertheless, a possible explanation behind this relationship is that when staff perceive that sufficient resources are available, they can devote more effort to exploratory activities, which in turn translates into higher radical-innovation outcomes. However, the opposite also applies: inadequate or misdirected resource allocation constrains the ability to pursue breakthrough ideas, thereby limiting radical-innovation performance.
2. Second, that continuous improvement measures, such as gathering lessons learned from innovation projects to improve innovation processes, are positively associated with radical innovation performance. The finding is unexpected, yet a plausible interpretation is that by iteratively improving innovation processes, staff can actively impact working methods to drive radical outcomes, through e.g. developing a rapid-prototyping process for concept validation. Consequently, the finding underscores the importance of a learning-oriented culture in which improvement is treated as a strategic capability. However, the robustness of this finding may be questioned, because the improvement construct was measured with a single item, raising concerns about content-validity since the item may not fully capture the full ISO area.
3. Third, that having a comprehensive innovation management system covering multiple ISO dimensions also is positively associated with radical innovation performance. This result provides an early indication of the effectiveness of adopting a comprehensive IMS such as ISO 56001, consistent with the findings of Gimenez Espin, et al. who studied the Spanish innovation management standards (2025). This result may also suggest that the comprehensiveness of the IMS brings performance benefits through supporting emergence in the system.

The two other tests reached the 0.05 significance level in only the Spearman-rank correlations. Customer-centricity was significantly related to two predictors: the overall Innovation-Management System (IMS) score and the ISO "Support" dimension. The first association (Spearman $\rho = 0.387$, $p = 0.038$) shows that staff within Saab Surveillance with

higher average scores across all ISO-requirement items tend to report stronger alignment with the company's strategy of "involving customers early and using their feedback to shape solutions that deliver real value." The second association (Spearman $\rho = 0.379$, $p = 0.042$) indicates that employees who perceive Saab Surveillance's innovation-management system to be more supportive (more time, budget, tools, etc.) also tend to rate their organization as more customer-centric. The first finding indicates that a more comprehensive, standards-based IMS is positively associated with employees' perception that the company is oriented toward the customer, and the second finding provides a more precise perspective, as it positively associates support for innovation to customer-centricity. A possible interpretation of these findings is that an ISO 56000-based IMS, with effective resource allocation toward innovation, can allow the organization to more actively involve and consider the customers in development. This result should however be interpreted with caution, because the significance only emerged in the Spearman test, and not in the Pearson robustness check. Furthermore, to assess which dimensions of the IMS contribute most to fulfilling the customer-centricity strategy, a linear regression could have been plotted. But due to the high number of variables considered versus the number of responses, such a regression would not result in an accurate analysis. This also applies to the other innovation performance dimensions studied.

Despite the significant correlations shown across these findings, it must be iterated that the results stem from a limited sample ($n \approx 29$), and should therefore be interpreted cautiously and as exploratory results. The three associations that were significant in either Spearman or Pearson should also be cautiously interpreted, as they have less statistical power than the associations that showed significance in both correlation analyses. Furthermore, when many tests are performed at the conventional $\alpha = 0.05$ level, the probability of obtaining at least one false-positive result (a Type I error) increases markedly, meaning that some of the significant correlations seen may simply reflect random fluctuation rather than a genuine relationship. Lastly, multiple results contrast Giménez-Espin et al. (2025) and Martinez-Costa et al.'s (2019), who found that the positive effects of individual items of standard showed no statistical significance. The findings could therefore be considered a novel contribution, or an artefact of the low number of respondents coming from a limited part of the organization. Thus, these results must be validated.

5.2.6. Spearman and Pearson Correlation Analyses to Examine the Relationship Between Perceived ISO-56001 Fulfilment and Innovation Performance – Limited Sample

To provide additional data and account for the potential limitation of copying managers' performance scores to employees, the same analysis was also conducted for only the middle managers and senior leadership ($n \approx 18$). The result of which can be seen in the table below, with the extended data table available in appendix B, section 11.2.3.

Table 13: Condensed result of the Spearman and Pearson correlation tests, featuring the limited sample of MM and SL respondents

DV	*Predictor*	*n*	*Spearman r*	*Spearman p-value*	*Pearson r*	*Pearson p-value*	*Significance*
Radical Innovation	Support	17	0.652	0.0046	0.663	0.0037	** p<0.01
Customer Involvement	Context	17	0.486	0.0478	0.468	0.0582	* p<0.05

As shown in Table 13 the limited sample produced far fewer $p < 0.05$ significant correlations. In this subsample, the only association statistically significant in both Spearman and Pearson was the positive link between Support for innovation and Radical-innovation performance (Spearman $r = 0.652$, $p = 0.0046$; Pearson $r = 0.663$, $p = 0.0037$). In addition, Customer Involvement was positively associated with Context in the Spearman analysis ($r = 0.486$, $p = 0.0478$) but did not reach significance in the Pearson test ($r = 0.468$, $p = 0.0582$). All other relationships that were significant in the full-employee sample, including Radical-innovation vs. Overall IMS, Customer-centricity vs. Overall IMS, Customer-centricity vs. Support, and Radical-innovation vs. Improvement were non-significant when only managers were considered.

These results suggest three, non-exclusive, explanations. First, the inclusion of all employees may have introduced additional data that helped detect relationships that were not robust at the higher organizational levels. Second, the copying of managers' performance scores to employees may have artificially inflated correlations in the full-sample analysis. Third, the persistence of the association between Support and Radical-innovation across both samples reinforces the initial conclusion that allocating tools, budget, and time to innovation is closely tied to radical-innovation outcomes. As with the full-sample findings, the manager-only data set is limited, and the multiple correlation tests increase the chance of false-positive (Type I) results. Therefore, these findings should be viewed as preliminary, exploratory evidence that needs to be confirmed with a more representative sample. After the Spearman and Pearson correlation tests, t-tests were performed.

5.2.7. Welch's Independent-Samples t-tests to Compare Innovation Management System Perceptions Between Hierarchical Levels

A Welch's t-test was run to compare the three respondent groups on every ISO-derived item and on the composite scores for each ISO area. In total, 77 pair-wise comparisons were performed. Using the conventional $\alpha = 0.05$ threshold, 13 of those comparisons yielded a raw p-value < 0.05, and only one of them survived a strict Bonferroni correction. The extended data table can be found in Appendix B, section 11.2.4. The bulk of the significant differences involve senior-leadership respondents. Compared with operational staff, senior-leadership differed on three factors, and compared with middle-management, senior-leadership differed on seven factors. In contrast, operational staff showed only four significant differences when compared with middle-management. Counting the raw-$p < 0.05$ results, both middle managers and senior-leadership differed from the other two respective groups on 10 of the 13 significant comparisons, while operational employees only differ from the other groups on 6 of the 13 comparisons (Figure 7). These disagreements suggest a possible misalignment and disagreement between staff levels in the organization, especially between senior leadership and middle managers. However, the raw significant results shall be interpreted as an area of future inquiry, and not as confirmatory.

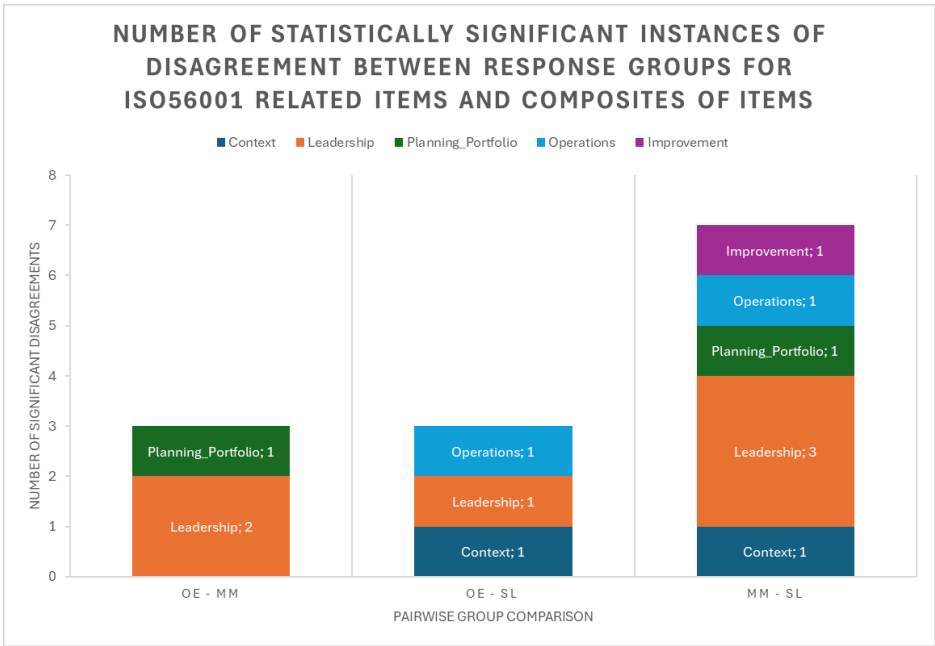


Figure 7: Column chart displaying the number of statistically significant instances of disagreement between response groups for ISO 56001 related items and composites of items

Furthermore, the ISO dimension that generated the most between-group disagreement is Leadership (Figure 8). For the two significant items (risk-taking for innovation and the tolerance of failure/experimentation), middle managers consistently reported the highest scores ($\mu = 4.100$ and 4.091, respectively), operational employees the second highest ($\mu = 3.062$ and 3.250, respectively), and senior leadership the lowest scores ($\mu = 2.143$ and 2.857, respectively). The ISO areas with least disagreement, i.e. those with no significant relationships between the group's responses, were Planning_Collaboration, Support, and Performance Evaluation, also seen in Figure 8. A possible implication of this is that the organization has a shared perception of these areas, in turn indicating high alignment.

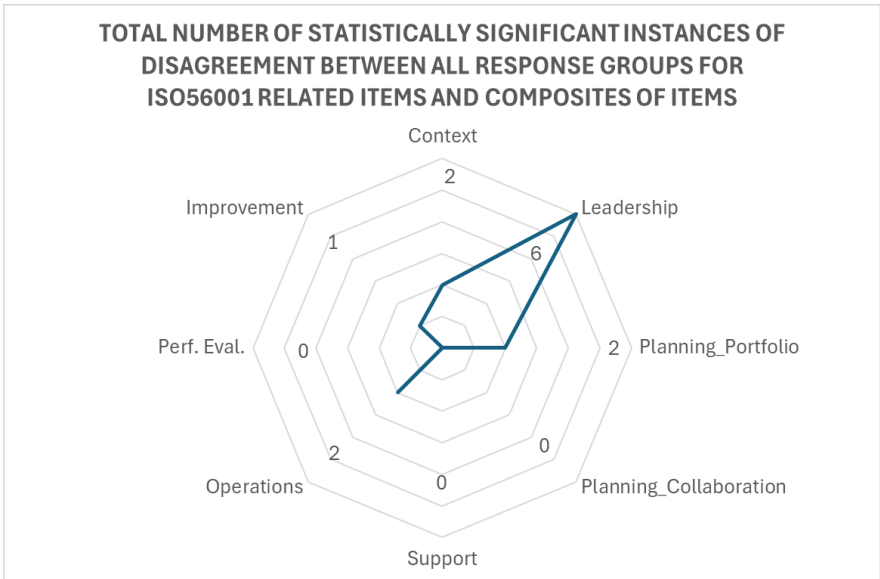


Figure 8: Total number of statistically significant instances of disagreement between all response groups for ISO 56001 related items and composites of items

These tests were also subjected to a Bonferroni Alpha threshold to control for the potential false positives (Type I error) that may arise when conducting multiple statistical tests simultaneously. With this analysis, only one result remained significant: The rating of "Risk-taking for innovation" by middle-management relative to senior leadership ($t = 5.59$, $p = 1.0 \times 10^{-4}$). For this item, middle management attained a mean of 4.100 (SD = 0.738), while senior leadership's mean was 2.143 (SD = 0.690). The underlying items of this result surveyed senior leadership "How often does senior leadership work with strategies aimed at improving employee attitudes to risk taking for innovation?", whereas middle managers were asked whether "in projects, it is okay to take risks when the goal is to be innovative". The discrepancy in scores thereby suggests that middle managers perceive that risk taking for innovation is accepted, despite senior leadership not widely encouraging it. In turn, this may suggest that it is primarily middle managers who are acting as drivers of an innovative culture, showing signs of transformational leadership qualities (Bass, 1985) that have the potential to improve innovation outcomes (Prajogo & Ahmed, 2006). However, because the subgroup sizes are modest ($n \approx 7–18$) and multiple testing inflates the chance of false-positive findings, the patterns revealed by these t-tests should be regarded as exploratory. More diverse samples are required before drawing firm conclusions about broader perception gaps within the hierarchies at Saab Surveillance. Disagreements shown through these t-tests may also stem from differences in item wording and Likert anchor options, acknowledged as a potential limitation. In a continued analysis, further t-tests were performed, although differences between business units were compared instead.

5.2.8. Welch's t-tests to Compare the Fighter Core Capabilities Unit (Dept 2) with All Other Business Units (Dept Rest) on ISO 56000 Dimensions

The results of the Welch's t-test comparing the responses of business unit 2 with other business units only resulted in one significant result – the perception of the budget allocated for innovation across the three respondent groups. Bonferroni Alpha significance was not reached, thus partially undermining the result. Nevertheless, Figure 9 presents a visualization of the exploratory finding.

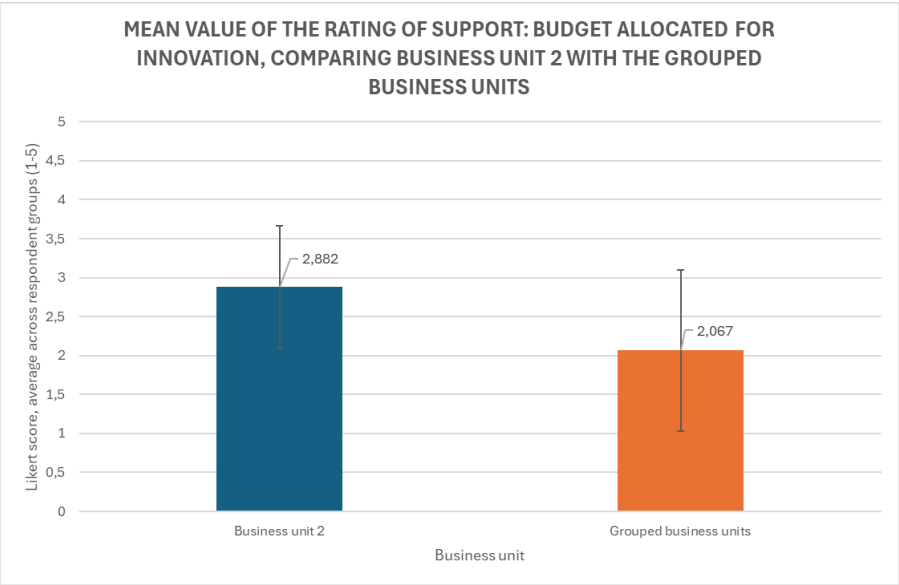


Figure 9: Diagram showing the mean value of how business unit 2 vs. other business units rated Support: budget allocated for innovation

The result indicates that the staff of business unit 2 perceive their resource allocation for innovation more positively with a mean of 2.882 (SD = 0.781), as opposed to the other business units who scored a mean of 2.067 (SD = 1.033). A potential interpretation is that business unit 2 either receives comparatively higher funding for innovation or benefits from more transparent budgeting processes, leading its staff to perceive the allocation more favorably than their counterparts in other units. However, given the modest sample and the relatively large standard deviations, the estimate can be considered imprecise and the statistical power limited, so the observed difference should be interpreted cautiously. The full output is featured in appendix B, section 11.2.5.

5.2.9. Welch's t-tests to Compare the Fighter Core Capabilities Unit (Dept 2) with All Other Business Units (Dept Rest) on Perceived Innovation Performance

The t-test between business unit 2 and the grouped business units on innovation performance dimensions also resulted in significance. When comparing Dept 2 against all other departments on perceived innovation performance outcomes, two differences emerged in the full sample: customer-centricity showed a Bonferroni-confirmed difference ($p = 0.002$), and radical innovation reached raw, non-Bonferroni significance ($p = 0.019$). Incremental innovation and innovation speed showed no significant differences across either sample. These results provide an indication that business unit 2 has greater performance in both radical innovation, and especially customer centricity.

However, the caveat is that these findings may be inflated due to the OE attribution procedure, described previously in section 4.2.3. This may be due to the reduced variation in responses that the attribution procedure creates, but also due to the sample sizes used in the test. These were considerably more uneven for the performance comparison ($n = 21–22$ vs. 14) than for the ISO 56000-area vs. department t-test (approximately $n = 19$ vs. 17), a direct consequence of the attribution constraint: OE attribution is only applied to departments where employees have a matching MM or SL performance score (Dept 2 and Dept 7). A number of OEs from other departments are therefore excluded from the performance analysis entirely, thus shrinking the "Rest" group compared to what it actually contributed to the ISO 56000 area tests.

This concern is further validated by the MM+SL-only t-test, seen in Table 14 below. When MM and SL scores were not attributed to OEs, none of the four outcomes reached significance – customer-centricity increased to $p = 0.145$ and radical innovation to $p = 0.211$. The apparent between-department differences in the OE included sample are therefore likely due to the attribution procedure rather than a genuine finding, and should be interpreted as such. To be noted is that this caveat only applies to the innovation performance dimensions, since the ISO dimension comparison between departments was not subject to the attribution method. The full output is featured in appendix B, section 11.2.5.

Table 14: Condensed result of the department dependent Welch's t-test, comparing Department 2 against other departments on innovation performance dimensions, including both OE+MM+SL results and MM+SL only results.

Test Type	*Group 1*	*Group 2*	*Perfor-mance dimension*	*n group 1*	*n group 2*	*p value*	*Bonf-erroni Alpha*	*Signi-ficance*
MM + SL and OE attributed	Dept2	Rest	Radical	21	14	0.0191	0.0125	* Raw only (p<0.05)
			Incremental	22	14	0.1756	0.0125	ns
			Customer	21	14	0.0024	0.0125	* Bonferroni (p<0.0125)
			Speed	22	14	0.0972	0.0125	ns
MM+SL only	Dept2	Rest	Radical	10	7	0.2109	0.0125	ns
			Incremental	11	7	0.3435	0.0125	ns
			Customer	10	7	0.1445	0.0125	ns
			Speed	11	7	0.5806	0.0125	ns

5.3. Qualitative Data: Workshop Results And Analysis

The qualitative and quantitative findings were presented in the workshop for validation and recommendations. A discussion among participants ensued about the validity of the findings, specific examples relating to the findings, and potential improvement actions. The following descriptive summary lists the core areas discussed.

Connection between leadership and support for innovation

The conversation revealed a recurring disconnect: many managers claim to have an innovative mindset and to encourage new ideas, yet they do not translate this belief into practice by allocating the appropriate resources. One participant noted that managers often do not know which concrete steps to take when an employee brings forward an idea; another countered that it is the manager's responsibility to learn the process and then guide the employee accordingly. Participants suggested that this gap could be narrowed through structured interventions such as education programs for managers, onboarding modules that spell out the expectations for supporting innovation, and the appointment of "innovation ambassadors" who actively help turn ideas into projects, in order to offload the responsibility from managers. The consensus was that a more systematic approach to supporting innovation is required. This topic naturally led the discussion toward human capital. Participants stressed that staff with an innovative mindset must be identified, valued, and enabled. One concrete proposal was to organize these individuals into cross-functional teams, thereby harnessing their potential more effectively.

Internal and External Collaboration and Confidentiality

The "not-invented-here" syndrome was discussed in the workshop. Employees are eager to devise their own solutions, but often feel too proud to adopt ideas generated elsewhere. This attitude hampers collaboration on both internal and external fronts. Internally, teams sometimes withhold information from other business units or neighboring projects, and many projects are labelled "high confidentiality" even when only moderate protection is needed, often because owners wish to guard their work from internal competitors. Externally, the perception that external solutions can also be developed in-house discourages genuine

partnership. When the discussion shifted to strategic goals, several participants emphasized that external collaboration is essential for improving radical innovation efforts, while incremental improvements can be developed in-house. Lastly, a tension emerged: the organization seeks quick, breakthrough ideas, yet current exploratory collaborations (e.g., university partnerships), which represent the majority of radical innovation efforts, operate on 10- to 15-year horizons, contradicting the speed-to-market strategy.

Balancing Exploitation and Exploration

Many participants observed that innovation is actively avoided in certain development projects. To meet budget and schedule constraints, teams cling to a predefined requirements list, viewing any change as a risk of "scope creep" that must be prevented. Consequently, suggestions for improvements are often set aside. This is seen as essential to meet customer requirements and fulfill orders. Nonetheless, participants also recognized the value of continual improvement. One example described a delivery project in which one out of eight teams was granted explicit freedom to experiment and take risks, providing a dedicated outlet for innovative ideas. Others proposed allocating separate time for improvement concepts or recording all ideas for later development when resources become available.

Measuring Innovation

All participants agreed that measuring innovation is difficult, which helps explain the low ISO 56001-compliance score observed. Traditional metrics such as ROI of innovations capture outcomes, but overlook the learning that often precedes radical innovation. Participants thereby argued that innovation should also be assessed through its process, e.g. by tracking insights gained from experiments or to what extent processes are followed. One participant suggested a "Return on Learnings (ROL)" metric to capture this process-oriented value. Furthermore, the idea of celebrating failures, for example through a "hall of shame," was raised as a way to acknowledge that even unsuccessful attempts generate useful knowledge. Finally, the group stressed the need to establish a clear strategic intent for measuring innovation, so that the organization knows what to track and why.

Ways of Working

It was noted that most exploitative projects already rely on agile methods, which supports development speed and efficiency. The participants noted that this framework can be repurposed for innovation work if the focus shifts from pure efficiency to learning and experimentation. However, the discussion also suggested that projects aimed at radical outcomes need to be separated from the core, target-driven organization to avoid the risk-minimizing focus of typical projects. This separation calls for distinct management approaches, dedicated resources, and protected space for exploration.

Strategic Scope

The final discussion centered on the product roadmap, the document intended to set the company's product development strategy. Participants acknowledged that Saab Surveillance has become better at defining its own strategic direction, independent of the Swedish Defense Materiel Administration's (FMV) requirements. Yet the current roadmap remains heavily weighted toward incremental innovation. The consensus was that the roadmap should be

refreshed to also encompass more radical and transformative solutions, thereby aligning its strategic scope with the organization's ambition for breakthrough innovation.

6. Synthesis

The qualitative and quantitative data collection yielded several key findings, which are summarized below. This section will compare and combine the most meaningful findings to synthesize a diagnostic of Saab Surveillance's innovation management system, and serve as a basis for the discussion.

Qualitative data – summarized interview findings

1. **Innovation portal potential appears under-explored**
   Surveillance has many skilled and creative employees that generate many ideas, yet the innovation portal seldom moves concepts beyond the idea stage; no idea has yet reached commercialization and follow-up or coaching seems minimal.

2. **Innovation labs are promising but operate in silos**
   The software and hardware labs foster rapid-prototyping and a "fail-fast" mindset, but are rarely able to collaborate with core projects or ideators from the portal, which may limit system-wide synergies.

3. **Resource allocation for innovation is sometimes ad-hoc and opaque**
   Formal mechanisms exist for earmarking resources for innovative ideas and development projects, and Saab allocates substantial funds toward R&D. However, in practice the allocation often hinges on the approval of individual managers, and some funds never fulfill their intended purpose. This suggests a gap between intention and execution that can lead to a fragmented process, perceived inequities, and limited traceability.

4. **There exists a great tension between exploitative and explorative work**
   The organization is seeking to become more explorative, but the operational reality where shorter-term project delivery targets naturally create a resource-allocation conflict with longer-term technology exploration.

5. **Internal and external collaboration is limited**
   Partnerships with universities, start-ups and organizations like Saab Ventures are a strength of the organization, yet internal communication gaps, "not-invented-here" attitudes, confidentiality restrictions, and siloed business-unit knowledge appear to hamper both internal and external collaboration.

6. **The culture has risk-averse inclinations**
   A procurement-driven, customer-order mindset has caused a short-term delivery focus. Furthermore, the tolerance for risk and experimentation appears low, especially among senior leadership, which may limit radical innovation.

7. **A shared innovation vocabulary is missing**
   Managers and staff interpret key concepts differently, which could impede coordinated actions and training.

8. **Incremental innovation is consistently achieved**
   Multiple interviewees highlight that incremental innovation outcomes, sprung from customer projects and continuous development of the existing product portfolio, are successful.

Quantitative data – summarized survey findings

1. **Saab Surveillance has moderate ISO 56001 maturity**
   The highest average scores are for Leadership (≈ 62 %) and Portfolio Management (≈ 61 %); the lowest is for Performance Evaluation (≈ 29 %). Support (≈ 39 %) and Improvement (≈ 44 %) fall below 50 %. Overall score suggests limited system conformity.

2. **Support for innovation shows a strong association with innovation performance outcomes**
   The "Support" dimension (time, budget, tools) is strongly associated with radical-innovation performance (Spearman $\rho \approx 0.63$, $p < 0.001$). Although a less robust finding, support is also associated with customer-centricity ($\rho \approx 0.38$, $p \approx 0.04$)., i.e. higher alignment with early-customer-involvement practices.

3. **Improvement mechanisms show a positive link to radical-innovation performance**
   Although measured with a single item, the "Improvement" construct showed a positive association with radical-innovation performance ($\rho \approx 0.48$, $p \approx 0.01$), hinting that lessons-learned loops add value.

4. **Business-unit 2 demonstrates a possible best-practice pattern**
   Staff in this unit perceived higher support for innovation and score better on customer-centricity and radical-innovation performance, indicating a potentially replicable model for resource allocation and working methods

5. **Leadership levels diverge on ISO 56001 compliance**
   Opinions about ISO 56001 compliance differ across hierarchical groups according to mean scores, and statistically significant relationships. In particular, middle managers and senior leaders disagree to a far greater extent than other levels, indicating a managerial misalignment that could undermine strategic decisions and reflect a cultural weakness in the organization.

6.1. Convergent Findings

The two methods of data collection broadly show concurrence. Below follows a comparison and analysis of the convergent findings, with the goal of synthesizing the data.

Overall ISO 56001 IMS adherence

Interviewees observed that Saab currently do not fully align with ISO 56001 principles across the system, lacking in areas such as Planning (especially collaboration), Support and Performance Evaluation. This was measured by the quantitative data, which provided a preliminary diagnosis of the current state of adherence to the standards' requirements. With multiple areas attaining 50% or less approval by surveyed employees, the organization shows potential for improvement. The survey analysis also reveals that this lack of comprehensiveness may negatively impact radical innovation performance – overall IMS fulfillment (ISO 56001 adherence) showed a significant association to radical innovation performance, consistent with Gimenez et al., 2025. Consequently, the organization's current innovation management practices can be improved to foster radical innovation performance and to operate a more holistic IMS.

Support for innovation

The qualitative observations showed that the resource allocation toward innovation is at times unorganized and under-utilized. The overall sentiment between interviewees was that dedicated resource allocation measures for innovation exist in multiple units and groups of the organization, but that resources are not always utilized as intended, resources are primarily directed toward incremental innovation, and the organization lacks awareness of what resources are available. Short-term delivery targets are straining the more explorative efforts, forcing resources to be allocated toward customer commitments, rather than innovation, in order to deliver on customer commitments. The quantitative findings converge with these observations: The ISO dimension Support showed poor alignment, with only a 39 % overall score. This sentiment was shared across the three levels surveyed, since no statistically significant differences between the ratings of Support from the respondent groups were found. The quantitative data also revealed that radical innovation performance was significantly associated with support for innovation, a notion that is supported by prevailing research (Klingebiel & Rammer, 2014). Therefore, the current resource allocation practices revealed by the data may be negatively impacting the organizations' radical innovation performance, and are indicative of a systematic challenge in the innovation management system.

Measuring innovation

Performance evaluation and continuous improvement mechanisms were areas with lacking development, as indicated by both the interviews and survey data. Interviewees indicated that measuring innovation and following up with KPIs is not done today, since accurately representing innovation success through measurements is complicated. They also indicated that innovative ideas that were given time and resources were not followed up – i.e. whether the ideas successfully moved through an innovation process toward commercialization. The survey data affirms this, with performance evaluation and improvement scoring 29% and 44%, respectively. The data specifically points to the efficacy of strengthening improvement, since it showed a statistically significant relationship with radical-innovation performance. However, both of these components are crucial for realizing innovation outcomes, since measuring

progress and improving processes can inform development and drive accountability (Taques, et al., 2021).

Collaboration in innovation

Collaboration was another area with possibilities for development, reported in both methods of data collection. Saab Surveillance actively seek collaborative innovation through partnerships with start-ups, mediated by Saab ventures, and with institutions and organizations. However, interviewees also noted that confidentiality measures degrade in- and outbound innovation efforts, that a "not-invented-here" attitude exists among staff which hampers both internal and external collaboration, and that business units and projects are siloed. In turn, this lowers the potential synergies that can be found when allowing people and organizations to communicate more openly. The quantitative data echoes these results. The planning_collaboration dimension attained a score of 45%, and the context dimension was rated at 53%. The latter, which captures how innovation activities are informed by internal and external signals, consists of two items: monitoring of internal factors (47 % agreement) and monitoring of external trends (58 % agreement). The lower rating for internal factors mirrors the reported silo issues between projects and business units. These data points did, however, not reach statistical significance ($p > .05$) in correlation tests with innovation performance dimensions, thus not providing any quantitative evidence of a direct effect on perceived performance.

End-to-end innovation operations

Survey results corroborate interview findings regarding the under-utilization of the innovation portal. Both data sources indicate that while the portal supports early-stage ideation and initial concept creation, it falters in guiding ideas through refined concept creation and subsequent commercialization. This trend aligns with the ISO 56000 operations dimension scores: identifying opportunities (53%), validating and testing concepts (55%), and commercialization (46%), which showed a less positive score for the commercialization of innovation. These findings point to a lack of an end-to-end innovation process, especially in the innovation portal, suggesting a deficiency in system comprehensiveness. However, to be recognized is that these results may reflect the respondents' view of the typical ideation to commercialization process in development, separate from the innovation portal.

Disagreements between respondent groups

The quantitative data showed indications of disagreement between respondent groups regarding perceived adherence to ISO 56001 requirements. As was shown in Figure 7, 13 pair-wise comparisons between the groups for survey items and composites showed statistical significance. A majority of which were between middle managers and senior leadership. Although the relationships should be interpreted cautiously due to their largely non-Bonferroni significance, these disagreements suggest a misalignment in the organization. The mean scores also reflect this, with middle managers consistently scoring items higher than operational employees and senior leadership, in that order. The qualitative data corroborates these observed associations to some extent. For example, in the case of the innovation portal, the senior leaders and middle managers evaluating ideas regarded the process and allocated resources as adequate, whereas idea submitters did not share this opinion. Taken together, this suggests a systemic misalignment across hierarchical levels which may negatively impact innovation performance.

Innovation performance dimensions

Lastly, the interviews and survey responses demonstrated congruence regarding innovation performance. Both data streams indicated that incremental innovation performance was satisfactory. Interviewees noted that existing products and systems undergo continuous improvement through well-established processes, often facilitated by customer engagement during development. This aligns with the survey data, where incremental performance received the highest score among all dimensions (3.67, SD = 0.84), and customer centricity was the second highest (3.18, SD = 1.01). Respondents described the latter as a core organizational capability, noting that much development occurs collaboratively within customer delivery projects. A similar alignment appeared in the assessment of radical innovation. Both interviewees and survey respondents (2.82, SD = 0.73) indicated that commercializing truly novel solutions remains less common. Conversely, speed to market scored lowest across the four performance categories (2.28, SD = 0.83). Interviewees highlighted this as a weakness, attributing it partially to the typical development process of the Swedish military materiel administration (FMV). This creates a notable tension, as the typical procurement process contradicts the core innovation strategy of speed to market.

6.2. Areas Of Divergence

However, multiple findings also indicated silence, were observed where a theme identified in one method was notably absent from the other, or non-convergence. These are analyzed in the current section.

Culture vs. support for innovation

The risk averse culture reported in the interviews is not supported by the quantitative findings. The interviews highlighted that experimentation is sometimes seen as a waste of time, that complete certainty is opted for in development, and that leadership, especially middle managers, frequently disregard allocating resources toward innovation. These results resonate with a culture geared for exploitation, and not exploration, highlighting a cultural gap in the system. It was also observed that senior leaders more aptly vouch for more explorative work. However, the broader survey data painted a different picture. The ISO 56000 dimension *leadership* surveyed respondents on items tangential to culture, and controversially, it showed the highest agreement among the ISO dimensions with a score of 62%. Among the items in leadership, the culture item stands out, with a score of 71%. Respondents indicated that they encourage their employees, or are themselves encouraged by their managers to be innovative. However, the responses of middle managers stand out: they consistently scored higher than other groups, and thereby increased the leadership dimension's overall score. For example, the rating of "Risk-taking for innovation" captures that SL seldom encourage risk taking for innovation, yet MM still perceive that risk taking for innovation is accepted. In turn, this suggests that MM display a higher degree of culture-supporting transformational leadership than SL.

This finding contrasts with the interview results, where SL stood out as drivers of an innovative culture. Middle managers seem to be displaying transformational leadership qualities by encouraging an innovative culture, but they do not allocate sufficient resources toward innovation. While further inquiry is needed to substantiate these findings, synthesized results suggest three primary causes: senior leadership may be withholding resources, managers may

lack the training or knowledge to support employees' innovative ideas, or a lack of follow-up mechanisms may cause them to de-prioritize innovation despite verbally endorsing it.

Business unit divergences

The quantitative data revealed that staff from business unit 2 had a more positive perception of support for innovation than other business units surveyed, according to a statistically significant test. Although the reported means show a generally low score for the area (Figure 9), BU2 scored almost one whole point greater than other departments. Furthermore, BU2 vs. other business units showed a statistically significant greater radical innovation ($p<0.05$) and customer-centricity performance ($p<0.0125$). The qualitative data did not contest these results, but not corroborate it either. Interviewees discussed varying degrees of support for innovation in different business units of Saab Surveillance, but not with specific mention of BU2. Further analysis of this indication must therefore be made – it is suggested that Saab Surveillance investigate the differences between the business units' ways of working.

7. Discussion

The present report set out to determine how Saab Surveillance currently conducts innovation, and how it compares to the recent ISO 56000 innovation management standards. A systems approach was used to frame the analysis, and a mixed-methods research design was employed. Data were gathered from internal documents, semi-structured interviews, and a questionnaire built on ISO 56001 requirements. This chapter serves to address the research questions, provide recommendations, highlight theoretical and practical contributions, and to discuss the limitations of the current study. Recommended actions appear as sub-headings with expanded details provided in a dedicated highlighted section.

7.1. Main Findings and Answers to Research Questions

This section will outline the main findings by answering the research questions. The first three questions specifically address the focal elements of the questions: RQ1 will give a summarized overview of the ISO 56001 adherence within each ISO dimension; RQ2 delves into how the dimensions of the systems approach, i.e., Comprehensiveness, Coherence and Correspondence, are seen in the organization; RQ3 addresses Leadership, Openness and Strategy – the three dimensions of systemic management argued for. Finally, the most comprehensive research question RQ4 will serve as a basis of recommendation for Saab Surveillance. This discussion structure is visualized in Figure 10.

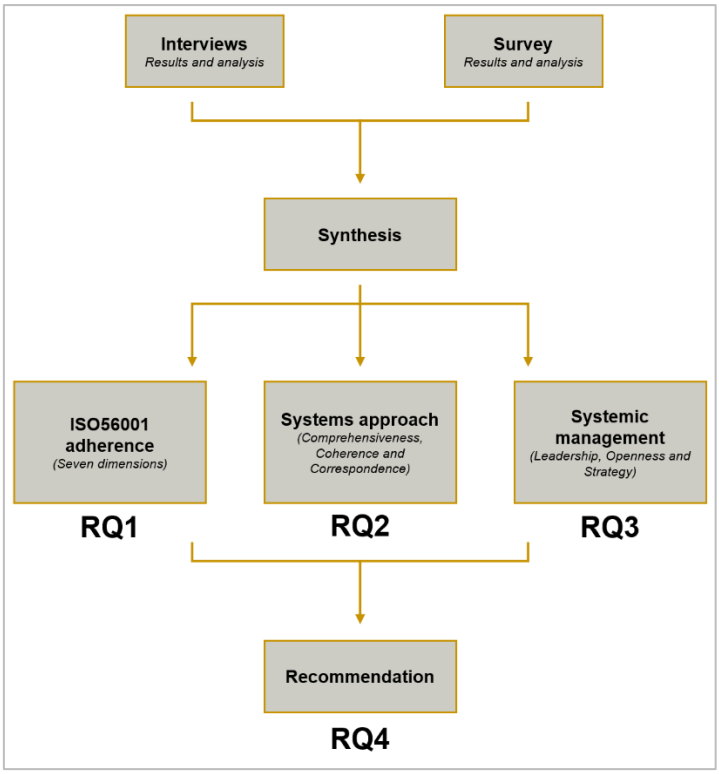


Figure 10: The data collection, analysis and discussion structure

7.1.1. Research Question 1: ISO 56001 Adherence

The first research question focused specifically on the requirements of an innovation management system according to ISO 56001.

RQ1: To what extent are ISO 56001 requirements fulfilled by Saab Surveillance's innovation management practices?

The analysis and synthesis reveal that ISO 56001 requirements are moderately fulfilled by Saab Surveillance. Leadership and portfolio planning show relatively strong alignment, but the organization falls short in the core operational, support, performance-evaluation and continual-improvement dimensions. The moderate analysis of environmental context, tensions in resource allocation, scattered innovation processes, and the absence of systematic measurement together indicate that the current innovation-management practices do not fully satisfy the ISO 56001 requirements. Although the current findings provide an indication of ISO 56001 requirement fulfillment, a more thorough analysis is recommended to more accurately assess adherence, and to identify specific, actionable improvement actions. The alignment to each ISO 56000 dimension is described below:

Context of the organization ≈53 %

The monitoring of contextual factors that may impact or shape the development of innovative solutions was rated with a moderate score. The organization shows a marginally greater alignment with analyzing the external context, such as political, market and technological factors, than the internal context, i.e. people, organizational culture and structure. However, both areas show potential for improvement.

Leadership ≈ 62 %

This area earns the highest score in the assessment, reflecting managers' work toward raising the tolerance for failure, risk taking, and promoting an innovative culture. However, the synthesized data reveals a gap between stated commitment to innovation and everyday practice, indicating that culture and leadership actions also must be strengthened.

Planning ≈ 61 and 45 %

The planning dimension was split into two fields due to its breadth in the standards' requirements. Analyzed separately, portfolio management (*≈ 61 %*) and collaboration (≈ 45 %) diverged in fulfillment, indicating that the organization has adequate methods for managing a portfolio of projects with different goals, but lacks methods and instructions for involving internal and external actors for collaboration. This indicates a readiness for pursuing both exploitation and exploration, but that collaboration must be given more support from the organization.

Support ≈ 39 %

The overall sentiment between employees was that dedicated resource allocation measures for innovation exist in multiple units and groups of the organization, but that resources are not always utilized as intended, resources are primarily directed toward core operations or incremental innovation, and the organization sometimes lacks awareness of what resources are available. Therefore, the support dimension is one with the lowest fulfillment in relation to ISO 56001.

Operations ≈ 52 %

The operative factors of innovation management attained a moderate score, spread across ideation, concept development and commercialization, the latter showing the lowest adherence. Low TRL solutions are actively researched, Saab Surveillance's innovation portal collects ideas, and software and hardware labs prototype concepts. However, these sub-systems operate in silos, and a formal process carrying solutions from ideation to commercialization does not exist, leading to lack of follow up when innovative solutions are developed. The assessment therefore reveals a disjointed innovation operations process.

Performance Evaluation ≈ 29 %

The performance-evaluation dimension received the lowest score, and interviews indicate that Saab Surveillance does not presently apply systematic monitoring, measurement or follow-up of its innovation activities. The synthesized data show that KPIs related to innovation are largely absent, and that results from innovation initiatives are not routinely captured or reviewed. The validation workshop also indicated that previous measurement efforts have focused on output indicators like ROI, thus overlooking valuable targets such as *Return on Learning* and knowledge development.

Improvement ≈ 44 %

The improvement dimension also shows low compliance with ISO 56001. As with the performance evaluation dimension, the evidence indicates that Saab Surveillance does not broadly capture systematic lessons from innovation initiatives and use them to improve innovation processes.

7.1.2. Research Question 2: Systems Approach

The second research question sought to answer whether Saab Surveillance manage their innovation efforts with a systems perspective:

RQ2: To what extent is innovation at Saab Surveillance managed with a systems approach according to the dimensions Comprehensiveness, Coherence and Correspondence?

The analysis below evaluates these three dimensions in detail. The results indicate that the systems design dimensions are moderately fulfilled, although there is room for improvement. This potential for improvement could potentially be realized by adopting a systems perspective in the organization, and by addressing ISO 56001 requirements.

Comprehensiveness

The analysis reveals that the IMS at Saab Surveillance falls short of the breadth required for a viable-systems approach. ISO 56001 compliance data show that several dimensions of the standard are only partially satisfied, indicating that the system does not span the full spectrum of innovation activities. In practice, the current workflow lacks an end-to-end pipeline that systematically channels ideas from initial ideation through concept development, selection, validation and finally commercialization. Procedures do exist for idea selection (the innovation portal) and for fast-prototype testing (the software and hardware labs), but these operate in isolation and are not linked into an end-to-end sequence. Consequently, internal interaction pathways, which are one of the hallmarks of a comprehensive sub-system, are weak or missing. The synthesized data also reveal a limitation in both sub-system and supra-system integration. Internal actors across business units are not readily involved when developing innovative solutions, and external partners are not effectively incorporated into the innovation process. This sometimes occurs because of confidentiality safeguards, but also due to a "not-invented-here" mindset exhibited by staff. This limits the exchange of resources, information and knowledge within and across organizational boundaries (Golinelli, 2005), counteracting the requisite variety needed to respond to environmental change (Ashby, 1956). In short, without a unified pipeline and more open internal-external interfaces, Saab Surveillance's IMS lacks the systemic breadth and connectivity essential for comprehensive, systems-based innovation management.

Coherence

The interview and survey data reveal that coherence at Saab Surveillance could be improved. First, communication is fragmented: even when budgets or time are formally allocated for innovation, teams often remain unaware of these resources, which reduces the interaction pathways necessary for coordinated responses to strategic shifts. Second, resource allocation is at times ad-hoc and dependent on manager sentiments; some managers divert spare funds form exploitation projects toward innovative ideas, others withhold innovation budgets to use them for upholding operational efficiency. This connects to the third area of lacking coherence: the inconsistent interpretation of strategy seen across hierarchical levels. Senior leadership, middle management and operational staff differ markedly in their perception of risk-taking and tolerance for failure, indicating that strategic signals have not spread uniformly through

the organization. Finally, the absence of a systematic performance-evaluation loop means that efforts toward fulfilling the innovation strategy cannot be monitored, hampering the feedback mechanism that would alert the system to misalignments and enable corrective action. These examples highlight how the strategic shifts toward more explorative work have not permeated all units and staff of the organization, leading to reduced strategic coherence.

Correspondence

The analysis shows that the strategic priorities set for innovation at Saab Surveillance are not consistently reflected in the performance objectives, resource allocations, and everyday work practices. Leadership advocates for faster, more radical innovation, but the inherent tension between meeting customer commitments and long-term focused innovation work means daily activities remain dominated by incremental, delivery-focused initiatives. Although resources are earmarked for explorative work, they are at times not applied as intended. Managers may divert funds to support the existing product line instead of the strategic goal of finding new, radical solutions. Statistical analysis reveals a significant positive association between support for innovation and radical-innovation performance, indicating that aligning the explorative strategy with resource allocation could yield measurable improvements. Therefore, strategic correspondence is crucial for achieving the explorative strategy.

A mismatch is also evident in the innovation portal, where ideas are collected and given limited resources for development, but rarely move toward commercialization, contradicting any strategic emphasis on bringing new solutions to market quickly. The workshop for validation and recommendations suggested that this issue can be mitigated through providing ideators with guided innovation calls that align with the current strategic goals. However these calls are rare. Participants do therefore not know where to target their ideas, making their proposals less relevant to the organization's strategy, in turn lowering correspondence. Lastly, the lack of a systematic performance-evaluation loop further limits system correspondence, similar to coherence. Because reliable metrics to track progress against strategically weighted objectives such as speed and disruptiveness are missing, the system cannot confirm that processes like rapid-prototyping labs actually serve the intended performance goals.

In sum, the strategic correspondence of the broader organization shows improvement potential, as strategic goals are not always matched by the concrete actions required for a truly systems-based approach.

7.1.3. Research Question 3: Systemic Management

The third research question related to how the three core components of systemic management, i.e. Leadership, Openness and Strategy, are handled:

RQ3: To what extent are the three systemic management dimensions Leadership, Openness and Strategy fulfilled at Saab Surveillance?

Strategy

The findings suggest that Saab Surveillance possesses a well-defined innovation strategy, and general strategy for pursuing both exploitation and exploration. In the innovation strategy, the organization defines key goals of innovation, such as speed to market, discovery through fast failure, and collaboration both internally and with external actors. The innovation strategy establishes new goals for such collaboration, such as having cross-functional teams and involving customers in development. The methods for achieving these strategies are also outlined, such as Saab AB wide performance evaluation, innovation processes, and forums for development. In the overall strategy, a balanced focus on delivering on customer commitments, improving working processes, and accelerating future capabilities through innovation can be found. Therefore, the Strategy dimension of systemic management is adequately fulfilled.

Leadership

As for leadership, it can be concluded that Saab Surveillance display an unsatisfactory degree of fulfillment. Disagreements between MM and SL highlight a misalignment in leadership which in turn divides the system and sends mixed signals to employees. Furthermore, results indicate that leaders, especially middle managers, are not able to operationalize the innovation strategy due to tensions between operations and innovation efforts, leading to challenges in dimensions like support for innovation and collaboration mechanisms. The strategy shall set the direction of the organization, but it is the leaders who must operationalize the strategies – e.g. through transformational leadership that inspires an innovative culture, granting employees' time to drive innovative ideas, and following up the performance of innovation initiatives. These actions serve to unify the system, and contributes to balancing exploitation and exploration. Many leaders do show commitment to these actions and seek to put them into effect, but this is not true for the whole organization, with many others focusing solely on exploitation. Results show that many of these leaders are heavily strained by the core operations that currently consume most resources, leading them to down-prioritize future oriented innovation projects that are out of their projects' scope. Others lack the tools and competencies needed to support innovative staff, as indicated by the workshop. This suggests a need for innovation ambassadors that may offload leaders in enabling innovation.

Although the actions of these leaders are defensible, they reduce the correspondence and interconnectedness of the IMS. In turn, more resources are awarded to the core operations instead of following the exploratory strategic initiatives. Therefore, Saab Surveillance possess the strategies for pursuing innovation, but the implementation by leaders is struggling, causing a lack of interconnectedness and unity in the system.

Openness

The Openness dimension currently spans from unsatisfactory to moderately satisfactory. As with Leadership, Openness is directly tied to the organization's strategic intent. The strategy explicitly calls for stronger collaboration, yet the practical mechanisms needed to embed that intent in day-to-day work are still missing. This shortfall appears both within the company and in its external partnerships.

Inside the organization, the absence of reliable collaboration channels blocks the combination and refinement of ideas. The essential role of open exchange across internal groups was highlighted by an employee, who noted that "Ideas are born where thoughts meet". However, projects and business units often operate in isolation, and there is no clear structure for sharing knowledge and lessons learned. Confidentiality requirements reinforce these silos, trapping valuable insight. Over time these controls have cultivated a closed culture; teams frequently withhold information from other units or neighboring projects, and many initiatives are labeled "high confidentiality" even when only moderate protection is warranted, often because owners aim to shield their work from internal competitors.

Externally, collaboration fuels exploratory activities such as technology scouting, knowledge development and the formation of strategic partnerships with universities, research institutes and other actors. However, partners regularly receive only limited information about the ultimate purpose or intended use of their research, often due to security concerns. Without a clear link to how their findings will be applied, the insights they generate risk remaining unused rather than being fed into new projects for testing and validation. Moreover, employees are uncertain about how to involve external stakeholders in the innovation process.

A common thread linking internal and external collaboration is the prevalence of a "not invented here" mindset among employees, which undermines willingness to share and adopt ideas from other sources. The combined effect of the internal and external openness challenges restricts the organization's capacity to turn knowledge into actionable innovation. These dynamics highlight significant room for improvement in the Openness dimension.

7.1.4. Research Question 4: Recommendation

The fourth research question, connecting systems theory, the innovation management standards, and their possible impact on future performance, read as follows:

RQ4: How can insights from ISO 56000 and systems theory guide improvements to Saab Surveillance's innovation management system in order to ensure the fulfillment of innovation strategies?

This research question broadly encompasses the purpose of the entire report, and the empirical observations have provided an exploratory foundation for answering this question. The data indicate three core areas, which upon improvement would strengthen Saab Surveillance's innovation management and drive compliance toward multiple ISO 56001 requirements. To move toward a systems approach that systemically integrates system components, improves idea to commercialization capabilities, and combines exploitation and long-term exploration, it is advised that the following areas are investigated for improvement.

Measure innovation to track exploratory innovation efforts, prompting manager accountability and the fulfillment of strategies

The performance evaluation of innovation performance is a factor limiting the innovation management at Surveillance. The lack of performance evaluation has bred a lack of accountability in managers, whom are aware of the new strategies aimed at increasing speed-to-market of new solutions and achieving more radical development, but do not enable these in practice. This is illustrated by middle managers' high verbal support for innovation but faltering practical support, such as coaching or resource allocation toward innovative ideas. Skilled and innovative employees are experiencing that resources are not adequately allocated toward innovation, discouraging development and improvement and hurting the innovation potential of the organization. Furthermore, this has decreased the coherence of the system. Strategy changes in one part of the organization do not translate across projects and down the organizational tree, reducing the possibilities for systemic transformation. Saab Surveillance is not the only company facing this struggle, and research on other companies indicates that merely allocating time and resources to innovation is insufficient; a formal performance-measurement system is needed to trigger the realization of innovation outcomes (Nilsson & Ritzén, 2014).

To enhance the innovation management system and strategic correspondence, the organization should create a set of innovation targets, based on measurable innovation KPIs that align with both the innovation strategy and the overall corporate strategy. These targets should not only include output such as financial returns, but also the inputs to the innovation process and the activities that drive innovation, internally and externally.

This approach ensures that factors like insights gained from experiments, the execution of innovation processes, and the resource utilization in innovation can be followed as success metrics. In turn, this can promote explorative innovation with a long-term perspective. As suggested in the workshop, a possible metric that captures non-financial performance is “Return on Learning” (ROL), which aims to capture knowledge development. Regular reporting should be ensured, and managers must process the information in internal audits to guide decisions, enforcing process improvement feedback loops. For example, a KPI dashboard acting across projects and business units can be established, and end-of-project follow-ups can be hosted to capture learnings that can improve future working processes. However, the measures reported by staff should be designed with the operational realities in mind, in order to mitigate the administrative burden that measurement may impose on staff. For further guidance regarding performance measurement and specific metrics, the organization may review the ISO56007 standard.

Design a specific framework for "high-trust” internal and external partnerships to enable collaboration without risking data leaks.

Currently, internal and external collaboration pathways suffer as a result of overly stringent confidentiality mechanisms. Staff label tasks as “high confidentiality” to guard their work from internal competitors, even when only moderate protection is needed. “Not-invented-here” attitudes have fostered a siloed culture and hamper systemic interactions between projects and business units within Surveillance. Externally, although collaborations are driving knowledge development and exploration, partners are not provided with sufficient context, resulting in findings that are difficult to translate into practical solutions. This impedes further development of research findings, slowing down the development of exploratory solutions.

It is therefore recommended that the organization investigates the confidentiality restrictions to potentially design a framework for "high-trust” internal and external partnerships that can enable closer collaboration both internally and externally. Evidence suggests that high-trust partnerships are manageable even amidst heightened security concerns (Reuven & Shamir, 2025). Achieving this balance could create a clear pathway for more actively integrating external research into internal development without compromising security. Providing more detailed requirement specifications and fostering closer collaboration would enable external institutions to deliver more practically applicable research. Internal actors would benefit from increased cross-functional collaboration between projects and BU’s, where synergies can be found from combining employees’ expertise. This action could serve to create the interconnectedness necessary to adopt a systems perspective and realize its associated performance benefits. By designing a framework for high-trust internal and external partnerships,

Saab Surveillance can become a more explorative organization that more readily leverages the benefits of openness and collaboration.

Implement clearer cross-functional pipelines: Connecting the idea gathering portal results directly with the hardware/software labs to prevent ideas from becoming stuck in silos.

The organization exhibits multiple innovation processes, such as idea collection portals and innovation labs. However, these are not linked in an end-to-end pipeline. To increase the conversion of ideas into projects that may result in commercialization, it is recommended to connect the idea gathering portal with the hardware and software labs.

To achieve this, the organization must establish clear collaboration pathways between the portal and the labs. This may be done through appointing dedicated innovation ambassadors or coaches, since leaders currently lack the time to support employees' innovation ideas. These individuals can ensure that the right people and groups connect across the end-to-end innovation process. Additionally, the organization may introduce transparent communication channels, such as open innovation calls hosted by project teams, to make ideas visible company-wide rather than restricting them to the strategy council or innovation portal. In turn, this will allow the organization to better utilize its repertoire of skilled employees who are passionate for innovating and solving problems. It will also allow the innovation portal to quickly increase the conversion of ideas into projects that can be used to drive both exploitative and explorative outcomes. By making the effort to establish a more comprehensive and interconnected innovation process, the organization may turn scattered ideas into coordinated, market-relevant solutions.

By addressing these recommendations, Saab Surveillance can work toward a more integrated management system that meets ISO 56001 requirements. This will enable the company to evolve into a more exploratory organization that systematically seeks, develops and commercializes future technologies while still meeting today's delivery targets.

7.2. Theoretical Contributions

This research provides several key theoretical contributions. Primarily, the theoretical framework bridges traditional systems theory with the standardization of innovation management systems (IMS). By integrating these fields, the study complements the seven dimensions of the ISO 56000 standards with a more holistic systems perspective, focused on interconnectedness between system components. The report contributes to theory by arguing that developing an effective innovation management system not only hinges on its components, but also on the continuous efforts to connect them, for instance through adhering to the system design principles and systemic management elements. The systemic management elements themselves may also be considered a theoretical contribution, since it is argued that leadership, openness, and strategy operate as the most critical drivers of an effective IMS.

Methodologically, this study introduces a robust approach for broadly assessing ISO requirement adherence. By concretizing abstract ISO requirements into surveyable metrics to map adherence, while simultaneously gathering in-depth qualitative data from practitioners, this mixed-methods framework offers a replicable model for future IMS research.

Regarding empirical outcomes, the findings reveal significant associations between specific ISO components and innovation performance – a link not previously established in studies examining standardized innovation management systems. This provides contrasting evidence to Giménez-Espin et al. (2025) and Martinez-Costa et al. (2019), who found no statistically significant positive effects for individual items within standards. However, aligning with those authors, this research confirms a significant positive relationship between a comprehensive IMS and radical innovation performance. Because the prior studies focused on the UNE 166000 standard rather than ISO 56000 and utilized much larger sample sizes, the findings of this report remain exploratory and suggest avenues for future quantitative investigation.

Finally, this study contributes to organizational research regarding large manufacturing firms, particularly within the defense sector. It exemplifies the complex challenge of combining operational efficiency for exploitation with exploratory innovation while navigating the confidentiality and security measures that often inherently restrict collaborative exploration.

7.3. Practical Contributions

As for the practical contributions, multiple findings are relevant for the focal company, and managers of similar organizations. It highlights the areas that the new innovation management standard ISO 56001 requires for adherence, and provides an example of how its dimensions can be assessed across an organization, but also addressed for improvement. It also introduces a systems mindset based on the dimensions Comprehensiveness, Coherence and Correspondence that can be adopted by practitioners seeking to transform their management approaches with regard to innovation and general management systems. Finally, it elaborates on components critical to achieving systemic management, namely Leadership, Openness, and Strategy, to provide a more concrete focus for practitioners looking to apply a systems approach to their standardized IMS. This case study is heavily rooted in the studied company, and results therefore offer limited generalizability. However, the results still provide valuable insights for organizations looking to structure and standardize their innovation efforts using system-based approaches, since they highlight crucial factors that need to be considered.

8. Limitations

Despite the contributions and findings presented in this report, the analysis must be regarded as exploratory, as previously noted. Several limitations, discussed in greater detail in the methodology chapter, affect the validity and generalizability of the results. The theoretical framework regarding standardized innovation management rests on a relatively thin literature base, since published studies on the potential performance benefits of such systems are scarce. Therefore, it was necessary to draw on articles from less prominent journals and to include research on innovation management standards other than ISO 56000.

Empirical data collection was constrained by limited time, budget and scope, which reduced both the breadth and depth of the investigation. Ideally, multiple organizations would have been studied to allow broader generalization of the theoretical concepts. Furthermore, a larger pool of interviewees and survey respondents would have provided a more accurate assessment of ISO 56001 adherence and of the company's innovation management practices. In addition, certain methodological decisions may have diminished the robustness of the data. The choice to bypass the Gioia methodology, the tailored wording of items toward their intended response groups, and the innovation performance scores attributed to operational employees reduce the robustness and potentially the accuracy of the data.

Another potential limitation of the empirical data collection was the way innovation performance was measured. It was assessed through four Likert type items, each addressing a different performance dimension. While this approach captures respondents' perceived performance, it is inherently subjective and reflects only one perspective on a construct that can be evaluated in many ways. Alternative or complementary measurement strategies could have strengthened the analysis. For example, a financial analysis of research and development expenditure, innovation related investment efficiency, or returns on innovation for Saab Surveillance would provide objective indicators. Another possibility would be to study a set of past innovation projects from inception to outcome, interviewing project participants and comparing the realized results with the processes employed. This would give greater insight into different working methods and processes.

Taken together, the limited literature foundation, the single case design, the modest sample size and specific methodological choices constrain the robustness and external validity of the findings. Future research should improve the theoretical framework as the field of standardized innovation management grows, incorporate multiple organizations to enhance generalizability, increase the number and diversity of respondents, adopt more rigorous qualitative techniques such as the Gioia approach, and triangulate perceived performance with objective financial or project-based indicators. These steps would provide a stronger empirical basis for evaluating the impact of standardized innovation management systems.

9. Future Work

While managing the tension between exploitation and exploration is widely recognized as a challenge, exploration itself constitutes a distinct and complex layer of difficulty. As explored in this report, exploration encompasses different types of innovation, such as incremental and radical – each demanding vastly different management practices and procedures. This internal complexity suggests that standards like ISO 56000 may be insufficient, as it treats innovation management as a uniform system rather than addressing these distinct types as separate subsystems with unique needs. For instance, incremental innovation relies heavily on leveraging existing knowledge, whereas radical innovation often necessitates the "unlearning" of current dominant knowledge. A forthcoming field in innovation management is the "system of systems" perspective, where an innovation management system is considered to be comprised of multiple distinct but interrelated systems (Chen, et al., 2025). This view more readily allows the management framework to be tailored to the specific innovation types, and to their distinct goals. Future investigations may therefore employ this perspective to tailor recommendations more precisely to the specific innovation objectives pursued.

Another aspect that could be investigated in further analyses is the intersection between the system design principles and the systemic management elements. The present report argues that both aspects are relevant in an innovation management system, yet doesn't fully explore the potential relationships between the six principles and elements. Therefore, this could be further investigated to evaluate potential interdependencies and synergies that may exist between them. These principles could also be explicitly tested against ISO 56001 dimension requirements and perceived innovation performance to quantitatively determine their role in designing an innovation management system.

10. Works Cited

Agrawal, A., Catalini, C., Goldfarb, A. & Luo, H., 2018. Slack Time and Innovation. *Organization Science,* 29(6), pp. 1056-1073.

Ashby, W. R., 1956. *An Introduction to Cybernetics*. 1st ed. London: Chapman and Hall, Ltd..

Bass, B. M., 1985. *Leadership and performance beyond expectations*. New York: Free Press.

Baxter, K., Courage, C. & Caine, K., 2015. Chapter 10 - Surveys. In: *Interactive Technologies. Understanding your Users: A Practical Guide to User Research Methods*. 2nd ed. s.l.:Morgan Kaufmann, pp. 264-301.

Benner, M. J. & Tushman, M. L., 2003. Exploitation, Exploration, and Process Management: The Productivity Dilemma Revisited. *The Academy of Management Review,* 28(2), pp. 238-256.

Bititci, U. S. et al., 2011. Managerial processes: business process that sustain performance. *International Journal of Operations & Production Management,* 31(8), pp. 851-887.

Bracio, K. & Szarucki, M., 2020. Mixed Methods Utilisation in Innovation Management Research: A Systematic Literature Review and Meta-Summary. *Journal of Risk and Financial Management,* 13(252).

Bran, A.-C. & Bălăşoiu, I.-G., 2025. Strategic Autonomy and Defense Procurement: European Union's Response to Geopolitical Uncertainties. *Proceedings of the International Conference on Business Excellence,* 19(1756-1764).

Cambridge Dictionary, 2026. [Online]
Available at: https://dictionary.cambridge.org/dictionary/english/systemic
[Accessed 24 May 2026].

Cambridge Dictionary, 2026. *Meaning of systematic in English*. [Online]
Available at: https://dictionary.cambridge.org/dictionary/english/systematic
[Accessed 24 May 2026].

Casper, W. C. et al., 2020. Selecting Response Anchors With Equal Intervals for Summated Rating Scales. *Journal of Applied Psychology,* 105(4), pp. 390-409.

Caverley, J. D. & Kapstein, E. B., 2025. The Atlantic Alliance: DivergingInterests, Converging Policies. *Survival,* 67(6), pp. 97-128.

Chen, Q., Chen, J. & Magnusson, M., 2025. Exploring Systems Approaches to Innovation Management From Second-Order Science in the West and China: System of Systems and TiXi. *Systems Research and Behavioral Science,* Volume 42, pp. 1098-1114.

Chen, Q., Chen, J., Magnusson, M. & Xiao, B., 2025. System-of-Systems Thinking for Complexity and Innovation—A Study Based on the Trajectory of Systems Thinking. *Systems Research and Behavioral Science,* Volume 0, pp. 1-20.

Chesbrough, H. & Crowther, A. K., 2006. Beyond high tech: early adopters of open innovation in other industries. *R&D management,* 36(3), pp. 229-236.

Cycyota, C. S. & Harrison, D. A., 2006. What (Not) to Expect When Surveying Executives: A Meta-Analysis of Top Manager Response Rates and Techniques Over Time. *Organizational Research Methods,* 9(2), pp. 133-160.

De Smet, A. et al., 2025. *Returning to the office? Focus more on practices and less on the policy,* New York: McKinsey & Company.

Dehoff, K., Dowdy, J. & Niehaus, J., 2013. *Managing a downturn: How the US defense industry can learn from its past,* s.l.: McKinsey & Company.

Demircioglu, M. A., Audretsch, D. B. & Slaper, T. F., 2019. Sources of innovation and innovation type: firm-level evidence from the United States. *Industrial and Corporate Change,* 28(6), pp. 1365-1379.

Dodgson, M., Gann, D. & Salter, A., 2006. The role of technology in the shift towards open innovation: the case of Procter & Gamble. *R&D management,* 36(3), pp. 333-346.

Gimenez Espin, J. A., Martinez-Costa, M. & Jimenez Jimenez, D., 2025. Effects of the UNE 166.002 standards on the incremental and radical product innovation and organizational performance. *European Journal of Innovation Management,* 28(3), pp. 948-977.

Gioia, D. A., Corley, K. G. & Hamilton, A. L., 2013. Seeking Qualitative Rigor in Inductive Research: Notes on the Gioia Methodology. *Organizational Research Methods,* 16(1), pp. 15-31.

Goffin, K., Åhlström, P., Bianchi, M. & Richtnér, A., 2019. Perspective: State-of-the-Art: The Quality of Case Study Research in Innovation Management. *Journal of Product Innovation Management,* 36(5), pp. 586-615.

Golinelli, . G. M., 2005. *L'approccio sistemico al governo dell'impresa. L'impresa sistema vitale.* 2nd ed. Padova: CEDAM.

Golinelli, G. M. et al., 2002. The firm as a viable system: managing inter-organisational relationships. Sinergie. 58, 65-98. *Sinergie – Italian Journal of Management,* 58(2), pp. 65-98.

He, Z.-L. & Wong, P.-K., 2004. Exploration vs. Exploitation: An Empirical Test of. *Organization Science,* 15(4), pp. 481-494.

Hinkin, T. R., 1998. A Brief Tutorial on the Development of Measures for Use in Survey Questionnaires. *Organizational Research Methods,* 1(1), pp. 104-121.

Hyland, J. & Karlsson, M., 2021. Letter from Standardization: Towards a Management System Standard for Innovation. *Journal of Innovation Management,* 9(1), pp. XI-XIX.

ISO, 2024. *Innovation management system – Requirements (ISO 56001:2024, IDT),* Geneva: International Organization for Standardization.

ISO, 2025. *Innovation management — Fundamentals and vocabulary (ISO 56000:2025, IDT),* s.l.: International Organization for Standardization).

ISO, 2025. *Innovation management — Fundamentals and vocabulary (ISO 56000:2025, IDT),* Geneva: International Organization for Standardisation.

ISO, n.d. *ISO/TC 279 Innovation Management*. [Online]
Available at: https://www.iso.org/committee/4587737.html
[Accessed 11 February 2026].

Jackson, M. C., 2003. *Systems thinking: creative holism for managers.* 1st ed. Chichester: Wiley.

Jalonen, H., 2011. The Uncertainty of Innovation: A Systematic Review of the Literature. *Journal of Management Research,* 4(1), pp. 1-34.

Jansen, J. J. P., Vera, D. & Crossan, M., 2009. Strategic leadership for exploration and exploitation: The moderating role of environmental dynamism. *The Leadership Quarterly,* Volume 20, pp. 5-18.

Jung, D. I., Chow, C. & Wu, A., 2003. The role of transformational leadership in enhancing organizational innovation: Hypotheses and some preliminary findings. *The Leadership Quarterly,* Volume 14, pp. 525-544.

Karlsson, M. & Magnusson, M., 2019. THE SYSTEMS APPROACH TO INNOVATION MANAGEMENT. In: J. Chen, A. Brem, E. Viardot & P. K. Wong, eds. *THE ROUTLEDGE COMPANION TO INNOVATION MANAGEMENT*. New York: Routledge, pp. 73-90.

Karstegl, N., Enjolras, M. & Mayer, F., 2025. Exploring innovation management systems: A systematic literature review and bibliometric analysis. *Journal of Engineering and Technology Management,* 76(101876).

Katz, D. & Kahn, R. L., 1978. *The Social Psychology of Organizations*. 2nd ed. New York: Wiley.

Kihlander, I., Magnusson, M. & Karlsson, M., 2022. Certification of Innovation Management Professionals: Reasons for and Results from Acquiring Certification. *Journal of Innovation Management,* 10(1), pp. 58-75.

Kihlander, I., Magnusson, M. & Karlsson, M., 2024. Critical Factors to Consider When Designing an Innovation Management System. *Research-Technology Management,* 67(3), pp. 34-43.

Klingebiel, R. & Rammer, C., 2014. Resource Allocation Strategy for Innovation Portfolio Management. *Strategic Management Journal,* 35(2), pp. 246-268.

Le, P. B. & Lei, H., 2019. Determinants of innovation capability: the roles of transformational leadership, knowledge sharing and perceived organizational support. *JOURNAL OF KNOWLEDGE MANAGEMENT,* 23(3), pp. 527-547.

Lerche, K. & Magnusson, M., 2025. Using a Standard-Based Innovation Management System Approach—A Catalyst for Innovation Legitimacy?. *R&D Management,* Volume 55, pp. 1313-1325.

Liang, X. et al., 2025. *TRENDS IN WORLD MILITARY EXPENDITURE, 2024,* Stockholm: Stockholm International Peace Research Institute (SIPRI).

Lichtenthaler, U., 2009. Outbound open innovation and its effect on fi rm performance: examining environmental infl uences. *R&D management,* 39(4), pp. 317-330.

Lim, W. M., 2025. What Is Qualitative Research? An Overview and Guidelines. *Australasian Marketing Journal,* 33(2), pp. 199-229.

Lindsten, P.-O., 2025. *Apparnas krig ny spelplan för SAAB*. [Online]
Available at: https://varldenom.com/sverige/apparnas-krig-ny-spelplan-for-saab/
[Accessed 25 01 2026].

March, J. G., 1991. Exploration and Exploitation in Organizational Learning. *Organization Science,* 2(1), pp. 71-87.

Martinez-Costa, M., Jimenez Jimenez, D. & del Pilar Castro-del-Rosario, Y., 2019. The performance implications of the UNE 166.000 standardised innovation management system. *European Journal of Innovation Management,* 22(2), pp. 281-301.

Mele, C., Pels, J. & Polese, F., 2010. A Brief Review of Systems Theories and Their Managerial Applications. *Service Science,* Volume 2, pp. 126-135.

Midgley, G. & Lindhult, E., 2021. A systems perspective on systemic innovation. *Systems research and behavioral science,* 38(5), pp. 635-670.

Mir, M. & Casadesus, M., 2011. Standardised innovation management systems: A case study of the Spanish Standard UNE 166002:2006. *INNOVAR,* 21(40), pp. 171-187.

Mir, M., Casadesus, M. & Honore Petnji, L., 2016. The impact of standardized innovation management systems on innovation capability and business performance: An empirical study. *Journal of Engineering and Technology Management,* Volume 41, pp. 26-44.

Mismetti, M., Bettinelli, C. & Rivelli, P., 2025. Survey-Based Research for Creativity and Innovation Management: Review and Recommendations. *Creativity and Innovation Management,* Volume 0, pp. 1-16.

Molina-Azorin, J. F., 2012. Mixed Methods Research in Strategic Management: Impact and Applications. *Organizational Research Methods,* 15(1), pp. 33-56.

Nagji, B. & Tuff, G., 2012. Managing Your Innovation Portfolio. *Harvard Business Review*, May, pp. 2-11.

Nilsson, S. & Ritzén, S., 2014. Exploring the Use of Innovation Performance Measurement to Build Innovation Capability in a Medical Device Company. *Creativity and Innovation Management,* 23(2), pp. 93-231.

OECD/Eurostat, 2018. *Guidelines for Collecting, Reporting and Using Data on Innovation, 4th Edition, The Measurement of Scientific, Technological and Innovation Activities,* Paris: OECD Publishing.

O'Reilly, C. A. & Tushman, M. L., 2004. The Ambidextrous Organization. *Harvard Business Review*, April, pp. 1-10.

O'Reilly, C. A. & Tushman, M. L., 2011. Organizational Ambidexterity in Action: HOW MANAGERS EXPLORE AND EXPLOIT. *California Management Review,* 53(4), pp. 5-20.

Prajogo, D. I. & Ahmed, P. K., 2006. Relationships between innovationstimulus, innovation capacity, and innovation performance. *R&D Management,* 36(5), pp. 499-515.

Reuven, N. & Shamir, E., 2025. The shift in technological dominance and the adaption of open innovation by the defence sector. *Defense & Security Analysis,* 41(3), pp. 392-415.

Saab, 2024. *Annual & Sustainability Report,* Stockholm: Saab.

Saab, 2025. *Innovation - Our Approach*. [Online]
Available at: https://www.saab.com/about/innovation/our-approach
[Accessed 02 February 2025].

Saab, 2025. *Interim Report Q4,* Stockholm: Saab.

Sattayaraksa, T. & Boon-itt, S., 2018. The roles of CEO transformational leadership and organizational factors on product innovation performance. *European Journal of Innovation Management,* 21(2), pp. 227-249.

SIPRI, 2025. *SIPRI Military Expenditure Database.* [Online]
Available at: https://www.sipri.org/databases/milex
[Accessed 05 February 2026].

Slack, N. & Lewis, M., 2011. *Operations Strategy*. 3rd ed. Essex: Pearson Education Limited.

Smith, M., Busi, M., Ball, P. & Van Der Meer, R., 2008. FACTORS INFLUENCING AN ORGANISATION'S ABILITY TO MANAGE INNOVATION: A STRUCTURED LITERATURE REVIEW AND CONCEPTUAL MODEL. *International Journal of Innovation Management,* 12(4), pp. 655-676.

Smith, W. K. & Tushman, M. L., 2005. Managing Strategic Contradictions: A Top Management Model for Managing Innovation Streams. *Organization Science,* 16(5), pp. 522-536.

Song, M. & Meier, K. J., 2022. Walking the Walk: Does Perceptual Congruence Between Managers and Employees Promote Employee Job Satisfaction?. *Review of Public Personnel Administration,* 42(2), pp. 195-225.

Spector, P. E., 1976. Choosing response categories for summated rating scales. *Journal of applied psychology,* 61(3), pp. 374-375.

Taques, F. H., López, M. G., Basso, L. F. & Areal, N., 2021. Indicators used to measure service innovation and manufacturing innovation. *Journal of Innovation & Knowledge,* Volume 6, pp. 11-26.

Teece, D. J., 2018. Dynamic capabilities as (workable) management systems theory. *Journal of Management & Organization,* 24(3), pp. 359-368.

Tessier, S., 2012. From Field Notes, to Transcripts, to Tape Recordings: Evolution or Combination?. *International Journal of Qualitative Methods,* 11(4), pp. 446-460.

Tidd, J., 2021. A REVIEW AND CRITICAL ASSESSMENT OF THE ISO56002INNOVATION MANAGEMENT SYSTEMS STANDARD:EVIDENCE AND LIMITATIONS. *International Journal of Innovation Management,* 25(1), pp. 1-17.

Turulja, L. & Bajgoric, N., 2019. Innovation, firms' performance and environmental turbulence: is there a moderator or mediator?. *European Journal of Innovation Management,* 22(1), pp. 213-232.

Van de Ven, A. H., 1986. Central Problems in the Management of Innovation. *Management Science,* 32(5), pp. 590-607.

Van de Ven, A. H., 2017. The innovation journey: you can't control it, but you can learn to maneuver it. *Innovation: Organization and Management,* 19(1), pp. 39-42.

Wang, C.-H., Chang, C.-H. & Shen, G. C., 2015. The effect of inbound open innovation on fi rm performance: Evidence from high-tech industry. *Technological Forecasting & Social Change,* Volume 99, pp. 222-230.

Wengraf, T., 2001. *Qualitative research interviewing: Biographic narrative and semi-structured methods.* 1st ed. London: Sage.

11. Appendices

The following chapter includes supplementary data from the quantitative and qualitative data collection.

11.1. Appendix A: Observation Validation

The following table (Table 15) presents the observations gathered from the interviews, that were subsequently validated with four interview participants.

Table 15: Validated observations from quantitative data

Name	Observation	Average
O1	Cross BU visibility of ongoing projects and research is low; people often do not know what other Business Units are working on, limiting synergies.	0.88
O2	After ideas have been evaluated, there is no formal continuation to the process to develop or commercialize the idea(s)	0.63
O3	The idea generation and development process lacks integration with other Saab functions. (innovation labs, Saab ventures, operational projects, etc.)	0.75
O4	Senior leaderships' explicit endorsement of innovation does not translate into higher perceived support for innovation among middle managers	0.88
O5	The overall resource allocation for the innovation pipeline (time, financial, etc.) is ad-hoc and unstructured	0.75
O6	Line managers are reluctant to allocate time for their employees to carry out innovation.	0.63
O7	Time for innovation work is not prioritized or protected in a mandated and structured way. (It depends a lot on managers' willingness to innovate)	1.00
O8	Sufficient time is not allocated toward innovation activities	0.75
O9	The organization lacks formal, measurable innovation objectives at the operational level	1.00
O10	The core organization lacks a culture geared toward exploration and innovation. (fail fast approaches, testing, prototyping, etc)	0.88
O11	The core organization is risk-minimizing	0.83
O12	Little personal coaching or mentorship is given to employees developing their ideas in the innovation portal	1.00
O13	An innovation portfolio is not systematically defined or prioritized, resulting in ad-hoc project selection	0.88
O14	There is a lack of support for innovation, like coaching, tools, and processes.	0.88
O15	There is competence development (training or coaching) that is specifically aimed at innovation skills, but employees are not aware	0.33
O16	There is lacking systematic corrective-action after an innovation project fails or stalls.	0.83
O17	Deep insights do not commonly precede projects / catalyze the starting of projects. (deep insights can be knowledge from technical and market research that lead to the creation of many ideas)	0.63
O18	External collaboration through partnerships often faces pushback due to security concerns	0.67
O19	Contextual circumstances that may be used as input to improve the innovation management system are not actively shared. (circumstances like internal issues, external market movements, etc.)	0.75
--	Average score - higher indicates agreement	**0.79**

11.2. Appendix B: Quantitative Data

The following appendix includes various outputs from the quantitative data analysis.

11.2.1. Descriptive Data – Determining ISO 56001 Adherence Through Mean Values

Table 16 presents the descriptive statistics from the questionnaire, while Table 17 shows the accumulated scores and average means across ISO dimensions.

Table 16: Descriptive statistics featuring the number of respondents, mean and standard deviations across the three respondent groups, per survey item

ISO Dimension	Item Label	OE n	OE M	OE SD	MM n	MM M	MM SD	SL n	SL M	SL SD	All n	All M	All SD
Context	External trends (political/market/tech)	17	3.41	1.12	11	3.64	1.12	6	2.5	0.55	34	3.32	1.09
Context	Internal factors (people/culture/structure)	17	2.65	1.17	11	3.27	0.9	7	2.86	1.07	35	2.89	1.08
Leadership	Culture of new ideas / psychological safety	18	4	0.84	10	3.9	1.1	7	3.29	0.76	35	3.83	0.92
Leadership	Risk-taking for innovation	16	3.06	1.34	10	4.1	0.74	7	2.14	0.69	33	3.18	1.26
Leadership	Tolerance of failure/experimentation	16	3.25	1	11	4.09	0.7	7	2.86	1.07	34	3.44	1.02
Planning Portfolio	Diverse project portfolio (risk/time horizon)	18	3.22	1.22	11	4.18	0.75	5	2.6	0.89	34	3.44	1.16
Planning Collaboration	Internal & external collaboration planning	14	3	1.18	10	2.7	0.67	7	2.57	0.98	31	2.81	0.98
Support	Time allocated for innovation	16	2.19	0.75	11	2.73	1.19	7	2	0.58	34	2.32	0.91
Support	Budget allocated for innovation	14	2.29	0.91	11	2.64	1.03	7	2.71	1.11	32	2.5	0.98
Support	Tools & technology availability	16	2.88	1.36	11	2.91	0.94	7	2.43	0.98	34	2.79	1.15
Operations	Identifying innovation insights/opportunities	16	3.19	0.91	10	3.2	1.23	7	2.86	1.07	33	3.12	1.02
Operations	Validating & testing concepts	17	3.24	0.9	9	3.67	0.87	7	2.43	1.13	33	3.18	1.01
Operations	Commercialisation of solutions	11	2.91	0.94	8	3	0.76	7	2.57	0.53	26	2.85	0.78
Performance	KPIs for innovation/R&D outcomes	NA	NA	NA	8	1.5	0.53	7	1.57	0.79	15	1.53	0.64
Performance	Continuous improvement of R&D processes	NA	NA	NA	11	2.64	1.03	7	2	0.58	18	2.39	0.92
Improvement	Lessons-learned captured & used	16	2.69	1.2	10	3.3	0.67	6	2	0.89	32	2.75	1.08
Innov Performance	Radical innovation	NA	NA	NA	11	2.73	0.65	6	3	0.89	17	2.82	0.73
Innov Performance	Incremental innovation	NA	NA	NA	11	3.73	0.79	7	3.57	0.98	18	3.67	0.84

Innov Perform-ance	Customer involvement	NA	NA	NA	10	3.1	1.1	7	3.29	0.95	17	3.18	1.01
Innov Perform-ance	Speed to market	NA	NA	NA	11	2.45	0.93	7	2	0.58	18	2.28	0.83

Table 17: Descriptive statistics featuring the accumulated number of respondents, average mean and standard deviations across the three respondent groups and survey items, per ISO dimension.

ISO_Area	OE accum-ulated n	OE M	OE SD	MM accum-ulated n	MM M	MM SD	SL accum-ulated n	SL M	SL SD	All n	All M
Context	34	3.03	NA	22	3.45	NA	13	2.69	NA	69	3.1
Leadership	50	3.46	NA	31	4.03	NA	21	2.76	NA	102	3.49
Planning_Portfolio	18	3.22	NA	11	4.18	NA	5	2.6	NA	34	3.44
Planning_Collab	14	3.00	NA	10	2.7	NA	7	2.57	NA	31	2.81
Support	46	2.46	NA	33	2.76	NA	21	2.38	NA	100	2.54
Operations	44	3.14	NA	27	3.3	NA	21	2.62	NA	92	3.07
Performance	24	2.33	NA	19	2.16	NA	14	1.79	NA	57	2.14
Improvement	16	2.69	NA	10	3.3	NA	6	2.0	NA	32	2.75
Innov_Performance	NA	NA	NA	43	3	NA	27	2.96	NA	70	2.99

11.2.2. Spearman And Pearson Correlation Tests – Extended Sample (OE, MM And SL)

The following table includes the comprehensive results of the extended Spearman and Pearson correlation test. Table 18 features the extended sample including MM, SL, and OEs with innovation performance scores attributed from managers

Table 18: Result of the Spearman and Pearson correlation tests, featuring the extended sample of OE, MM and SL respondents

DV	Predictor	n	Spearman_r	p_spearman	Pearson_r	p_pearson	Significance
Radical	Context	29	0.087	0.6549	0.054	0.7821	ns
Radical	Leadership	29	0.321	0.0895	0.306	0.107	ns
Radical	Planning_Portfolio	27	-0.081	0.6891	-0.021	0.9176	ns
Radical	Planning_Collab	26	0.298	0.1387	0.24	0.2372	ns
Radical	Support	29	0.629	3.00E-04	0.593	7.00E-04	** Both significant (p<0.01)
Radical	Operations	28	0.214	0.2737	0.255	0.191	ns
Radical	Performance	29	0.259	0.175	0.192	0.3193	ns
Radical	Improvement	27	0.485	0.0104	0.396	0.0411	** Both significant (p<0.05)
Incremental	Context	30	0.118	0.5363	0.109	0.5651	ns
Incremental	Leadership	30	0.202	0.2847	0.215	0.2542	ns
Incremental	Planning_Portfolio	28	-0.003	0.9871	-0.151	0.4436	ns
Incremental	Planning_Collab	27	0.194	0.3333	0.245	0.218	ns
Incremental	Support	30	0.189	0.3163	0.175	0.3554	ns

Incremental	Operations	29	0.103	0.5959	0.168	0.384	ns
Incremental	Performance	30	0.301	0.1055	0.234	0.2128	ns
Incremental	Improvement	28	0.365	0.0562	0.241	0.2164	ns
Customer	Context	29	0.331	0.0795	0.321	0.0895	ns
Customer	Leadership	29	-0.025	0.8991	-0.023	0.9047	ns
Customer	Planning_Portfolio	27	-0.026	0.8993	-0.049	0.8088	ns
Customer	Planning_Collab	26	0.244	0.2305	0.27	0.1823	ns
Customer	Support	29	0.379	0.0425	0.318	0.0927	* Spearman only (p<0.05)
Customer	Operations	28	0.031	0.8746	0.123	0.5333	ns
Customer	Performance	29	0.145	0.4521	0.2	0.2978	ns
Customer	Improvement	28	0.294	0.1288	0.258	0.1849	ns
Speed	Context	30	0.133	0.4834	0.249	0.1848	ns
Speed	Leadership	30	0.226	0.2308	0.32	0.0843	ns
Speed	Planning_Portfolio	28	0.155	0.4309	0.181	0.3563	ns
Speed	Planning_Collab	27	0.205	0.3057	0.171	0.3924	ns
Speed	Support	30	0.04	0.8348	0.276	0.1399	ns
Speed	Operations	29	0.26	0.1725	0.348	0.0643	ns
Speed	Performance	30	0.156	0.4106	0.171	0.3659	ns
Speed	Improvement	28	-0.116	0.5558	0.06	0.7615	ns
Radical	Overall_IMS	29	0.382	0.0407	0.382	0.0409	** Both sig (p<0.05)
Incremental	Overall_IMS	30	0.277	0.1381	0.281	0.1325	ns
Customer	Overall_IMS	29	0.387	0.0383	0.28	0.1409	* Spearman only (p<0.05)
Speed	Overall_IMS	30	0.082	0.6672	0.322	0.0825	ns

11.2.3. Spearman And Pearson Correlation Tests – Limited Sample (MM And SL Only)

The following table includes the comprehensive results of the limited Spearman and Pearson correlation test. Table 19 shows the analysis where OEs are excluded, only including MM and SL responses.

Table 19: result of the Spearman and Pearson correlation tests, featuring the limited sample of MM and SL respondents

DV	Predictor	n	Spearman_r	p_value	Pearson_r	p_value_pearson	Significance
Radical	Context	17	0.137	0.5993	0.088	0.7358	ns
Radical	Leadership	17	0.409	0.1029	0.425	0.0888	ns
Radical	Planning_Portfolio	15	0.011	0.9682	0.086	0.7602	ns
Radical	Planning_Collab	16	0.417	0.1078	0.343	0.1931	ns
Radical	Support	17	0.652	0.0046	0.663	0.0037	** Both significant (p<0.01)
Radical	Operations	16	0.363	0.1671	0.326	0.2179	ns

Radical	Performance	17	0.259	0.3145	0.228	0.3782	ns
Radical	Improvement	15	0.433	0.1068	0.454	0.0889	ns
Incremental	Context	18	0.102	0.6865	0.157	0.5335	ns
Incremental	Leadership	18	0.265	0.2881	0.293	0.2383	ns
Incremental	Planning_Portfolio	16	-0.026	0.9244	-0.18	0.5057	ns
Incremental	Planning_Collab	17	0.307	0.2314	0.358	0.1579	ns
Incremental	Support	18	0.113	0.6545	0.198	0.4317	ns
Incremental	Operations	17	0.21	0.4194	0.205	0.4299	ns
Incremental	Performance	18	0.344	0.1627	0.296	0.2325	ns
Incremental	Improvement	16	0.297	0.2643	0.298	0.2631	ns
Customer	Context	17	0.486	0.0478	0.468	0.0582	* Spearman only (p<0.05)
Customer	Leadership	17	-0.105	0.6889	-0.046	0.8613	ns
Customer	Planning_Portfolio	15	0.064	0.8208	0.059	0.8337	ns
Customer	Planning_Collab	16	0.347	0.1876	0.379	0.1473	ns
Customer	Support	17	0.231	0.3717	0.25	0.334	ns
Customer	Operations	16	0.068	0.8038	0.184	0.4951	ns
Customer	Performance	17	0.221	0.3946	0.234	0.3663	ns
Customer	Improvement	16	0.154	0.5699	0.166	0.5381	ns
Speed	Context	18	0.319	0.197	0.333	0.1772	ns
Speed	Leadership	18	0.388	0.1114	0.458	0.056	ns
Speed	Planning_Portfolio	16	0.266	0.3202	0.181	0.5012	ns
Speed	Planning_Collab	17	0.352	0.1654	0.315	0.2174	ns
Speed	Support	18	0.374	0.1261	0.483	0.0422	ns
Speed	Operations	17	0.363	0.1519	0.418	0.0953	ns
Speed	Performance	18	0.193	0.443	0.265	0.2872	ns
Speed	Improvement	16	0.138	0.611	0.229	0.3945	ns
Radical	Overall_IMS	17	0.327	0.1997	0.414	0.0981	ns
Incremental	Overall_IMS	18	0.223	0.3744	0.313	0.2054	ns
Customer	Overall_IMS	17	0.304	0.2353	0.276	0.284	ns
Speed	Overall_IMS	18	0.325	0.1876	0.429	0.0756	ns

11.2.4. Role Dependent Welch's T-Tests Vs. ISO 56000 Dimensions

This section includes the data from the Welch's t-tests comparing different roles on their scores for individual survey items, and composites of survey items (Table 20).

Table 20: Result of Welch's t-test analysis of discrepancies in recipient group responses to survey items and composites

ISO dimension	Item Label	Group 1	Group 2	n Group1	n Group2	Mean Group 1	Mean Group2	SD Group1	SD Group2	t statistic	p value raw	Bonferroni alpha	Significance
Innovation	New solutions / low TRL commercialization	Middle Management	Senior Leadership	11	6	2.727	3	0.647	0.894	-0.659	0.5286	0.00096	ns
Innovation	Existing solutions / high TRL improvement	Middle Management	Senior Leadership	11	7	3.727	3.571	0.786	0.976	0.355	0.7291	0.00096	ns
Innovation	Customer involvement & feedback	Middle Management	Senior Leadership	10	7	3.1	3.286	1.101	0.951	-0.371	0.716	0.00096	ns
Innovation	Speed to market	Middle Management	Senior Leadership	11	7	2.455	2	0.934	0.577	1.276	0.2203	0.00096	ns
Context	External trends (political/market/tech)	Operational	Middle Management	17	11	3.412	3.636	1.121	1.12	-0.518	0.6098	0.00096	ns
Context	Internal factors (people/culture/structure)	Operational	Middle Management	17	11	2.647	3.273	1.169	0.905	-1.59	0.1244	0.00096	ns
Context	External trends (political/market/tech)	Operational	Senior Leadership	17	6	3.412	2.5	1.121	0.548	2.59	0.0184	0.00096	* Raw only (p<0.05)
Context	Internal factors (people/culture/structure)	Operational	Senior Leadership	17	7	2.647	2.857	1.169	1.069	-0.426	0.6778	0.00096	ns
Context	External trends (political/market/tech)	Middle Management	Senior Leadership	11	6	3.636	2.5	1.12	0.548	2.806	0.0133	0.00096	* Raw only (p<0.05)
Context	Internal factors (people/culture/structure)	Middle Management	Senior Leadership	11	7	3.273	2.857	0.905	1.069	0.853	0.4116	0.00096	ns
Leadership	Culture of new ideas / psychological safety	Operational	Middle Management	18	10	4	3.9	0.84	1.101	0.25	0.8062	0.00096	ns
Leadership	Risk-taking for innovation	Operational	Middle Management	16	10	3.062	4.1	1.34	0.738	-2.541	0.018	0.00096	* Raw only (p<0.05)
Leadership	Tolerance of failure/experimentation	Operational	Middle Management	16	11	3.25	4.091	1	0.701	-2.569	0.0166	0.00096	* Raw only (p<0.05)
Leadership	Culture of new ideas / psychological safety	Operational	Senior Leadership	18	7	4	3.286	0.84	0.756	2.055	0.062	0.00096	ns
Leadership	Risk-taking for innovation	Operational	Senior Leadership	16	7	3.062	2.857	1.34	1.069	0.391	0.7014	0.00096	ns
Leadership	Tolerance of failure/experimentation	Operational	Senior Leadership	16	7	3.25	2.143	1	0.69	3.064	0.0072	0.00096	* Raw only (p<0.05)
Leadership	Culture of new ideas / psychological safety	Middle Management	Senior Leadership	10	7	3.9	3.286	1.101	0.756	1.364	0.1926	0.00096	ns
Leadership	Risk-taking for innovation	Middle Management	Senior Leadership	10	7	4.1	2.143	0.738	0.69	5.592	0.0001	0.00096	* Bonferroni (p<0.00096)
Leadership	Tolerance of failure/experimentation	Middle Management	Senior Leadership	11	7	4.091	2.857	0.701	1.069	2.706	0.0235	0.00096	* Raw only (p<0.05)
Planning_Portfolio	Diverse project portfolio (risk/time horizon)	Operational	Middle Management	18	11	3.222	4.182	1.215	0.751	-2.628	0.014	0.00096	* Raw only (p<0.05)
Planning_Portfolio	Diverse project portfolio (risk/time horizon)	Operational	Senior Leadership	18	5	3.222	2.6	1.215	0.894	1.265	0.2391	0.00096	ns
Planning_Portfolio	Diverse project portfolio (risk/time horizon)	Middle Management	Senior Leadership	11	5	4.182	2.6	0.751	0.894	3.442	0.0116	0.00096	* Raw only (p<0.05)
Planning_Collab	Internal & external collaboration planning	Operational	Middle Management	14	10	3	2.7	1.177	0.675	0.789	0.4387	0.00096	ns
Planning_Collab	Internal & external collaboration planning	Operational	Senior Leadership	14	7	3	2.571	1.177	0.976	0.884	0.3911	0.00096	ns
Planning_Collab	Internal & external collaboration planning	Middle Management	Senior Leadership	10	7	2.7	2.571	0.675	0.976	0.302	0.7691	0.00096	ns

Support	Time allocated for innovation	Operational	Middle Management	16	11	2.188	2.727	0.75	1.191	-1.333	0.202	0.00096	ns
Support	Budget allocated for innovation	Operational	Middle Management	14	11	2.286	2.636	0.914	1.027	-0.889	0.3843	0.00096	ns
Support	Tools & technology availability	Operational	Middle Management	16	11	2.875	2.909	1.36	0.944	-0.077	0.9393	0.00096	ns
Support	Time allocated for innovation	Operational	Senior Leadership	16	7	2.188	2	0.75	0.577	0.652	0.5245	0.00096	ns
Support	Budget allocated for innovation	Operational	Senior Leadership	14	7	2.286	2.714	0.914	1.113	-0.881	0.3985	0.00096	ns
Support	Tools & technology availability	Operational	Senior Leadership	16	7	2.875	2.429	1.36	0.976	0.89	0.3868	0.00096	ns
Support	Time allocated for innovation	Middle Management	Senior Leadership	11	7	2.727	2	1.191	0.577	1.731	0.1036	0.00096	ns
Support	Budget allocated for innovation	Middle Management	Senior Leadership	11	7	2.636	2.714	1.027	1.113	-0.149	0.8838	0.00096	ns
Support	Tools & technology availability	Middle Management	Senior Leadership	11	7	2.909	2.429	0.944	0.976	1.031	0.3217	0.00096	ns
Operations	Identifying innovation insights/opportunities	Operational	Middle Management	16	10	3.188	3.2	0.911	1.229	-0.028	0.9782	0.00096	ns
Operations	Validating & testing concepts	Operational	Middle Management	17	9	3.235	3.667	0.903	0.866	-1.19	0.2503	0.00096	ns
Operations	Commercialization of solutions	Operational	Middle Management	11	8	2.909	3	0.944	0.756	-0.233	0.8187	0.00096	ns
Operations	Identifying innovation insights/opportunities	Operational	Senior Leadership	16	7	3.188	2.857	0.911	1.069	0.712	0.4925	0.00096	ns
Operations	Validating & testing concepts	Operational	Senior Leadership	17	7	3.235	2.429	0.903	1.134	1.676	0.1269	0.00096	ns
Operations	Commercialization of solutions	Operational	Senior Leadership	11	7	2.909	2.571	0.944	0.535	0.967	0.3478	0.00096	ns
Operations	Identifying innovation insights/opportunities	Middle Management	Senior Leadership	10	7	3.2	2.857	1.229	1.069	0.611	0.5506	0.00096	ns
Operations	Validating & testing concepts	Middle Management	Senior Leadership	9	7	3.667	2.429	0.866	1.134	2.396	0.0355	0.00096	* Raw only (p<0.05)
Operations	Commercialization of solutions	Middle Management	Senior Leadership	8	7	3	2.571	0.756	0.535	1.279	0.224	0.00096	ns
Perf-ormance	KPIs for innovation/R&D outcomes	Operational	Middle Management	9	8	2	1.5	0.866	0.535	1.449	0.1701	0.00096	ns
Perf-ormance	Continuous improvement of R&D processes	Operational	Middle Management	15	11	2.533	2.636	1.06	1.027	-0.249	0.8054	0.00096	ns
Perf-ormance	KPIs for innovation/R&D outcomes	Operational	Senior Leadership	9	7	2	1.571	0.866	0.787	1.034	0.3192	0.00096	ns
Perf-ormance	Continuous improvement of R&D processes	Operational	Senior Leadership	15	7	2.533	2	1.06	0.577	1.524	0.1439	0.00096	ns
Perf-ormance	KPIs for innovation/R&D outcomes	Middle Management	Senior Leadership	8	7	1.5	1.571	0.535	0.787	-0.203	0.8433	0.00096	ns
Perf-ormance	Continuous improvement of R&D processes	Middle Management	Senior Leadership	11	7	2.636	2	1.027	0.577	1.68	0.1125	0.00096	ns
Improve-ment	Lessons-learned captured & used	Operational	Middle Management	16	10	2.688	3.3	1.195	0.675	-1.668	0.1084	0.00096	ns
Improve-ment	Lessons-learned captured & used	Operational	Senior Leadership	16	6	2.688	2	1.195	0.894	1.457	0.1705	0.00096	ns
Improve-ment	Lessons-learned captured & used	Middle Management	Senior Leadership	10	6	3.3	2	0.675	0.894	3.074	0.0143	0.00096	* Raw only (p<0.05)
Innovation	All items in Innovation (composite mean)	Middle Management	Senior Leadership	11	7	2.992	2.964	0.393	0.529	0.121	0.906	0.002	ns
Context	All items in Context (composite mean)	Operational	Middle Management	18	11	3.083	3.455	0.989	0.879	-1.052	0.3037	0.002	ns
Context	All items in Context (composite mean)	Operational	Senior Leadership	18	7	3.083	2.714	0.989	0.756	1.001	0.3334	0.002	ns
Context	All items in Context (composite mean)	Middle Management	Senior Leadership	11	7	3.455	2.714	0.879	0.756	1.899	0.0777	0.002	ns
Leadership	All items in Leadership (composite mean)	Operational	Middle Management	18	11	3.509	4.03	0.86	0.658	-1.837	0.0779	0.002	ns

Leadership	All items in Leadership (composite mean)	Operational	Senior Leadership	18	7	3.509	2.762	0.86	0.763	2.12	0.0549	0.002	ns
Leadership	All items in Leadership (composite mean)	Middle Management	Senior Leadership	11	7	4.03	2.762	0.658	0.763	3.625	0.0037	0.002	* Raw only (p<0.05)
Support	All items in Support (composite mean)	Operational	Middle Management	17	11	2.471	2.758	0.913	0.87	-0.836	0.4122	0.002	ns
Support	All items in Support (composite mean)	Operational	Senior Leadership	17	7	2.471	2.381	0.913	0.756	0.248	0.8079	0.002	ns
Support	All items in Support (composite mean)	Middle Management	Senior Leadership	11	7	2.758	2.381	0.87	0.756	0.971	0.3478	0.002	ns
Operations	All items in Operations (composite mean)	Operational	Middle Management	17	10	3.127	3.25	0.676	0.914	-0.369	0.7174	0.002	ns
Operations	All items in Operations (composite mean)	Operational	Senior Leadership	17	7	3.127	2.619	0.676	0.448	2.156	0.0457	0.002	* Raw only (p<0.05)
Operations	All items in Operations (composite mean)	Middle Management	Senior Leadership	10	7	3.25	2.619	0.914	0.448	1.884	0.0809	0.002	ns
Perf-ormance	All items in Performance (composite mean)	Operational	Middle Management	16	11	2.375	2.273	0.992	0.905	0.277	0.7839	0.002	ns
Perf-ormance	All items in Performance (composite mean)	Operational	Senior Leadership	16	7	2.375	1.786	0.992	0.636	1.706	0.1056	0.002	ns
Perf-ormance	All items in Performance (composite mean)	Middle Management	Senior Leadership	11	7	2.273	1.786	0.905	0.636	1.339	0.1995	0.002	ns

11.2.5. Department Dependent Welch's T-Tests

The following tables show the differences between department 2 and other departments in the Welch's t-test. Table 21 compares the departments on survey items related to ISO 56001 requirements, while Table 22 compares innovation performance dimensions.

Table 21: Result of the department dependent Welch's t-test, comparing Department 2 against other departments on survey items and composites

ISO Dimension	Item Label	Group 1	Group 2	n Dept2	n Rest	Mean Dept2	Mean Rest	SD Dept2	SD Rest	t statistic	p value raw	Bonferroni alpha	Significance
Context	External trends (political/market/tech)	Dept2	Rest	17	17	3.059	3.588	1.144	1.004	-1.434	0.1614	0.00313	ns
Context	Internal factors (people/culture/structure)	Dept2	Rest	19	16	3.053	2.688	1.129	1.014	1.007	0.3212	0.00313	ns
Leadership	Culture of new ideas / psychological safety	Dept2	Rest	19	16	3.684	4	0.885	0.966	-1.001	0.3247	0.00313	ns
Leadership	Risk-taking for innovation	Dept2	Rest	19	14	3.105	3.286	1.329	1.204	-0.407	0.6869	0.00313	ns
Leadership	Tolerance of failure/experimentation	Dept2	Rest	19	15	3.421	3.467	1.017	1.06	-0.127	0.9	0.00313	ns
Planning Portfolio	Diverse project portfolio (risk/time horizon)	Dept2	Rest	17	17	3.118	3.765	1.166	1.091	-1.67	0.1047	0.00313	ns
Planning Collaboration	Internal & external collaboration planning	Dept2	Rest	16	15	2.688	2.933	0.946	1.033	-0.69	0.4961	0.00313	ns
Support	Time allocated for innovation	Dept2	Rest	18	16	2.333	2.312	0.84	1.014	0.065	0.9488	0.00313	ns
Support	Budget allocated for innovation	Dept2	Rest	17	15	2.882	2.067	0.781	1.033	2.494	0.0194	0.00313	* Raw only (p<0.05)
Support	Tools & technology availability	Dept2	Rest	17	17	2.882	2.706	1.054	1.263	0.442	0.6613	0.00313	ns
Operations	Identifying innovation insights/opportunities	Dept2	Rest	17	16	2.941	3.312	0.966	1.078	-1.04	0.3068	0.00313	ns
Operations	Validating & testing concepts	Dept2	Rest	16	17	2.875	3.471	0.957	1.007	-1.741	0.0915	0.00313	ns
Operations	Commercialisation of solutions	Dept2	Rest	13	13	3	2.692	0.707	0.855	1	0.3276	0.00313	ns
Performance	KPIs for innovation/R&D outcomes	Dept2	Rest	12	12	1.833	1.583	0.937	0.515	0.81	0.4292	0.00313	ns
Performance	Continuous improvement of R&D processes	Dept2	Rest	17	16	2.412	2.5	0.939	1.033	-0.256	0.7995	0.00313	ns
Improvement	Lessons-learned captured & used	Dept2	Rest	16	16	3.062	2.438	0.998	1.094	1.689	0.1017	0.00313	ns
Context	Context (composite mean)	Dept2	Rest	19	17	3.079	3.176	0.961	0.918	-0.311	0.7575	0.00625	ns
Leadership	Leadership (composite mean)	Dept2	Rest	19	17	3.404	3.657	0.85	0.925	-0.852	0.4004	0.00625	ns
Planning Portfolio	Planning_Portfolio (composite mean)	Dept2	Rest	17	17	3.118	3.765	1.166	1.091	-1.67	0.1047	0.00625	ns
Planning Collaboration	Planning_Collab (composite mean)	Dept2	Rest	16	15	2.688	2.933	0.946	1.033	-0.69	0.4961	0.00625	ns
Support	Support (composite mean)	Dept2	Rest	18	17	2.667	2.412	0.732	0.983	0.866	0.3933	0.00625	ns
Operations	Operations (composite mean)	Dept2	Rest	17	17	2.892	3.225	0.618	0.821	-1.338	0.1911	0.00625	ns
Performance	Performance (composite mean)	Dept2	Rest	17	17	2.206	2.235	0.885	0.954	-0.093	0.9263	0.00625	ns
Improvement	Improvement (composite mean)	Dept2	Rest	16	16	3.062	2.438	0.998	1.094	1.689	0.1017	0.00625	ns

Table 22: Result of the department dependent Welch's t-test, comparing Department 2 against other departments on innovation performance dimensions, including both OE+MM+SL results and MM+SL only results.

ISO dimen-sion	Item Label	Group 1	Group 2	n Dept2	n Rest	Mean Dept2	Mean Rest	SD Dept2	SD Rest	t statistic	p value raw	Bonf-erroni alpha	Signi-ficance
Performance Outcome (incl. OE attributed)	Radical	Dept2	Rest	21	14	3	2.619	0.548	0.366	2.465	0.0191	0.0125	* Raw only (p<0.05)

Performance Outcome (incl. OE attributed)	Incremental	Dept2	Rest	22	14	3.818	3.548	0.603	0.549	1.387	0.1756	0.0125	ns
Performance Outcome (incl. OE attributed)	Customer	Dept2	Rest	21	14	3.5	2.69	0.57	0.756	3.411	0.0024	0.0125	* Bonferroni ($p<0.0125$)
Performance Outcome (incl. OE attributed)	Speed	Dept2	Rest	22	14	2.182	2.548	0.518	0.674	-1.73	0.0972	0.0125	ns
Performance Outcome (MM+SL only)	Radical	Dept2	Rest	10	7	3	2.571	0.816	0.535	1.307	0.2109	0.0125	ns
Performance Outcome (MM+SL only)	Incremental	Dept2	Rest	11	7	3.818	3.429	0.874	0.787	0.981	0.3435	0.0125	ns
Performance Outcome (MM+SL only)	Customer	Dept2	Rest	10	7	3.5	2.714	0.85	1.113	1.574	0.1445	0.0125	ns
Performance Outcome (MM+SL only)	Speed	Dept2	Rest	11	7	2.182	2.429	0.751	0.976	-0.57	0.5806	0.0125	ns

11.3. Appendix C: Interview Guide

1. What is your role?
2. What is your academic and working background?
3. How do you work with innovation?
4. Where does innovation happen at Saab?
5. What are the goals of innovation?
6. Is innovation measured in some way?
7. How are innovative ideas handled?
8. How does Saab collaborate in innovation?

11.4. Appendix D: Questionnaire

This section presents the full questionnaire distributed to the organization. First, the introduction questions are shown, and then the questions pertaining to ISO56001 and innovation performance (Table 23).

11.4.1. Introduction Section

To which department at Saab Surveillance do you belong?

BU Airborne Early Warning
BU Fighter Core Capabilities
BU Naval Combat Systems
BU Safety and Security Solutions
BU Surface Sensor Solutions
BU Digital Battlespace Solutions
Other

For how long have you worked at Saab?

Less than 1 year
Up to 3 years
Up to 10 years
Up to 15 years
More than 15 years
More than 30 years

Have you previously received any Innovation Management Training?

No, I am not familiar with the field
I joined one lecture or workshop about it
I have some knowledge about it from multiple lectures and/or training
I have worked with innovation management for 1-4 years
I have worked with innovation management for 5+ years

What is your primary mode of work? Please select the option that is most consistent over a one year period

Remote
On site 1-2 days a week
On site 3-4 days a week
On site 5 days a week

What is your role?

Operational level employee
Middle / project manager
Senior Leadership

11.4.2. ISO 56001 Requirement And Innovation Performance Questions

Table 23: The comprehensive survey, including the questions and Likert Scale anchors provided to the three respondent groups

ISO dimension	Topic	Operational Employees	Middle Managers	Senior Leadership
Context	External trends (political/market/tech)	**In the projects I recently participated in, my team considered political, market and technological trends when developing the solution** I don't know Strongly disagree Disagree Neutral Agree Strongly agree	**In the projects I recently participated in, my team considered political, market and technological trends when developing the solution** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**How thoroughly are emerging external trends that could affect Saab's innovation work captured and analyzed? (External trends like political, market and technological factors)** I don't know Very poorly Poorly Moderately Well Very well
Context	Internal factors (people/culture/structure)	**In the projects I recently participated in, my team considered how our people,**	**In the projects I recently participated in, my team considered our people,**	**How thoroughly are internal factors that could affect Saab's innovation work**

		culture and organizational structures could affect us when developing the solution I don't know Strongly disagree Disagree Neutral Agree Strongly agree	**culture and organizational structures when developing the solution** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**captured and analyzed? (Internal factors like people, organizational culture and structure)** I don't know Very poorly Poorly Moderately Well Very well
Leadership	Culture of new ideas / psychological safety	**My manager encourages me to be innovative and supports a culture where new ideas are welcomed.** I don't know Strongly disagree Disagree Neutral Agree Strongly agree	**To what extent do you promote a culture in your project / team where new ideas are openly welcomed and acted upon?** I don't know Very limited extent Limited extent Neutral Large extent Very large extent	**To what extent does senior leadership promote a culture where new ideas are openly welcomed and acted upon?** I don't know Very limited extent Limited extent Neutral Large extent Very large extent
Leadership	Risk-taking for innovation	**In projects, it is okay to take risks when the goal is to be innovative.** I don't know Strongly disagree Disagree Neutral Agree Strongly agree	**In projects, it is okay to take risks when the goal is to be innovative** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**How often does senior leadership work with strategies aimed at improving employee attitudes to risk taking for innovation?** I don't know Never Rarely Occasionally Frequently Always
Leadership	Tolerance of failure/experimentation	**If something goes wrong in my project because we tested a new technology, way of working, or similar, that is okay.** I don't know Strongly disagree Disagree Neutral Agree Strongly agree	**If something goes wrong in my project because we tested a new technology, way of working, or similar, that is okay** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**How well does senior leadership encourage fast failure and experimentation to employees working in projects?** I don't know Very poorly Poorly Moderately Well Very well
Planning_Portfolio	Diverse project portfolio (risk/time horizon)	**In my department, we have projects with different objectives (such as short-term or long term payback, higher risk or lower risk, improving what we have vs. developing something entirely new, etc.)** I don't know Strongly disagree Disagree Neutral Agree Strongly agree	**In my department, we have projects with different objectives. (such as short-term or long-term payback, higher risk or lower risk, developing what we have vs. developing something entirely new, etc.)** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**To what extent does senior leadership manage a portfolio where projects are compared based on various criteria? (Criteria may include risk versus return, value versus effort, degrees of novelty, types of innovations, technology readiness levels (TRL), etc.)** I don't know Very limited extent Limited extent Neutral Large extent Very large extent
Planning_Collab	Internal & external collaboration planning	**When we want to collaborate with, or need support from internal AND/OR external actors, we know what to do. (Internal: other Saab projects and functions, experts). (External: research groups, universities, companies etc).** I don't know Strongly disagree Disagree Neutral Agree Strongly agree	**We have an established plan for how to involve internal AND/OR external actors who could contribute to projects. (Internal: other Saab projects and functions, experts). (External: research groups, universities, companies etc).** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**To what extent does senior leadership ensure that project teams know how to involve internal AND/OR external actors for collaboration? (Internal: other Saab projects and functions, experts). (External: research groups, universities, companies etc).** I don't know Very limited extent Limited extent Neutral Large extent Very large extent
Support	Time allocated for innovation	**When I propose an innovation activity, enough time is provided to, for example, develop or test it.** I don't know Strongly disagree Disagree	**When my employees plan & propose an innovation activity, enough time is provided to, for example, develop or test it.** I don't know Strongly disagree	**How effectively does senior leadership allocate and protect adequate time across the organization for employees to be innovative?** I don't know Very ineffectively

		Neutral Agree Strongly agree	Disagree Neither agree nor disagree Agree Strongly agree	Ineffectively Neither effectively nor ineffectively Effectively Very effective
Support	Budget allocated for innovation	**When I propose an innovation activity, enough financial budget is provided to, for example, develop or test it.** I don't know Strongly disagree Disagree Neutral Agree Strongly agree	**When my employees plan & propose an innovation activity, enough financial budget is provided to, for example, develop or test it.** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**How effectively does senior leadership allocate and protect adequate financial budget across the organization for employees to be innovative?** I don't know Very ineffectively Ineffectively Neither effectively nor ineffectively Effectively Very effective
Support	Tools & technology availability	**The technology and tools that I need in order to be innovative are readily available. (For example, rapid-prototyping tools, simulation and data-analytics tools, hardware materials)** I don't know Strongly disagree Disagree Neutral Agree Strongly agree	**The technology and tools that my team needs in order to be innovative are readily available (for example, rapid-prototyping tools, simulation and data-analytics tools, hardware materials)** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**How effectively does senior leadership provide and maintain the technology platforms, tools and physical infrastructure that can enable innovation across all business units (for example, rapid-prototyping tools, simulation and data-analytics tools, hardware materials)?** I don't know Very ineffectively Ineffectively Neither effectively nor ineffectively Effectively Very effective
Operations	Identifying innovation insights/opportunities	**We actively seek and identify insights that can be used to generate ideas for innovation. (for example, internal and external trends, customer needs, technological research etc.)** I don't know Strongly disagree Disagree Neutral Agree Strongly agree	**We actively seek and identify insights that can be used to generate ideas for innovation (for example, internal and external trends, customer needs, technological research etc.)** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**To what extent do your employees work with identifying opportunities for innovation that can be used to generate ideas (for example, internal and external trends, customer needs, technological research etc.)?** I don't know Very limited extent Limited extent Neutral Large extent Very large extent
Operations	Validating & testing concepts	**We actively seek to validate and test concepts in order to gain new knowledge about them and reduce the uncertainty of the concepts.** I don't know Strongly disagree Disagree Neutral Agree Strongly agree	**We actively seek to validate and test concepts in order to gain new knowledge about them and reduce the uncertainty of the concepts.** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**To what extent has senior leadership installed processes for testing concepts, which can be used to learn more about innovative concepts to evaluate which to develop further?** I don't know Very limited extent Limited extent Neutral Large extent Very large extent
Operations	Commercialization of solutions	**We take multiple steps to achieve the effective commercialization of solutions, such as implementing production and distribution plans, promoting them to customers, and considering the IP of solutions.** I don't know Strongly disagree Disagree Neutral Agree Strongly agree	**We take multiple steps to achieve the effective commercialization of solutions, such as implementing production and distribution plans, promoting them to customers, and considering the IP of solutions.** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**To what extent does senior leadership take steps to ensure the effective commercialization of solutions, such as implementing production and distribution plans, promoting them to customers, and considering the IP of solutions?** I don't know Very limited extent Limited extent Neutral Large extent Very large extent
Performance	KPIs for innovation/R&D outcomes	**We use defined key performance indicators (KPI) to follow up the**	**We use defined key performance indicators (KPI) to follow up the**	**To what extent does senior leadership use RD and innovation KPIs to follow up**

		outcomes of innovation and RD projects. (e.g. financial, customer value, societal outcomes) I don't know Strongly disagree Disagree Neutral Agree Strongly agree	**outcomes of innovation and R&D projects (e.g. financial, customer value, societal outcomes)** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**innovation outcomes and performance? (e.g. financial, customer value, societal outcomes)** I don't know Very limited extent Limited extent Neutral Large extent Very large extent
Performance	Continuous improvement of R&D processes	**I have experienced that R&D and innovation processes improve/change/adapt continuously** I don't know Strongly disagree Disagree Neutral Agree Strongly agree	**I have experienced that R&D and innovation processes improve/change/adapt continuously** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**To what extent does senior leadership use R&D and innovation KPIs to make improvements to the innovation processes?** I don't know Very limited extent Limited extent Neutral Large extent Very large extent
Improvement	Lessons-learned captured & used	**My team and I are encouraged by senior leaders to record lessons-learned from each project and actively use them to improve and plan for future work.** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**My team and I are encouraged by senior leaders to record lessons-learned from each project and actively use them to improve and plan for future work.** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**To what extent does senior leadership ensure that lessons-learned from innovation projects are captured and used to improve innovation processes?** I don't know Very limited extent Limited extent Neutral Large extent Very large extent
Innov_Performance	Radical innovation	NA	**Saab Surveillance is successful in creating and commercializing innovation from completely new solutions (like projects with low Technology Readiness Level, TRL)** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**Saab Surveillance is successful in creating and commercializing innovation from completely new solutions (like projects with low Technology Readiness Level, TRL)** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree
Innov_Performance	Incremental innovation	NA	**Saab Surveillance is successful in creating and commercializing innovation from projects aiming to improve existing solutions (like projects with high Technology Readiness Level, TRL)** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**Saab Surveillance is successful in creating and commercializing innovation from projects aiming to improve existing solutions (like projects with high Technology Readiness Level, TRL)** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree
Innov_Performance	Customer involvement	NA	**Saab Surveillance involves customers early and uses their feedback to shape solutions that deliver real value.** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**Saab Surveillance involves customers early and uses their feedback to shape solutions that deliver real value.** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree
Innov_Performance	Speed to market	NA	**Saab Surveillance is able to quickly develop new solutions and capabilities, and to quickly bring them to market.** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree	**Saab Surveillance is able to quickly develop new solutions and capabilities, and to quickly bring them to market.** I don't know Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree

11.5. Appendix E: Generative AI Statement

This statement discloses the assistance provided by generative artificial intelligence tools during the preparation of this master's thesis. Generative AI systems were primarily used for linguistic refinement after the author had already drafted all textual content independently. Additionally, the software assisted in developing and debugging R code used for survey data analysis. While the author wrote the initial code, the AI provided support in optimization and error correction. All final decisions regarding content and methodology remain the sole responsibility of the author.

11.6. Appendix F: R Code Used for the Quantitative Data Analysis

This section presents all R code used in the quantitative data analysis in order to maintain transparency.

HARMAN'S SINGLE-FACTOR TEST

```
==========================================================
library(readxl) library(psych)

# ── 1. LOAD & RECODE ──────────────────────────────────────────────────────
df <- read_excel("Innovation-Management-Assessment-V5-Raw-Data.xlsx", sheet = "Values")
recode_likert <- function(x) { x <- suppressWarnings(as.numeric(x)) x[x == 1] <- NA
item_cols <- 11:66 df[, item_cols] <- lapply(df[, item_cols], recode_likert) role_col <- df[[10]]

# ── 2. HARMAN'S SINGLE-FACTOR TEST ────────────────────────────────────────
run_harman <- function(group_data, label) {
group_data <- group_data[, colSums(!is.na(group_data)) > 1, drop = FALSE]
if (ncol(group_data) < 2) { cat(label, ": Not enough columns for Harman test.\n")
return(invisible(NA)) }
cor_mat <- cor(group_data, use = "pairwise.complete.obs")
valid <- rowSums(is.na(cor_mat)) == 0 cor_mat <- cor_mat[valid, valid]
if (nrow(cor_mat) < 2) { cat(label, ": Insufficient valid correlations — skipping.\n")
return(invisible(NA)) }
pca <- principal(cor_mat, nfactors = 1, rotate = "none", n.obs = nrow(group_data)) pct <-
round(pca$values[1] / sum(pca$values) * 100, 1)
cat(sprintf("%-30s first factor = %5.1f%%", label, pct)) if (pct < 50) cat(" → CMB not a primary
concern.\n") else cat(" → CMB may be present — interpret with caution.\n")
return(invisible(pct)) }
cat("=== Harman's Single-Factor Test ===\n") run_harman(df[role_col == 2, 27:46], "Middle
Managers (n=11):") run_harman(df[role_col == 3, 47:66], "Senior Leadership (n=7):") cat("\n")
```

CRONBACH ALPHA

```
==========================================================
library(readxl)
library(psych)
library(dplyr)

# ── 1. LOAD & RECODE ──────────────────────────────────────────────────────
df <- read_excel("Innovation-Management-Assessment-V5-Raw-Data.xlsx",
                 sheet = "Values")
recode_likert <- function(x) {
  x <- suppressWarnings(as.numeric(x))
  x[x == 1] <- NA
  x <- x - 1
  return(x)
}
item_cols <- 11:66
df[, item_cols] <- lapply(df[, item_cols], recode_likert)
role_col <- df[[10]]
```

```
# ── 2. PARALLEL ITEM MAP (OE / MM / SL) ──────────────────────────────
parallel_map <- list(
  Context = list(
    item1 = list(OE = 11, MM = 31, SL = 51),
    item2 = list(OE = 12, MM = 32, SL = 52)
  ),
  Leadership = list(
    item1 = list(OE = 13, MM = 33, SL = 53),
    item2 = list(OE = 14, MM = 34, SL = 54),
    item3 = list(OE = 15, MM = 35, SL = 55)
  ),
  Planning_Portfolio = list(
    item1 = list(OE = 16, MM = 36, SL = 56)
  ),
  Planning_Collab = list(
    item1 = list(OE = 17, MM = 37, SL = 57)
  ),
  Support = list(
    item1 = list(OE = 18, MM = 38, SL = 58),
    item2 = list(OE = 19, MM = 39, SL = 59),
    item3 = list(OE = 20, MM = 40, SL = 60)
  ),
  Operations = list(
    item1 = list(OE = 21, MM = 41, SL = 61),
    item2 = list(OE = 22, MM = 42, SL = 62),
    item3 = list(OE = 23, MM = 43, SL = 63)
  ),
  Performance = list(
    item1 = list(OE = 24, MM = 44, SL = 64),
    item2 = list(OE = 25, MM = 45, SL = 65)
  ),
  Improvement = list(
    item1 = list(OE = 26, MM = 46, SL = 66)
  )
)

role_labels <- c("OE", "MM", "SL")
role_nums   <- c(OE = 1, MM = 2, SL = 3)

# ── 3. HELPER: BUILD STACKED ITEM MATRIX FOR ONE AREA ────────────────
build_matrix <- function(area_items) {
  n_respondents <- nrow(df)
  n_items       <- length(area_items)
  mat <- matrix(NA_real_, nrow = n_respondents, ncol = n_items)

  for (j in seq_len(n_items)) {
    item_def <- area_items[[j]]
    for (role in role_labels) {
      role_rows     <- which(role_col == role_nums[role])
      mat[role_rows, j] <- as.numeric(df[[item_def[[role]]]][role_rows])
    }
  }
  colnames(mat) <- paste0("item", seq_len(n_items))
  as.data.frame(mat)
```

```
}

# ── 4. COMPUTE ALPHA PER AREA ──────────────────────────────────────────
cat("=== Cross-Group Cronbach's Alpha (n=39, stacked parallel items) ===\n")
cat("Assumption: items are parallel measures asked from different role perspectives\n\n")

alpha_results <- data.frame()

for (area_name in names(parallel_map)) {
  items   <- parallel_map[[area_name]]
  n_items <- length(items)

  if (n_items < 2) {
    cat(sprintf("%-20s : Single item — alpha N/A\n", area_name))
    alpha_results <- rbind(alpha_results, data.frame(
      ISO_Area    = area_name,
      n_items     = n_items,
      n_complete  = NA,
      Alpha       = NA,
      Note        = "Single item"
    ))
    next
  }

  mat       <- build_matrix(items)
  valid_mat  <- na.omit(mat)
  n_complete <- nrow(valid_mat)

  if (n_complete < 3) {
    cat(sprintf("%-20s : Insufficient complete cases (n=%d)\n",
                area_name, n_complete))
    next
  }

  has_variance <- sapply(valid_mat, function(x) length(unique(x)) > 1)
  valid_mat    <- valid_mat[, has_variance, drop = FALSE]

  if (ncol(valid_mat) < 2) {
    cat(sprintf("%-20s : Not enough items with variance\n", area_name))
    next
  }

  a_res <- alpha(valid_mat, check.keys = TRUE)
  a_val <- round(a_res$total$raw_alpha, 3)

  quality <- ifelse(a_val >= 0.70, "Acceptable",
                ifelse(a_val >= 0.60, "Questionable",
                       ifelse(a_val >= 0.50, "Poor", "Unacceptable")))

  cat(sprintf("%-20s : Alpha = %.3f  (n=%d, %d items)  [%s]\n",
              area_name, a_val, n_complete, ncol(valid_mat), quality))

  alpha_results <- rbind(alpha_results, data.frame(
    ISO_Area   = area_name,
    n_items    = ncol(valid_mat),
```

```
    n_complete = n_complete,
    Alpha      = a_val,
    Note       = quality
  ))
}

# ── 5. EXPORT ──────────────────────────────────────────────────────────────
write.csv(alpha_results, "crossgroup_alpha.csv", row.names = FALSE)
cat("\nSaved: crossgroup_alpha.csv\n")
```

DESCRIPTIVE DATA – DETERMINING ISO 56001 ADHERENCE THROUGH MEAN VALUES

```
==============================================================
library(readxl)
library(dplyr)

# ── 1. LOAD & RECODE ──────────────────────────────────────────────────────
df <- read_excel("Innovation-Management-Assessment-V5-Raw-Data.xlsx",
                 sheet = "Values")

recode_likert <- function(x) {
  x <- suppressWarnings(as.numeric(x))
  x[x == 1] <- NA
  x <- x - 1
  return(x)
}

df[, 11:ncol(df)] <- lapply(df[, 11:ncol(df)], recode_likert)
role_col <- as.numeric(df[[10]])

oe_mask <- role_col == 1
mm_mask <- role_col == 2
sl_mask <- role_col == 3

# ── 2. ITEM DEFINITIONS ──────────────────────────────────────────────────

items <- list(
  list(area = "Context", label = "External trends (political/market/tech)",
       oe = 11, mm = 31, sl = 51),
  list(area = "Context", label = "Internal factors (people/culture/structure)",
       oe = 12, mm = 32, sl = 52),

  list(area = "Leadership", label = "Culture of new ideas / psychological safety",
       oe = 13, mm = 33, sl = 53),
  list(area = "Leadership", label = "Risk-taking for innovation",
       oe = 14, mm = 34, sl = 54),
  list(area = "Leadership", label = "Tolerance of failure/experimentation",
       oe = 15, mm = 35, sl = 55),

  list(area = "Planning_Portfolio", label = "Diverse project portfolio (risk/time horizon)",
       oe = 16, mm = 36, sl = 56),
  list(area = "Planning_Collab",    label = "Internal & external collaboration planning",
       oe = 17, mm = 37, sl = 57),
```

```
  list(area = "Support", label = "Time allocated for innovation",
       oe = 18, mm = 38, sl = 58),
  list(area = "Support", label = "Budget allocated for innovation",
       oe = 19, mm = 39, sl = 59),
  list(area = "Support", label = "Tools & technology availability",
       oe = 20, mm = 40, sl = 60),

  list(area = "Operations", label = "Identifying innovation insights/opportunities",
       oe = 21, mm = 41, sl = 61),
  list(area = "Operations", label = "Validating & testing concepts",
       oe = 22, mm = 42, sl = 62),
  list(area = "Operations", label = "Commercialisation of solutions",
       oe = 23, mm = 43, sl = 63),


  list(area = "Performance", label = "KPIs for innovation/R&D outcomes",
       oe = 24, mm = 44, sl = 64),
  list(area = "Performance", label = "Continuous improvement of R&D processes",
       oe = 25, mm = 45, sl = 65),

  list(area = "Improvement", label = "Lessons-learned captured & used",
       oe = 26, mm = 46, sl = 66),

  # Innovation Performance - MM + SL only, not asked of OE
  list(area = "Innov_Performance", label = "Radical innovation",
       oe = NA, mm = 27, sl = 47),
  list(area = "Innov_Performance", label = "Incremental innovation",
       oe = NA, mm = 28, sl = 48),
  list(area = "Innov_Performance", label = "Customer involvement",
       oe = NA, mm = 29, sl = 49),
  list(area = "Innov_Performance", label = "Speed to market",
       oe = NA, mm = 30, sl = 50)
)

# ── 3. COMPUTE ITEM-LEVEL DESCRIPTIVES ──────────────────────────────
desc_row <- function(item) {
  get_vals <- function(mask, col) {
    if (is.na(col)) return(numeric(0))
    na.omit(as.numeric(df[[col]][mask]))
  }

  oe_v  <- get_vals(oe_mask, item$oe)
  mm_v  <- get_vals(mm_mask, item$mm)
  sl_v  <- get_vals(sl_mask, item$sl)
  all_v <- c(oe_v, mm_v, sl_v)

  fmt <- function(v) {
    if (length(v) == 0) return(c(n = NA, M = NA, SD = NA))
    c(n  = length(v),
      M  = round(mean(v), 2),
      SD = round(sd(v),   2))
  }

  oe_s  <- fmt(oe_v)
  mm_s  <- fmt(mm_v)
```

```
  sl_s  <- fmt(sl_v)
  all_s <- fmt(all_v)

  data.frame(
    ISO_Area   = item$area,
    Item_Label = item$label,
    OE_n  = oe_s["n"],  OE_M  = oe_s["M"],  OE_SD  = oe_s["SD"],
    MM_n  = mm_s["n"],  MM_M  = mm_s["M"],  MM_SD  = mm_s["SD"],
    SL_n  = sl_s["n"],  SL_M  = sl_s["M"],  SL_SD  = sl_s["SD"],
    All_n = all_s["n"], All_M = all_s["M"],  All_SD = all_s["SD"],
    stringsAsFactors = FALSE, row.names = NULL
  )
}

results <- do.call(rbind, lapply(items, desc_row))
rownames(results) <- NULL

# ── 4. COMPUTE AREA-LEVEL DESCRIPTIVES (SUMPRODUCT) ──────────────

area_row <- function(area_name, area_items_df) {

  sumproduct_M <- function(m_col, n_col) {
    m_vals <- area_items_df[[m_col]]
    n_vals <- area_items_df[[n_col]]
    valid  <- !is.na(m_vals) & !is.na(n_vals)
    if (sum(valid) == 0) return(NA)
    round(sum(m_vals[valid] * n_vals[valid]) / sum(n_vals[valid]), 2)
  }

  sum_n <- function(n_col) {
    vals <- na.omit(area_items_df[[n_col]])
    if (length(vals) == 0) return(NA)
    sum(vals)
  }

  data.frame(
    ISO_Area   = area_name,
    Item_Label = "(area average)",
    OE_n  = sum_n("OE_n"),  OE_M  = sumproduct_M("OE_M", "OE_n"),  OE_SD  = NA,
    MM_n  = sum_n("MM_n"),  MM_M  = sumproduct_M("MM_M", "MM_n"),  MM_SD  = NA,
    SL_n  = sum_n("SL_n"),  SL_M  = sumproduct_M("SL_M", "SL_n"),  SL_SD  = NA,
    All_n = sum_n("All_n"), All_M = sumproduct_M("All_M", "All_n"), All_SD = NA,
    stringsAsFactors = FALSE, row.names = NULL
  )
}

areas <- unique(sapply(items, `[[`, "area"))
area_results <- do.call(rbind, lapply(areas, function(a) {
  area_items_df <- results[results$ISO_Area == a, ]
  area_row(a, area_items_df)
}))
rownames(area_results) <- NULL

# ── 5. PRINT & EXPORT ────────────────────────────────────────────
cat("=== ITEM-LEVEL DESCRIPTIVES ===\n")
```

```
print(results, row.names = FALSE)
write.csv(results, "item_descriptives.csv", row.names = FALSE)

cat("\n=== AREA-LEVEL DESCRIPTIVES ===\n")
print(area_results, row.names = FALSE)
write.csv(area_results, "area_descriptives.csv", row.names = FALSE)

cat("\nSaved: item_descriptives.csv and area_descriptives.csv\n")
cat("Total items:", nrow(results), "\n")
cat("Total areas:", nrow(area_results), "\n")
```

SPEARMAN AND PEARSON CORRELATION TESTS – EXTENDED SAMPLE

==========

```
library(readxl)
library(dplyr)

# ── 1. LOAD & RECODE ──────────────────────────────────────────────────
df <- read_excel("Innovation-Management-Assessment-V5-Raw-Data.xlsx",
                 sheet = "Values")

recode_likert <- function(x) {
  x <- suppressWarnings(as.numeric(x))
  x[x == 1] <- NA
  x <- x - 1
  return(x)
}

item_cols <- 11:66
df[, item_cols] <- lapply(df[, item_cols], recode_likert)
role_col <- df[[10]]

# ── 2. CLASSIFY DEPARTMENTS ──────────────────────────────────────────
df$dept_num  <- df[[5]]
df$dept_text <- ifelse(is.na(df[[6]]), "", as.character(df[[6]]))

df$dept <- mapply(function(num, text) {
  if (is.na(num)) return(NA_character_)
  if (num == 7) {
    if (grepl("design",       tolower(text))) return("7_design")
    if (grepl("surveillance", tolower(text))) return("7_common_surv")
    return("7_other")
  }
  return(as.character(as.integer(num)))
}, df$dept_num, df$dept_text)

# ── 3. COLUMN MAPS ────────────────────────────────────────────────────

col_map_validity <- list(
  Operational = list(
    Context     = c(11, 12),
    Leadership  = c(13, 14, 15),
    Planning    = c(16, 17),
```

```
    Support     = c(18, 19, 20),
    Operations  = c(21, 22, 23),
    Performance = c(24, 25),
    Improvement = c(26)
  )
)

col_map <- list(
  Operational = list(
    Context           = c(11, 12),
    Leadership        = c(13, 14, 15),
    Planning_Portfolio = c(16),
    Planning_Collab   = c(17),
    Support           = c(18, 19, 20),
    Operations        = c(21, 22, 23),
    Performance       = c(24, 25),
    Improvement       = c(26)
  ),
  MM = list(
    Context           = c(31, 32),
    Leadership        = c(33, 34, 35),
    Planning_Portfolio = c(36),
    Planning_Collab   = c(37),
    Support           = c(38, 39, 40),
    Operations        = c(41, 42, 43),
    Performance       = c(44, 45),
    Improvement       = c(46)
  ),
  SL = list(
    Context           = c(51, 52),
    Leadership        = c(53, 54, 55),
    Planning_Portfolio = c(56),
    Planning_Collab   = c(57),
    Support           = c(58, 59, 60),
    Operations        = c(61, 62, 63),
    Performance       = c(64, 65),
    Improvement       = c(66)
  )
)

iso_areas <- c("Context", "Leadership",
               "Planning_Portfolio", "Planning_Collab",
               "Support", "Operations", "Performance", "Improvement")

perf_cols <- list(
  MM = list(Radical = 27, Incremental = 28, Customer = 29, Speed = 30),
  SL = list(Radical = 47, Incremental = 48, Customer = 49, Speed = 50)
)

perf_outcomes <- c("Radical", "Incremental", "Customer", "Speed")

# ── 4. DEPT-LEVEL ATTRIBUTED PERFORMANCE ──────────────────────────
valid_depts <- c("2", "7_design")
dept_perf   <- list()
```

```
for (dept in valid_depts) {
  dept_mask        <- !is.na(df$dept) & df$dept == dept
  dept_perf[[dept]] <- list()
  for (outcome in perf_outcomes) {
    mm_vals  <- df[[perf_cols$MM[[outcome]]]][dept_mask & role_col == 2]
    sl_vals  <- df[[perf_cols$SL[[outcome]]]][dept_mask & role_col == 3]
    all_vals <- na.omit(c(mm_vals, sl_vals))
    dept_perf[[dept]][[outcome]]              <- if (length(all_vals) > 0) mean(all_vals) else NA_real_
    dept_perf[[dept]][[paste0(outcome, "_n")]] <- length(all_vals)
  }
}

cat("=== Attributed performance scores by department ===\n")
for (dept in valid_depts) {
  cat(sprintf("Dept %s:\n", dept))
  for (outcome in perf_outcomes) {
    cat(sprintf("  %-12s M = %.3f  (n = %d managers)\n",
                outcome,
                dept_perf[[dept]][[outcome]],
                dept_perf[[dept]][[paste0(outcome, "_n")]]))
  }
}
cat("\n")

# ── 5. OE INCLUSION ──────
oe_mask       <- role_col == 1
oe_row_indices <- which(oe_mask)

oe_areas_valid <- sapply(oe_row_indices, function(i) {
  sum(sapply(col_map_validity$Operational, function(cols) {
    any(!is.na(as.numeric(df[i, cols])))
  }))
})

oe_include_logical <- df$dept[oe_mask] %in% valid_depts & oe_areas_valid == 7
oe_include_idx     <- oe_row_indices[oe_include_logical]

cat("=== OE inclusion summary ===\n")
cat(sprintf("Total OEs in dataset: %d\n", sum(oe_mask)))
cat(sprintf("OEs in matchable depts (2, 7_design): %d\n",
            sum(oe_mask & df$dept %in% valid_depts)))
cat(sprintf("OEs excluded: %d\n",
            sum(oe_mask & df$dept %in% valid_depts) - length(oe_include_idx)))
cat(sprintf("OEs included in analysis: %d\n\n", length(oe_include_idx)))

oe_rows <- lapply(oe_include_idx, function(i) {
  dept_i     <- df$dept[i]
  iso_scores <- sapply(iso_areas, function(area) {
    vals <- as.numeric(df[i, col_map$Operational[[area]]])
    if (all(is.na(vals))) NA_real_ else mean(vals, na.rm = TRUE)
  })
  perf_scores <- sapply(perf_outcomes, function(o) dept_perf[[dept_i]][[o]])
  cbind(
    data.frame(role_group  = "Operational",
               dept        = dept_i,
```

```
            perf_source = "Attributed (unit MM+SL mean)",
            stringsAsFactors = FALSE),
   as.data.frame(t(iso_scores)),
   as.data.frame(t(setNames(perf_scores, perf_outcomes)))
  )
})

oe_df <- do.call(rbind, oe_rows)

# ── 6. MM + SL SUBSET ──────────────────────────────────────────────────
mm_sl_rows <- lapply(which(role_col %in% c(2, 3)), function(i) {
 role_i <- role_col[i]
 grp    <- if (role_i == 2) "MM" else "SL"
 iso_scores <- sapply(iso_areas, function(area) {
  vals <- as.numeric(df[i, col_map[[grp]][[area]]])
  if (all(is.na(vals))) NA_real_ else mean(vals, na.rm = TRUE)
 })
 perf_scores <- sapply(perf_outcomes, function(o) {
  as.numeric(df[[perf_cols[[grp]][[o]]]][i])
 })
 cbind(
  data.frame(role_group  = if (role_i == 2) "Middle Management"
            else "Senior Leadership",
            dept        = df$dept[i],
            perf_source = "Own response",
            stringsAsFactors = FALSE),
  as.data.frame(t(iso_scores)),
  as.data.frame(t(setNames(perf_scores, perf_outcomes)))
 )
})

mm_sl_df <- do.call(rbind, mm_sl_rows)

# ── 7. COMBINE──────────────────────────────────────────────────────────
analysis_df      <- rbind(oe_df, mm_sl_df)
rownames(analysis_df) <- NULL

n_oe <- sum(analysis_df$role_group == "Operational")
n_mm <- sum(analysis_df$role_group == "Middle Management")
n_sl <- sum(analysis_df$role_group == "Senior Leadership")

cat(sprintf(
 "Analysis sample: %d OE (attributed) + %d MM + %d SL = %d total\n\n",
 n_oe, n_mm, n_sl, nrow(analysis_df)
))

# ── 8. OVERALL IMS COMPOSITE ──────────────────────────────────────────
analysis_df$Planning_Combined <- rowMeans(
 analysis_df[, c("Planning_Portfolio", "Planning_Collab")], na.rm = TRUE
)
analysis_df$Planning_Combined[is.nan(analysis_df$Planning_Combined)] <- NA

ims_areas_7 <- c("Context", "Leadership", "Planning_Combined",
                 "Support", "Operations", "Performance", "Improvement")
```

```
analysis_df$Overall_IMS <- rowMeans(analysis_df[, ims_areas_7], na.rm = TRUE)
analysis_df$Overall_IMS[is.nan(analysis_df$Overall_IMS)] <- NA

# ── 9. ISO AREAS vs PERFORMANCE ──────────────────────────────────────────
cat("=== ISO Sub-Area vs Innovation Performance ===\n\n")

results_iso <- data.frame()

for (dv in perf_outcomes) {
  for (pred in iso_areas) {
    x     <- analysis_df[[pred]]
    y     <- analysis_df[[dv]]
    valid <- complete.cases(x, y)
    n_v   <- sum(valid)

    if (n_v < 4) {
      results_iso <- rbind(results_iso, data.frame(
        DV = dv, Predictor = pred, n = n_v,
        Spearman_r = NA, p_spearman = NA,
        Pearson_r  = NA, p_pearson  = NA,
        Significance = "Insufficient data",
        stringsAsFactors = FALSE
      ))
      next
    }

    sp  <- cor.test(x[valid], y[valid], method = "spearman", exact = FALSE)
    pe  <- cor.test(x[valid], y[valid], method = "pearson")
    sig <- ifelse(sp$p.value < 0.01 & pe$p.value < 0.01, "** p<0.01",
                  ifelse(sp$p.value < 0.05 & pe$p.value < 0.05, "**Both sig (p<0.05)",
                         ifelse(sp$p.value < 0.05, "* Spearman only (p<0.05)", "ns")))

    results_iso <- rbind(results_iso, data.frame(
      DV           = dv,
      Predictor    = pred,
      n            = n_v,
      Spearman_r   = round(sp$estimate, 3),
      p_spearman   = round(sp$p.value,  4),
      Pearson_r    = round(pe$estimate, 3),
      p_pearson    = round(pe$p.value,  4),
      Significance = sig,
      stringsAsFactors = FALSE
    ))
  }
}

print(results_iso, row.names = FALSE)

# ── 10. OVERALL IMS vs PERFORMANCE ──────────────────────────────────────
cat("\n=== Overall IMS vs Innovation Performance ===\n\n")

results_ims <- data.frame()

for (dv in perf_outcomes) {
  x     <- analysis_df$Overall_IMS
```

```
  y     <- analysis_df[[dv]]
  valid <- complete.cases(x, y)
  n_v   <- sum(valid)
  if (n_v < 4) next

  sp  <- cor.test(x[valid], y[valid], method = "spearman", exact = FALSE)
  pe  <- cor.test(x[valid], y[valid], method = "pearson")
  sig <- ifelse(sp$p.value < 0.05 & pe$p.value < 0.05,
                "** Both sig (p<0.05)",
                ifelse(sp$p.value < 0.05,
                       "*  Spearman only (p<0.05)", "ns"))

  results_ims <- rbind(results_ims, data.frame(
    Predictor    = "Overall_IMS",
    DV           = dv,
    n            = n_v,
    Spearman_r   = round(sp$estimate, 3),
    p_spearman   = round(sp$p.value,  4),
    Pearson_r    = round(pe$estimate, 3),
    p_pearson    = round(pe$p.value,  4),
    Significance = sig,
    stringsAsFactors = FALSE
  ))
}

print(results_ims, row.names = FALSE)

# ── 11. EXPORT ──────────────────────────────────────────────────────────
results_iso$Analysis <- "ISO_Area_vs_Performance"
results_ims$Analysis <- "Overall_IMS_vs_Performance"

combined_results <- rbind(
  results_iso[, c("Analysis", "DV", "Predictor", "n",
                  "Spearman_r", "p_spearman",
                  "Pearson_r",  "p_pearson", "Significance")],
  results_ims[, c("Analysis", "DV", "Predictor", "n",
                  "Spearman_r", "p_spearman",
                  "Pearson_r",  "p_pearson", "Significance")]
)

write.csv(combined_results,
          "extended_all_correlations_planning_split.csv",
          row.names = FALSE)

cat(sprintf("\nSaved: extended_all_correlations_planning_split.csv (%d rows)\n",
            nrow(combined_results)))
```

SPEARMAN AND PEARSON CORRELATION TESTS – LIMITED SAMPLE

==========

```
library(readxl)
library(psych)
library(dplyr)
```

```
# ── 1. LOAD & RECODE ──────────────────────────────────────────────────────
df <- read_excel("Innovation-Management-Assessment-V5-Raw-Data.xlsx",
                 sheet = "Values")

recode_likert <- function(x) {
  x <- suppressWarnings(as.numeric(x))
  x[x == 1] <- NA
  x <- x - 1
  return(x)
}

item_cols <- 11:66
df[, item_cols] <- lapply(df[, item_cols], recode_likert)
role_col <- df[[10]]

# ── 2. ISO COMPOSITE MAP (MM + SL, Planning split) ────────────────────────
iso_cols <- list(
  MM = list(
    Context           = c(31, 32),
    Leadership        = c(33, 34, 35),
    Planning_Portfolio = c(36),
    Planning_Collab   = c(37),
    Support           = c(38, 39, 40),
    Operations        = c(41, 42, 43),
    Performance       = c(44, 45),
    Improvement       = c(46)
  ),
  SL = list(
    Context           = c(51, 52),
    Leadership        = c(53, 54, 55),
    Planning_Portfolio = c(56),
    Planning_Collab   = c(57),
    Support           = c(58, 59, 60),
    Operations        = c(61, 62, 63),
    Performance       = c(64, 65),
    Improvement       = c(66)
  )
)

areas <- c("Context", "Leadership", "Planning_Portfolio", "Planning_Collab",
           "Support", "Operations", "Performance", "Improvement")

# ── 3. BUILD ANALYSIS DATAFRAME (MM + SL) ─────────────────────────────────
mm_sl_mask <- role_col %in% c(2, 3)
mm_mask    <- role_col == 2
sl_mask    <- role_col == 3

analysis_df <- data.frame(role = role_col[mm_sl_mask])

mm_rows <- which(mm_sl_mask)[role_col[mm_sl_mask] == 2]
sl_rows <- which(mm_sl_mask)[role_col[mm_sl_mask] == 3]
mm_idx  <- which(role_col[mm_sl_mask] == 2)
sl_idx  <- which(role_col[mm_sl_mask] == 3)

for (area in areas) {
```

```
  scores <- rep(NA_real_, sum(mm_sl_mask))
  mm_data <- df[mm_rows, iso_cols$MM[[area]], drop = FALSE]
  sl_data <- df[sl_rows, iso_cols$SL[[area]], drop = FALSE]
  scores[mm_idx] <- rowMeans(mm_data, na.rm = TRUE)
  scores[sl_idx] <- rowMeans(sl_data, na.rm = TRUE)
  scores[is.nan(scores)] <- NA
  analysis_df[[area]] <- scores
}

# ── 4. DEPENDENT VARIABLES ──────────────────────────────────────────────
dv_names   <- c("Radical_Innovation", "Incremental_Innovation",
                "Customer_Involvement", "Speed_to_Market")
mm_dv_cols <- c(27, 28, 29, 30)
sl_dv_cols <- c(47, 48, 49, 50)

mm_rows_full <- which(role_col == 2)
sl_rows_full <- which(role_col == 3)

for (i in seq_along(dv_names)) {
  vec <- rep(NA_real_, sum(mm_sl_mask))
  vec[mm_idx] <- df[[mm_dv_cols[i]]][mm_rows_full]
  vec[sl_idx] <- df[[sl_dv_cols[i]]][sl_rows_full]
  analysis_df[[dv_names[i]]] <- vec
}

analysis_df$Role_Dummy <- ifelse(analysis_df$role == 3, 1, 0)

cat("MM n =", sum(analysis_df$role == 2),
    "| SL n =", sum(analysis_df$role == 3),
    "| Total =", nrow(analysis_df), "\n\n")

# ── 5. SPEARMAN + PEARSON CORRELATIONS ──────────────────────────────────
predictors <- c(areas, "Role_Dummy")
results    <- data.frame()

for (dv in dv_names) {
  for (pred in predictors) {
    x     <- analysis_df[[pred]]
    y     <- analysis_df[[dv]]
    valid <- complete.cases(x, y)
    n_v   <- sum(valid)
    if (n_v < 4) {
      results <- rbind(results, data.frame(
        DV = dv, Predictor = pred, n = n_v,
        Spearman_r = NA, p_value = NA,
        Pearson_r = NA, p_value_pearson = NA,
        Significance = "Insufficient data"
      ))
      next
    }
    sp  <- cor.test(x[valid], y[valid], method = "spearman", exact = FALSE)
    pe  <- cor.test(x[valid], y[valid], method = "pearson")
    sig <- ifelse(sp$p.value < 0.01, "** p<0.01",
                  ifelse(sp$p.value < 0.05, "* p<0.05", "ns"))
    results <- rbind(results, data.frame(
```

```
      DV             = dv,
      Predictor      = pred,
      n              = n_v,
      Spearman_r     = round(sp$estimate, 3),
      p_value        = round(sp$p.value, 4),
      Pearson_r      = round(pe$estimate, 3),
      p_value_pearson = round(pe$p.value, 4),
      Significance   = sig
    ))
  }
}

cat("=== Correlations: ISO Areas vs Innovation Performance ===\n")
print(results, row.names = FALSE)

write.csv(results, "correlation_results_planning_split.csv", row.names = FALSE)
cat("\nSaved: correlation_results_planning_split.csv\n")

# ── 6. OVERALL IMS vs PERFORMANCE (Planning split excluded from IMS) ─
analysis_df$Planning_Combined <- rowMeans(
  analysis_df[, c("Planning_Portfolio", "Planning_Collab")], na.rm = TRUE)
analysis_df$Planning_Combined[is.nan(analysis_df$Planning_Combined)] <- NA

ims_areas <- c("Context", "Leadership", "Planning_Combined",
               "Support", "Operations", "Performance", "Improvement")
analysis_df$Overall_IMS <- rowMeans(analysis_df[, ims_areas], na.rm = TRUE)
analysis_df$Overall_IMS[is.nan(analysis_df$Overall_IMS)] <- NA

ims_results <- data.frame()
for (dv in dv_names) {
  x     <- analysis_df$Overall_IMS
  y     <- analysis_df[[dv]]
  valid <- complete.cases(x, y)
  n_v   <- sum(valid)
  if (n_v < 4) next
  sp  <- cor.test(x[valid], y[valid], method = "spearman", exact = FALSE)
  pe  <- cor.test(x[valid], y[valid], method = "pearson")
  sig <- ifelse(sp$p.value < 0.01, "** p<0.01",
                ifelse(sp$p.value < 0.05, "* p<0.05", "ns"))
  ims_results <- rbind(ims_results, data.frame(
    Predictor = "Overall_IMS", DV = dv, n = n_v,
    Spearman_r = round(sp$estimate, 3), p_value = round(sp$p.value, 4),
    Pearson_r = round(pe$estimate, 3), p_value_pearson = round(pe$p.value, 4),
    Significance = sig
  ))
}
cat("\n=== Overall IMS vs Performance ===\n")
print(ims_results, row.names = FALSE)
write.csv(ims_results, "ims_correlations_planning_split.csv", row.names = FALSE)
cat("Saved: ims_correlations_planning_split.csv\n")
```

ROLE DEPENDENT WELCH'S T-TEST VS. ISO56000 DIMENSION

==========

```
library(readxl)
library(dplyr)

# ── 1. LOAD & RECODE ──────────────────────────────────────────────────────
df <- read_excel("Innovation-Management-Assessment-V5-Raw-Data.xlsx", sheet = "Values")

recode_likert <- function(x) {
  x <- suppressWarnings(as.numeric(x))
  x[x == 1] <- NA
  x <- x - 1
  return(x)
}

item_cols <- 11:66
df[, item_cols] <- lapply(df[, item_cols], recode_likert)
role_col <- df[[10]]

# ── 2. COLUMN INDEX MAP ──────────────────────────────────────────────────
op <- list(
  Context           = c(11, 12),
  Leadership        = c(13, 14, 15),
  Planning_Portfolio  = c(16),
  Planning_Collab     = c(17),
  Support           = c(18, 19, 20),
  Operations        = c(21, 22, 23),
  Performance       = c(24, 25),
  Improvement       = c(26)
)

mm <- list(
  Innovation        = c(27, 28, 29, 30),
  Context           = c(31, 32),
  Leadership        = c(33, 34, 35),
  Planning_Portfolio  = c(36),
  Planning_Collab     = c(37),
  Support           = c(38, 39, 40),
  Operations        = c(41, 42, 43),
  Performance       = c(44, 45),
  Improvement       = c(46)
)

sl <- list(
  Innovation        = c(47, 48, 49, 50),
  Context           = c(51, 52),
  Leadership        = c(53, 54, 55),
  Planning_Portfolio  = c(56),
  Planning_Collab     = c(57),
  Support           = c(58, 59, 60),
  Operations        = c(61, 62, 63),
  Performance       = c(64, 65),
  Improvement       = c(66)
)

col_map <- list(
```

```
  "Operational"       = op,
  "Middle Management"  = mm,
  "Senior Leadership"  = sl
)

role_values <- list(
  "Operational"       = 1,
  "Middle Management"  = 2,
  "Senior Leadership"  = 3
)

# ── 3. ITEM LABELS ──────────────────────────────────────────────────────────
item_labels <- list(
  Context           = c("External trends (political/market/tech)",
                        "Internal factors (people/culture/structure)"),
  Leadership        = c("Culture of new ideas / psychological safety",
                        "Risk-taking for innovation",
                        "Tolerance of failure/experimentation"),
  Planning_Portfolio = c("Diverse project portfolio (risk/time horizon)"),
  Planning_Collab    = c("Internal & external collaboration planning"),
  Support           = c("Time allocated for innovation",
                        "Budget allocated for innovation",
                        "Tools & technology availability"),
  Operations        = c("Identifying innovation insights/opportunities",
                        "Validating & testing concepts",
                        "Commercialisation of solutions"),
  Performance       = c("KPIs for innovation/R&D outcomes",
                        "Continuous improvement of R&D processes"),
  Improvement       = c("Lessons-learned captured & used"),
  Innovation        = c("New solutions / low TRL commercialisation",
                        "Existing solutions / high TRL improvement",
                        "Customer involvement & feedback",
                        "Speed to market")
)

# ── 4. HELPER FUNCTION──────────────────────────────────────────────────────
run_ttest <- function(vec1, vec2, test_type, area, item_num,
                      item_label, g1_name, g2_name) {
  vec1 <- vec1[!is.na(vec1)]
  vec2 <- vec2[!is.na(vec2)]
  if (length(vec1) < 2 | length(vec2) < 2) {
    return(data.frame(
      Test_Type = test_type, ISO_Area = area,
      Item_Number = as.character(item_num), Item_Label = item_label,
      Group_1 = g1_name, Group_2 = g2_name,
      n_Group1 = length(vec1), n_Group2 = length(vec2),
      Mean_Group1 = NA, Mean_Group2 = NA,
      SD_Group1 = NA, SD_Group2 = NA,
      t_statistic = NA, p_value_raw = NA
    ))
  }
  result <- t.test(vec1, vec2, var.equal = FALSE)
  data.frame(
    Test_Type    = test_type,
    ISO_Area     = area,
```

```
    Item_Number  = as.character(item_num),
    Item_Label   = item_label,
    Group_1      = g1_name,
    Group_2      = g2_name,
    n_Group1     = length(vec1),
    n_Group2     = length(vec2),
    Mean_Group1  = round(mean(vec1), 3),
    Mean_Group2  = round(mean(vec2), 3),
    SD_Group1    = round(sd(vec1), 3),
    SD_Group2    = round(sd(vec2), 3),
    t_statistic  = round(result$statistic, 3),
    p_value_raw  = round(result$p.value, 4)
  )
}

# ── 5. AREA LISTS ──────────────────────────────────────────────────────────
all_areas <- c("Innovation", "Context", "Leadership",
               "Planning_Portfolio", "Planning_Collab",
               "Support", "Operations", "Performance", "Improvement")

group_areas <- list(
  "Operational"       = c("Context","Leadership","Planning_Portfolio",
                          "Planning_Collab","Support","Operations",
                          "Performance","Improvement"),
  "Middle Management" = c("Innovation","Context","Leadership",
                          "Planning_Portfolio","Planning_Collab",
                          "Support","Operations","Performance","Improvement"),
  "Senior Leadership" = c("Innovation","Context","Leadership",
                          "Planning_Portfolio","Planning_Collab",
                          "Support","Operations","Performance","Improvement")
)

pairs <- list(
  c("Operational",       "Middle Management"),
  c("Operational",       "Senior Leadership"),
  c("Middle Management", "Senior Leadership")
)

# ── 6. ITEM-LEVEL TESTS ────────────────────────────────────────────────────
item_rows <- list()

for (area in all_areas) {
  for (pair in pairs) {
    g1 <- pair[1]; g2 <- pair[2]
    if (!(area %in% group_areas[[g1]]) | !(area %in% group_areas[[g2]])) next
    idx_g1  <- col_map[[g1]][[area]]
    idx_g2  <- col_map[[g2]][[area]]
    n_items <- min(length(idx_g1), length(idx_g2))
    mask1   <- role_col == role_values[[g1]]
    mask2   <- role_col == role_values[[g2]]
    for (i in seq_len(n_items)) {
      v1    <- df[[idx_g1[i]]][mask1]
      v2    <- df[[idx_g2[i]]][mask2]
      label <- item_labels[[area]][i]
      row   <- run_ttest(v1, v2, "Item-Level", area, i, label, g1, g2)
```

```
      item_rows <- append(item_rows, list(row))
    }
  }
}

# ── 7. COMPOSITE (AREA-LEVEL) TESTS ──────────────────────────────────────
comp_rows <- list()

for (area in all_areas) {
  for (pair in pairs) {
    g1 <- pair[1]; g2 <- pair[2]
    if (!(area %in% group_areas[[g1]]) | !(area %in% group_areas[[g2]])) next
    mask1   <- role_col == role_values[[g1]]
    mask2   <- role_col == role_values[[g2]]
    idx_g1  <- col_map[[g1]][[area]]
    idx_g2  <- col_map[[g2]][[area]]
    comp1   <- rowMeans(df[mask1, idx_g1, drop = FALSE], na.rm = TRUE)
    comp2   <- rowMeans(df[mask2, idx_g2, drop = FALSE], na.rm = TRUE)
    comp1[is.nan(comp1)] <- NA
    comp2[is.nan(comp2)] <- NA
    label <- paste0("All items in ", area, " (composite mean)")
    row   <- run_ttest(comp1, comp2, "Composite (Area-Level)",
                       area, "ALL", label, g1, g2)
    comp_rows <- append(comp_rows, list(row))
  }
}

# ── 8. COMBINE & BONFERRONI ──────────────────────────────────────────────
results   <- bind_rows(item_rows, comp_rows)
alpha_lvl <- 0.05

n_item <- sum(results$Test_Type == "Item-Level")
n_comp <- sum(results$Test_Type == "Composite (Area-Level)")
bonf_item <- alpha_lvl / n_item
bonf_comp <- alpha_lvl / n_comp

cat("Item-level tests:", n_item, "→ Bonferroni α =", round(bonf_item, 5), "\n")
cat("Composite tests: ", n_comp, "→ Bonferroni α =", round(bonf_comp, 5), "\n")

results <- results %>%
  mutate(
    Bonferroni_alpha = case_when(
      Test_Type == "Item-Level"           ~ round(bonf_item, 5),
      Test_Type == "Composite (Area-Level)"~ round(bonf_comp, 5)
    ),
    Significance = case_when(
      is.na(p_value_raw)                  ~ "N/A",
      p_value_raw < Bonferroni_alpha      ~ paste0("* Bonferroni (p<",
                                             round(Bonferroni_alpha, 5), ")"),
      p_value_raw < alpha_lvl             ~ "* Raw only (p<0.05)",
      TRUE                                ~ "ns"
    )
  )
```

```
# ── 9. EXPORT ──────────────────────────────────────────────────────────────
write.csv(results, "ttest_results_planning_split.csv", row.names = FALSE)
cat("Saved: ttest_results_planning_split.csv\n")
cat("Total tests:", nrow(results), "\n")
```

DEPARTMENT DEPENDENT WELCH'S T-TESTS VS. ISO 56000 DIMENSIONS AND PERCEIVED INNOVATION PERFORMANCE

```
============================================================

library(readxl)
library(dplyr)

# ── 1. LOAD & RECODE ──────────────────────────────────────────────────────
df <- read_excel("Innovation-Management-Assessment-V5-Raw-Data.xlsx",
                 sheet = "Values")

recode_likert <- function(x) {
  x <- suppressWarnings(as.numeric(x))
  x[x == 1] <- NA
  x <- x - 1
  return(x)
}

item_cols <- 11:66
df[, item_cols] <- lapply(df[, item_cols], recode_likert)

role_col <- df[[10]]
dept_col <- df[[5]]

# ── 2. DEPARTMENT CLASSIFICATION ──────────────────────────────────────────
dept_text <- ifelse(is.na(df[[6]]), "", as.character(df[[6]]))

df$dept <- mapply(function(num, text) {
  if (is.na(num)) return(NA_character_)
  if (num == 7) {
    if (grepl("design",       tolower(text))) return("7_design")
    if (grepl("surveillance", tolower(text))) return("7_common_surv")
    return("7_other")
  }
  return(as.character(as.integer(num)))
}, dept_col, dept_text)

is_dept2 <- !is.na(dept_col) & dept_col == 2
is_rest  <- !is.na(dept_col) & dept_col != 2

cat("=== Department Groups ===\n")
cat("Dept 2: n =", sum(is_dept2), "\n")
cat("Rest:   n =", sum(is_rest),  "\n\n")

# ── 3. COLUMN INDEX MAP (Planning split) ──────────────────────────────────
op <- list(
```

```
  Context = c(11, 12), Leadership = c(13, 14, 15),
  Planning_Portfolio = c(16), Planning_Collab = c(17),
  Support = c(18, 19, 20), Operations = c(21, 22, 23),
  Performance = c(24, 25), Improvement = c(26)
)
mm <- list(
  Context = c(31, 32), Leadership = c(33, 34, 35),
  Planning_Portfolio = c(36), Planning_Collab = c(37),
  Support = c(38, 39, 40), Operations = c(41, 42, 43),
  Performance = c(44, 45), Improvement = c(46)
)
sl <- list(
  Context = c(51, 52), Leadership = c(53, 54, 55),
  Planning_Portfolio = c(56), Planning_Collab = c(57),
  Support = c(58, 59, 60), Operations = c(61, 62, 63),
  Performance = c(64, 65), Improvement = c(66)
)
col_map    <- list("1" = op, "2" = mm, "3" = sl)
role_values <- c("1" = 1, "2" = 2, "3" = 3)

# ── 4. ITEM LABELS ──────────────────────────────────────────────────────────
item_labels <- list(
  Context           = c("External trends (political/market/tech)",
                        "Internal factors (people/culture/structure)"),
  Leadership        = c("Culture of new ideas / psychological safety",
                        "Risk-taking for innovation",
                        "Tolerance of failure/experimentation"),
  Planning_Portfolio = c("Diverse project portfolio (risk/time horizon)"),
  Planning_Collab    = c("Internal & external collaboration planning"),
  Support           = c("Time allocated for innovation",
                        "Budget allocated for innovation",
                        "Tools & technology availability"),
  Operations        = c("Identifying innovation insights/opportunities",
                        "Validating & testing concepts",
                        "Commercialisation of solutions"),
  Performance       = c("KPIs for innovation/R&D outcomes",
                        "Continuous improvement of R&D processes"),
  Improvement       = c("Lessons-learned captured & used")
)

iso_areas    <- c("Context", "Leadership", "Planning_Portfolio",
                  "Planning_Collab", "Support", "Operations",
                  "Performance", "Improvement")
perf_outcomes <- c("Radical", "Incremental", "Customer", "Speed")

# ── 5. HELPERS: ISO SCORE EXTRACTION ────────────────────────────────────────
get_scores <- function(area) {
  scores <- rep(NA_real_, nrow(df))
  for (r in c("1", "2", "3")) {
    mask <- role_col == role_values[r]
    if (!area %in% names(col_map[[r]])) next
    cols <- col_map[[r]][[area]]
    vals <- rowMeans(df[mask, cols, drop = FALSE], na.rm = TRUE)
    vals[is.nan(vals)] <- NA
    scores[mask] <- vals
```

```
  }
  scores
}

get_item_scores <- function(area, item_i) {
  scores <- rep(NA_real_, nrow(df))
  for (r in c("1", "2", "3")) {
    mask <- role_col == role_values[r]
    if (!area %in% names(col_map[[r]])) next
    cols <- col_map[[r]][[area]]
    if (item_i > length(cols)) next
    scores[mask] <- as.numeric(df[[cols[item_i]]][mask])
  }
  scores
}

# ── 6. ATTRIBUTED PERFORMANCE SCORES FOR OE ──────────────────────────
perf_direct_cols <- list(
  MM = list(Radical = 27, Incremental = 28, Customer = 29, Speed = 30),
  SL = list(Radical = 47, Incremental = 48, Customer = 49, Speed = 50)
)
valid_depts <- c("2", "7_design")

dept_perf <- list()
for (dept in valid_depts) {
  dept_mask <- !is.na(df$dept) & df$dept == dept
  dept_perf[[dept]] <- list()
  for (outcome in perf_outcomes) {
    mm_vals  <- df[[perf_direct_cols$MM[[outcome]]]][dept_mask & role_col == 2]
    sl_vals  <- df[[perf_direct_cols$SL[[outcome]]]][dept_mask & role_col == 3]
    all_vals <- na.omit(c(mm_vals, sl_vals))
    dept_perf[[dept]][[outcome]] <- if (length(all_vals) > 0) mean(all_vals) else NA_real_
    dept_perf[[dept]][[paste0(outcome, "_n")]] <- length(all_vals)
  }
}

cat("=== Attributed performance scores by department ===\n")
for (dept in valid_depts) {
  cat(sprintf("Dept %s:\n", dept))
  for (outcome in perf_outcomes) {
    cat(sprintf("  %-12s M = %.3f (n = %d managers)\n", outcome,
                dept_perf[[dept]][[outcome]],
                dept_perf[[dept]][[paste0(outcome, "_n")]]))
  }
}
cat("\n")

get_perf_scores <- function(outcome) {
  scores <- rep(NA_real_, nrow(df))

  mm_mask <- role_col == 2
  scores[mm_mask] <- as.numeric(df[[perf_direct_cols$MM[[outcome]]]][mm_mask])

  sl_mask <- role_col == 3
  scores[sl_mask] <- as.numeric(df[[perf_direct_cols$SL[[outcome]]]][sl_mask])
```

```
  oe_mask <- role_col == 1 & df$dept %in% valid_depts
  scores[oe_mask] <- sapply(which(oe_mask), function(i) {
    dept_perf[[df$dept[i]]][[outcome]]
  })

  scores
}

# ── 7. HELPER: WELCH T-TEST → ROW ──────────────────────────────────────
run_ttest <- function(vec1, vec2, test_type, area, item_num, item_label) {
  vec1 <- vec1[!is.na(vec1)]
  vec2 <- vec2[!is.na(vec2)]
  if (length(vec1) < 2 | length(vec2) < 2) {
    return(data.frame(
      Test_Type = test_type, ISO_Area = area,
      Item_Number = as.character(item_num), Item_Label = item_label,
      Group_1 = "Dept2", Group_2 = "Rest",
      n_Dept2 = length(vec1), n_Rest = length(vec2),
      Mean_Dept2 = NA, Mean_Rest = NA,
      SD_Dept2 = NA, SD_Rest = NA,
      t_statistic = NA, p_value_raw = NA
    ))
  }
  res <- t.test(vec1, vec2, var.equal = FALSE)
  data.frame(
    Test_Type = test_type, ISO_Area = area,
    Item_Number = as.character(item_num), Item_Label = item_label,
    Group_1 = "Dept2", Group_2 = "Rest",
    n_Dept2 = length(vec1), n_Rest = length(vec2),
    Mean_Dept2  = round(mean(vec1),    3),
    Mean_Rest   = round(mean(vec2),    3),
    SD_Dept2    = round(sd(vec1),      3),
    SD_Rest     = round(sd(vec2),      3),
    t_statistic = round(res$statistic, 3),
    p_value_raw = round(res$p.value,   4)
  )
}

# ── 8. ITEM-LEVEL TESTS ────────────────────────────────────────────────
item_rows <- list()
for (area in iso_areas) {
  for (i in seq_len(length(item_labels[[area]]))) {
    scores <- get_item_scores(area, i)
    item_rows <- append(item_rows,
                        list(run_ttest(scores[is_dept2], scores[is_rest],
                                       "Item-Level", area, i, item_labels[[area]][i])))
  }
}

# ── 9. COMPOSITE (AREA-LEVEL) TESTS ────────────────────────────────────
comp_rows <- list()
for (area in iso_areas) {
  scores <- get_scores(area)
  comp_rows <- append(comp_rows,
```

```
                list(run_ttest(scores[is_dept2], scores[is_rest],
                               "Composite (Area-Level)", area,
                               "ALL", paste0(area, " (composite mean)"))))
}

# ── 10. PERFORMANCE OUTCOME TESTS ──────────────────────────────────────

perf_rows_full <- list()
for (outcome in perf_outcomes) {
  scores <- get_perf_scores(outcome)
  perf_rows_full <- append(perf_rows_full,
                           list(run_ttest(scores[is_dept2], scores[is_rest],
                                          "Performance Outcome (incl. OE attributed)", outcome,
                                          "1", outcome)))
}

is_mm_sl <- role_col %in% c(2, 3)

perf_rows_mmsl <- list()
for (outcome in perf_outcomes) {
  mm_col <- perf_direct_cols$MM[[outcome]]
  sl_col <- perf_direct_cols$SL[[outcome]]

  scores_mmsl <- rep(NA_real_, nrow(df))
  scores_mmsl[role_col == 2] <- as.numeric(df[[mm_col]][role_col == 2])
  scores_mmsl[role_col == 3] <- as.numeric(df[[sl_col]][role_col == 3])

  v1 <- scores_mmsl[is_dept2 & is_mm_sl]
  v2 <- scores_mmsl[is_rest  & is_mm_sl]

  perf_rows_mmsl <- append(perf_rows_mmsl,
                           list(run_ttest(v1, v2,
                                          "Performance Outcome (MM+SL only)", outcome,
                                          "1", outcome)))
}

# ── 11. COMBINE & BONFERRONI ──────────────────────────────────────────
results   <- bind_rows(item_rows, comp_rows,
                       perf_rows_full, perf_rows_mmsl)
alpha_lvl <- 0.05

n_item      <- sum(results$Test_Type == "Item-Level")
n_comp      <- sum(results$Test_Type == "Composite (Area-Level)")
n_perf_full <- sum(results$Test_Type == "Performance Outcome (incl. OE attributed)")
n_perf_mmsl <- sum(results$Test_Type == "Performance Outcome (MM+SL only)")

bonf_item      <- alpha_lvl / n_item
bonf_comp      <- alpha_lvl / n_comp
bonf_perf_full <- alpha_lvl / n_perf_full
bonf_perf_mmsl <- alpha_lvl / n_perf_mmsl

cat("Item-level tests:                ", n_item,      "→ Bonferroni α =", round(bonf_item,      5), "\n")
cat("Composite tests:                 ", n_comp,      "→ Bonferroni α =", round(bonf_comp,      5), "\n")
cat("Performance outcome (incl. OE):  ", n_perf_full, "→ Bonferroni α =", round(bonf_perf_full, 5),
"\n")
```

```
cat("Performance outcome (MM+SL only):  ", n_perf_mmsl, "→ Bonferroni α =",
round(bonf_perf_mmsl, 5), "\n\n")

results <- results %>%
  mutate(
    Bonferroni_alpha = case_when(
      Test_Type == "Item-Level"                          ~ round(bonf_item,      5),
      Test_Type == "Composite (Area-Level)"              ~ round(bonf_comp,      5),
      Test_Type == "Performance Outcome (incl. OE attributed)" ~ round(bonf_perf_full, 5),
      Test_Type == "Performance Outcome (MM+SL only)"    ~ round(bonf_perf_mmsl, 5)
    ),
    Significance = case_when(
      is.na(p_value_raw)            ~ "N/A",
      p_value_raw < Bonferroni_alpha ~ paste0("* Bonferroni (p<",
                                      round(Bonferroni_alpha, 5), ")"),
      p_value_raw < alpha_lvl       ~ "* Raw only (p<0.05)",
      TRUE                          ~ "ns"
    )
  )
# ── 12. PRINT SIGNIFICANT RESULTS ──────────────────────────────────────────
cat("=== Significant Results (raw p<0.05) ===\n")
sig <- results %>%
  filter(Significance != "ns", Significance != "N/A") %>%
  select(Test_Type, ISO_Area, Item_Label,
         Mean_Dept2, Mean_Rest, t_statistic,
         p_value_raw, Significance)
print(sig, row.names = FALSE)

# ── 13. EXPORT
──────────────────────────────────────────────────────────────────────────
write.csv(results, "dept_comparison_results_New.csv", row.names = FALSE)
cat("\nSaved: dept_comparison_results_New.csv\n")
cat("Total tests:", nrow(results), "\n")
```

TRITA-ITM-EX 2026:204
Stockholm, Sweden 2026

www.kth.se